SEVEN OPEN UNIT EQUILATERAL TRIANGLES DO NOT COVER A REGULAR UNIT HEXAGON

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ABSTRACT. We prove that seven open unit equilateral triangles cannot cover a regular unit hexagon. The center and vertices distinguish the covering triangles. Boundary gaps and midpoint coverage reduce the proof to trace-length, area-loss, and finite-enclosure obstructions. The enclosure arguments use six points for a selected boundary gap, four points for a supported midpoint, and nine points when the vertex triangles cover the boundary. A neighboring-ray bound avoids type distinctions along nonsupercritical paths. The two remaining disk-tangent inequalities are verified by an exact polynomial certificate over $\mathbb{Q}(\sqrt{3})$.

CONTENTS

1. INTRODUCTION AND PROOF OUTLINE

1.1. **Notation, theorem, and canonical labeling.** All distances, angles, and areas are Euclidean. A unit equilateral triangle has side length one; its open version is its interior. We write $[x, y]$ for the closed segment, $\mathrm{conv}K$ for the convex hull, and $\mathrm{int}K$ and ∂K for the interior and boundary. The relative interior $\mathrm{relint}\, K$ is its interior within its affine span. The measure $\mathcal{H}^1$ is one-dimensional Hausdorff measure, which agrees with length on segments. We use $[t]_+ = \max\{t, 0\}$ and $\mathrm{cross}(x, y) = x_1y_2 - x_2y_1$ for vectors in $\mathbb{R}^2$. A *trace* on a set E means its intersection with E.

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Let

$$O = (0,0), \qquad V_i = \left(\cos\frac{i\pi}{3}, \sin\frac{i\pi}{3}\right), \qquad H = \operatorname{conv}\{V_0, \dots, V_5\},$$

where indices lie in $\mathbb{Z}/6\mathbb{Z}$. Write

$$e_{i,i+1} = [V_i, V_{i+1}], \qquad r_i = [O, V_i], \qquad M_i = \frac{1}{2}V_i,$$

and put

$$S = \partial H \cup \bigcup_{i=0}^{5} r_i.$$

We call S the *hexagon skeleton.* For $X \in \mathbb{R}^2$ and $r \geq 0$, write $\mathbb{B}(X,r) = \{Y : \|Y - X\| \leq r\}$ for the closed disk.

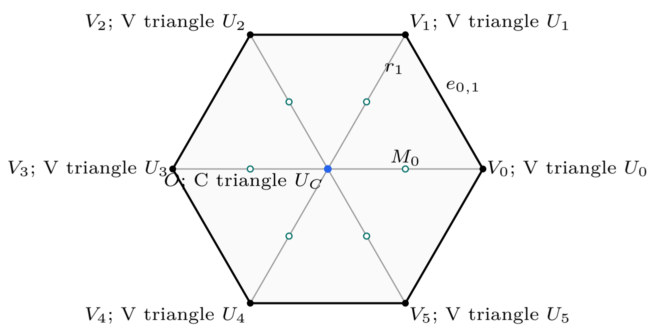


FIGURE 1. The center, vertices, radial segments, and radial midpoints of H.

For a nonempty compact set $K \subset \mathbb{R}^2$, define its *equilateral enclosure side* by

$$\Lambda(K) := \inf\{s > 0 : K \subset T \text{ for a closed equilateral triangle } T \text{ of side } s\}. \tag{1}$$

Then

$$\mathcal{H}^1(\partial H) = 6, \qquad \mathcal{H}^1\left(\bigcup_{i=0}^{5} r_i\right) = 6, \qquad \mathcal{H}^1(S) = 12. \tag{2}$$

Theorem 1.1 (Main theorem)**.** *The hexagon H cannot be covered by seven open equilateral triangles of side length one.*

Corollary 1.2 (Soifer's Hexagon Problem)**.** *For every $L > 1$, the regular hexagon $H_L = LH$ cannot be covered by seven closed unit equilateral triangles.*

By lemma 2.3, a hypothetical cover has a unique labeling

$$U_C, U_0, \dots, U_5$$

such that

$$O \in U_C, \qquad V_i \in U_i.$$

We call U_C the *C triangle* and U_i the *V triangle at V_i*. Set

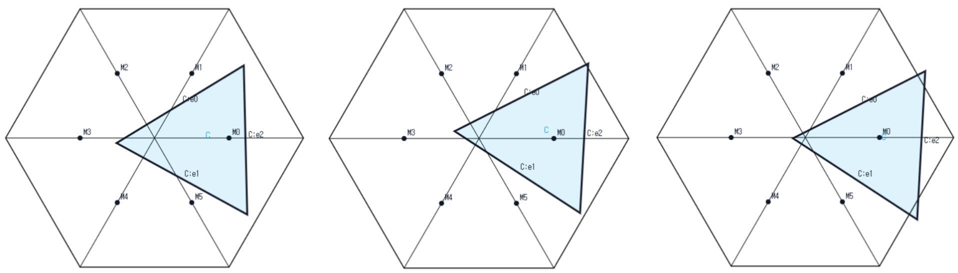
$$T_C = \overline{U_C}, \qquad T_i = \overline{U_i}.$$

A *role* means one of these labeled triangles. Statements about an arrangement of roles assume $O \in U_C$ and $V_i \in U_i$; coverage of H, S, or ∂H is an additional hypothesis, stated when needed. A *skeleton cover* covers S, not necessarily all of H. An edge is *center-free* when it is disjoint from U_C.

1.2. **Finite classifications.** Set

$$N_E(T_C) := \left|\left\{i \in \mathbb{Z}/6\mathbb{Z} : \mathcal{H}^1(T_C \cap e_{i,i+1}) > 0\right\}\right|.$$

Proposition 2.4 gives $N_E(T_C) \in \{0,1,2\}$; we call these types CE0, CE1, CE2, respectively.



(a) CE0 (b) CE1 (c) CE2

FIGURE 2. Examples of the three C triangle classes. From left to right, the C triangle has positive-length trace on zero, one, and two edges of ∂H; point contacts do not count. The panels illustrate the classification.

For the V triangle T_i, let $o(T_i)$ be the number of its triangle vertices outside H and put

$$n(T_i) = \left|\left\{j \in \{i-1, i+1\} : \mathcal{H}^1(T_i \cap r_j) > 0\right\}\right|,$$

with indices modulo 6. The raw pair $(3,0)$ says that all three vertices of T_i lie outside H; it does not say that T_i is disjoint from H, since $V_i \in \operatorname{int}(T_i) \cap H$. Proposition 2.6 replaces every such triangle by a same-orientation translate with exactly the same trace in H, still containing V_i in its interior, and with at most two vertices outside H. We make this exact-trace normalization before defining any reach. The normalized exhaustive types used below are

$(o(T_i), n(T_i))$	established type
$(1,0)$	Vd0
$(2,0)$	Vd0
$(1,1)$	Vd1
$(1,2)$	Vd2
$(2,1)$	T3-like.

The actual maximal reaches are

$$A_i := \max\{x \in [0,1] : V_i + x(V_{i-1} - V_i) \in T_i\},$$
$$B_i := \max\{x \in [0,1] : V_i + x(V_{i+1} - V_i) \in T_i\},$$
$$C_i := \max\{x \in [0,1] : V_i + x(O - V_i) \in T_i\}.$$

The same numbers are the suprema of the corresponding reaches in the open triangle U_i. The V triangle T_i is *supercritical* when $A_i + B_i > 1$, and *nonsupercritical* when $A_i + B_i \le 1$. Put

$$N_+ := \left|\{i \in \mathbb{Z}/6\mathbb{Z} : A_i + B_i > 1\}\right|. \tag{3}$$

Lowercase reaches are used only after a selection satisfying

$$0 \le a_i \le A_i, \qquad 0 \le b_i \le B_i, \qquad 0 \le c_i \le C_i.$$

They never define N_+.

Parametrize $e_{i,i+1}$ from V_i to V_{i+1} by

$$X_i(x) := V_i + x(V_{i+1} - V_i), \qquad 0 \le x \le 1.$$

Define the *boundary gap*

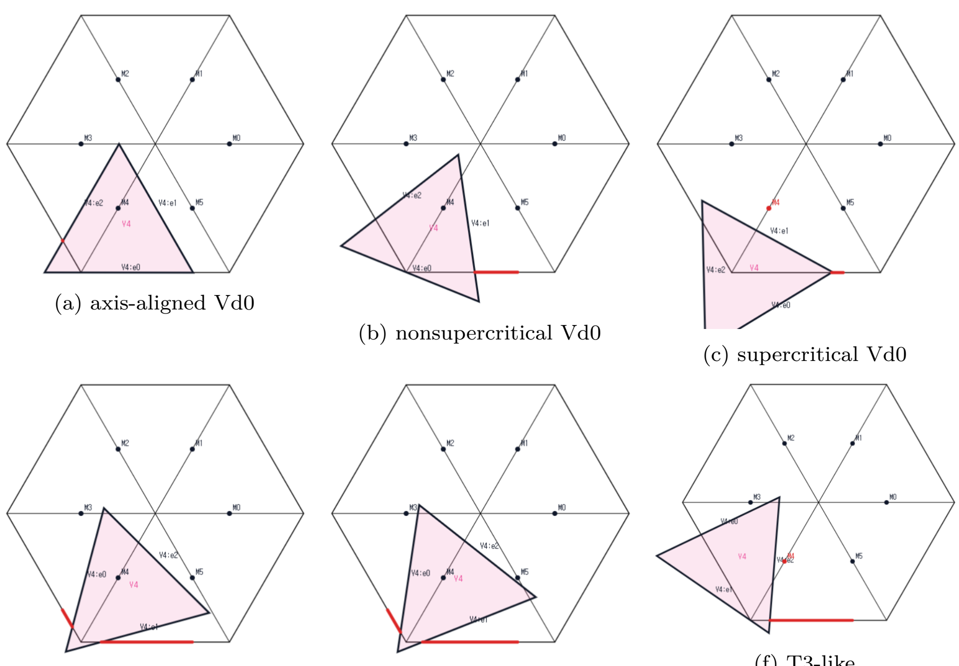
$$J_i := e_{i,i+1} \setminus (U_i \cup U_{i+1}), \qquad N_{\mathrm{gap}} := |\{i : J_i \neq \varnothing\}|.$$

Lemma 2.10 gives $J_i = X_i([B_i, 1-A_{i+1}])$ when $B_i + A_{i+1} \le 1$, and $J_i = \varnothing$ otherwise. Equality is a singleton gap, not an overlap.

A V role has *positive adjacent-radial support*, or simply *positive support*, when $n(T_i) > 0$. Put

$$N_{\mathrm{sp}} := |\{i : n(T_i) > 0\}|.$$

Positive-support roles are nonsupercritical, so the number of roles that are supercritical or have positive adjacent-radial support is $N_+ + N_{\mathrm{sp}}$. The skeleton budget excludes nonzero-gap covers with $N_+ + N_{\mathrm{sp}} \ge 3$.



(a) axis-aligned Vd0 (b) nonsupercritical Vd0 (c) supercritical Vd0

(d) Vd1 (e) Vd2 (f) T3-like

FIGURE 3. Examples of the normalized V triangle types. The top row shows an axis-aligned Vd0, a nonsupercritical Vd0 with $A_i + B_i \le 1$, and a supercritical Vd0 with $A_i + B_i > 1$. The bottom row shows Vd1, Vd2, and T3-like. The panels illustrate the classification.

1.3. **Exhaustive routing.** When the C triangle has type CE1 or CE2, proposition 2.15 will show that it contains exactly one of $M_0, \dots, M_5$; this is the *C midpoint*. If it is M_k, the role at V_k is called *center-based*; a supercritical role there is *center-aligned*, and a role at another vertex is *away* or *off-center*. A V role that contains an adjacent midpoint openly is called its *supplier* or *rescuer*. A *gap-free nonsupercritical path* is a consecutive sequence of nonsupercritical V roles whose shared edges have empty gaps.

We use BC for the selected-gap six-point construction, D for the supported-midpoint four-point construction, and F for the zero-gap nine-point construction, all defined in Section 5. N0 denotes theorem 5.10, the assertion that every skeleton cover has a supercritical V role.

Zero gaps mean that the V triangles cover ∂H, and the proof splits according to N_+. With a gap, the C triangle is CE1 or CE2; N0 and the skeleton budget leave one supercritical role and at most one positive-support role. Its position relative to the C midpoint determines the remaining obstruction.

N_{gap}	N_+	Additional condition	Closing mechanism
0	0	Arbitrary V types	Strict overlap
0	1	Arbitrary V types	Nine-point F
0	≥ 2	Arbitrary V types	Cyclic area loss
≥ 1	0	Arbitrary V types	N0, through BC
≥ 1	≥ 1	$N_+ \geq 2$ or $N_+ + N_{\text{sp}} \geq 3$	Skeleton budget
≥ 1	1	$N_{\text{sp}} \leq 1$	Midpoint supplier: BC, D, replacement, or Vd2 length

TABLE 1. Disjoint routes for the original open cover. Nonzero gaps force CE1 or CE2; both classes share the same placement reduction. Replacement preserves the skeleton and ends at N0, without preserving number of gaps.

The rows are disjoint and exhaustive: separate zero from nonzero gaps, then $N_+ = 0$, and finally the displayed high-count condition. Lemma 5.11 resolves the last row. BC uses the same selected-gap witness in CE1 and CE2, with separate candidate calculations.

1.4. **Organization of the proof.** Section 2 proves the structural classification, strict handoff selection, and midpoint facts. Appendix A proves the signed-center normal form used by the later arguments. Section 3 gives the complete trace-length argument. Section 4 proves the local and cyclic area-loss estimates. Section 5 gives the nonzero-gap and zero-gap enclosure obstructions. Section 6 then applies the zero-gap alternatives before the nonzero-gap midpoint-supplier reduction of Table 1 and closes the proof. Appendices A–E supply the corresponding solved optimizations; Appendix F is the exact polynomial certificate for the two mixed zero-gap intersections.

2. Common Geometry and Structural Reductions

We prove the structural facts used by all three mechanisms. The coordinate calculations are in Appendix A.

2.1. **Open and closed formulations.**

Lemma 2.1 (Compact covering margin)**.** *If a nonempty compact set K is covered by finitely many open unit equilateral triangles U_ν, then moving every side line inward by the same sufficiently small positive distance gives a cover by closed equilateral triangles of side less than one. In particular, $K \subset U$ implies $\Lambda(K) < 1$, with Λ defined in* (1)*.*

Proof. The continuous function $m(x) = \max_\nu \operatorname{dist}(x, \mathbb{R}^2 \setminus U_\nu)$ has positive minimum η on K. Move every side line inward by $0 < \rho < \min\{\eta, \sqrt{3}/6\}$. The resulting closed triangles have side $1 - 2\sqrt{3}\rho$, and each $x \in K$ remains in a triangle realizing $m(x) \geq \eta > \rho$. Conversely, if $\Lambda(K) < 1$, choose a closed enclosure of side below one and enlarge it about its incenter to side one. The smaller closed triangle lies in the interior of the enlarged triangle. □

Proposition 2.2 (Compactness and scaling equivalence)**.** *The following assertions are equivalent.*

(1) *H is covered by seven open equilateral triangles of side* 1*.*
(2) *For some $\varepsilon > 0$, H is covered by seven closed equilateral triangles of side $1 - \varepsilon$.*
(3) *For some $L > 1$, $H_L := LH$ is covered by seven closed equilateral triangles of side* 1*.*

Proof. Apply lemma 2.1 to obtain (1)⇒(2). Enlarging the smaller closed triangles about their incenters gives (2)⇒(1); dilation by $(1-\varepsilon)^{-1}$ identifies (2) and (3). □

Lemma 2.3 (Distinct triangles)**.** *The seven triangles can be labelled so that*

$$O \in U_C, \qquad V_i \in U_i \quad (0 \leq i \leq 5),$$

and these seven triangles are distinct. Consequently

$$O \in \operatorname{int}(T_C), \qquad V_i \in \operatorname{int}(T_i).$$

Proof. The seven points $O, V_0, \ldots, V_5$ are pairwise at distance at least 1, whereas two points of an open unit equilateral triangle are at distance strictly less than its diameter 1. Selecting a covering triangle for each point therefore gives a bijection. □

2.2. **C triangle and V triangle classifications.**

Point contacts have zero trace length, so they do not create an additional positive boundary trace in the definition of $N_E(T_C)$.

Proposition 2.4 (Exhaustive center types)**.** *The closed C triangle T_C has exactly one of the types* CE0, CE1, CE2*; equivalently,*

$$N_E(T_C) \leq 2.$$

Proof. If $N_E(T_C) \geq 3$, two of the traced edges are nonadjacent in the boundary six-cycle. Choose a relative-interior point from each positive-length trace. By lemma A.1, those points are more than unit distance apart, so a unit equilateral triangle cannot contain both. □

Lemma 2.5 (Local wedge)**.** *Every original V triangle has $1 \leq o(T_i) \leq 3$. If $\mathcal{H}^1(T_i \cap r_j) > 0$ for some $j \in \{i-1, i+1\}$, at least one vertex of T_i belongs to H; if both adjacent radial traces have positive length, at least two vertices belong to H.*

Proof. This is the wedge-incidence calculation in lemma A.3. The incidence statement concerns the closures; openness is used separately when forcing points. □

Raw roles with $(o(T_i), n(T_i)) = (3, 0)$ admit an exact-trace normalization.

Proposition 2.6 (Exact-trace normalization of raw $(3, 0)$ roles)**.** *Let U be an original open V triangle at V_i, put $T = \overline{U}$, and suppose $(o(T), n(T)) = (3, 0)$. There is a same-orientation translate $\widehat{U}$, with closure $\widehat{T}$, such that*

$$V_i \in \widehat{U}, \qquad (o(\widehat{T}), n(\widehat{T})) = (2, 0), \tag{4}$$

and

$$T \cap H = \widehat{T} \cap H, \qquad U \cap H = \widehat{U} \cap H. \tag{5}$$

In particular, all three actual maximal reaches and the inequality $A_i + B_i > 1$ are unchanged.

Proof. The two-orientation affine calculation and the trace-preserving translation are given in lemma A.4. □

We apply this translation simultaneously and reuse U_i, T_i for the normalized triangles.

Proposition 2.7 (Exhaustive normalized vertex types)**.** *Every normalized V triangle has exactly one of the four types* Vd0, Vd1, Vd2, T3-like *in the exact dictionary of the introduction.*

Proof. Lemma 2.5 gives $o \leq 2$ when $n \geq 1$ and $o \leq 1$ when $n = 2$, while $o \geq 1$. Proposition 2.6 replaces the only additional raw pair, $(3, 0)$, by $(2, 0)$. □

The next construction is a trace majorant, not a new open covering triangle.

Proposition 2.8 (T3-like closed-trace domination)**.** *Let T be the closure of an original* T3-like *V triangle at V_i. There is a same-orientation translate $\widehat{T}$ such that*

$$V_i \in \operatorname{relint}(\textit{a side of } \widehat{T}), \qquad T \cap H \subsetneq \widehat{T} \cap H,$$

and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$, while

$$\operatorname{int}(T) \cap (H \setminus \{V_i\}) \subseteq \operatorname{int}(\widehat{T}).$$

Since V_i becomes a boundary point, $\widehat{T}$ is only a closed-trace majorant.

Proof. See the two-orientation feasibility calculation in lemma A.5. □

2.3. **Initial reaches and boundary gaps.**

Lemma 2.9 (Initial reach segments)**.** *On the three segments from V_i toward V_{i-1}, V_{i+1}, O, the closed parameter sets are respectively*

$$[0, A_i], \qquad [0, B_i], \qquad [0, C_i],$$

and the open parameter sets are

$$[0, A_i), \qquad [0, B_i), \qquad [0, C_i).$$

Thus the closed maxima equal the corresponding open-triangle suprema.

Proof. The other endpoint of each segment is a distinguished point at distance one from V_i, and is not even in T_i: otherwise moving V_i slightly away from it inside U_i would exceed diameter one. Thus $0 < A_i, B_i, C_i < 1$. Each intersection with a segment issuing from V_i is convex. Closedness of T_i, openness of U_i, and $T_i = \overline{U_i}$ give the stated endpoints. □

Lemma 2.10 (Exhaustiveness of the gap split)**.** *One has*

$$J_i = \begin{cases} X_i([B_i, 1 - A_{i+1}]), & B_i + A_{i+1} \le 1, \\ \varnothing, & B_i + A_{i+1} > 1. \end{cases}$$

Every nonempty gap, including a singleton, is missed by all open V triangles. Under a hypothetical cover it lies in U_C and creates a positive-length C triangle trace. Consequently a closed C triangle T_C of type CE0, CE1, CE2 *permits respectively at most* $0, 1, 2$ *gaps, and*

$$N_{\mathrm{gap}} = 0 \iff U_0, \dots, U_5 \text{ cover } \partial H.$$

Proof. By lemma 2.9, the incident open traces have parameters $[0, B_i)$ and $(1 - A_{i+1}, 1]$. Their complement is the stated closed interval. A nonincident V triangle cannot contain a relative-interior point of the edge: that point is more than unit distance from its distinguished vertex. Hence a gap lies in U_C; openness turns even a singleton V-gap into a positive-length C trace. Proposition 2.4 gives the count and the final equivalence. □

Lemma 2.11 (T3-like V triangles are nonsupercritical)**.** *Every original* T3-like *V triangle satisfies*

$$A_i + B_i \le 1. \tag{6}$$

Moreover $M_i \notin T_i$. In particular, a T3-like role is not counted by N_+ and cannot supply its own midpoint.

Proof. Both assertions follow from the Type-II calculation in lemma A.5. □

2.4. **Strict handoffs and midpoint forcing.**

Proposition 2.12 (Strict boundary handoffs)**.** *Assume that $U_0, \dots, U_5$ cover ∂H. There are choices*

$$\xi_i \in (1 - A_{i+1}, B_i), \qquad a_i := 1 - \xi_{i-1}, \qquad b_i := \xi_i,$$

such that

$$0 < a_i < A_i, \qquad 0 < b_i < B_i, \qquad a_i^2 + a_i b_i + b_i^2 < 1.$$

The selected anchors lie in U_i. If exactly one actual V triangle is supercritical, every strict selection has that same unique selected supercritical index. If at least two actual V triangles are supercritical, some simultaneous strict selection retains at least two.

Proof. The overlap intervals and the cyclic selector are proved in lemma A.6. Geometrically, each ξ_i lies in the overlap of two adjacent open edge traces, which makes both associated reach demands strict. □

Lemma 2.13 (Boundary deficit and overlap identities)**.** *For the actual reaches put* $s_i = 1 - A_i - B_i$ *and* $\omega_i = B_i + A_{i+1} - 1$. *Then*

$$A_{i+1} - A_i = s_i + \omega_i, \qquad B_i - B_{i+1} = s_{i+1} + \omega_i.$$

On a nonsupercritical gap-free path the A_i *increase and the* B_i *decrease strictly. Weak handoffs give the weak version. Around the full cycle,* $\sum_i \omega_i = -\sum_i s_i$.

Proof. Expand the definitions and telescope. Nonsupercriticality gives $s_i \geq 0$, and a gap-free edge has $\omega_i > 0$ because its incident traces are open. A supercritical role has $s_i < 0$; propagation through it is not licensed. ☐

Lemma 2.14 (Self-midpoint obstruction)**.** *Let* u, v *be unit vectors meeting at* $120°$, *and let* $a, b \in [0, 1]$. *If a closed unit equilateral triangle contains*

$$0, \qquad au, \qquad bv, \qquad \frac{u+v}{2},$$

then $a + b \leq 1$. *Consequently*

$$M_i \in T_i \Longrightarrow A_i + B_i \leq 1.$$

If the boundary points and midpoint are contained with positive margin, the conclusion is strict.

Proof. This is the admissible-set corollary lemma A.12. ☐

Proposition 2.15 (CE1/CE2 exactly-one-midpoint property)**.** *If* $O \in \operatorname{int}(T_C)$ *and* T_C *is* CE1 *or* CE2, *then*

$$|T_C \cap \{M_0, \ldots, M_5\}| = 1.$$

This midpoint belongs to U_C. *Both possible positive boundary traces are on the two hexagon edges incident to the same vertex.*

Proof. Normalize one positive boundary trace. The signed-center optimization proposition A.14 gives the exact midpoint set $\{M_0\}$, with radial exit $d_0^C > 1/2$ and positive side slacks at M_0. Its boundary traces are confined to $e_{5,0}, e_{0,1}$. Undoing the symmetry proves the assertions. ☐

The interior hypothesis is essential: $\operatorname{conv}\{O, V_0, V_1\}$ has a boundary edge overlap and contains both M_0 and M_1, but O is not interior to it.

3. Trace-Length Mechanism

For a closed triangle T and a rectifiable target E, define

$$L_E(T) := \mathcal{H}^1(T \cap E).$$

Lemma 3.1 (Connected open-cover budget)**.** *Let* X *be compact and connected, with a finite Borel measure* μ *positive on every nonempty relatively open subset. Let* $U_1, \ldots, U_m$ *be open sets covering* X, *put* $T_j = \overline{U_j}$, *and assume at least two of the relative pieces* $U_j \cap X$ *are nonempty. Then*

$$\mu(X) < \sum_j \mu(U_j \cap X) \leq \sum_j \mu(T_j \cap X).$$

Consequently closed-piece upper bounds whose sum is at most $\mu(X)$ *already contradict the open cover.*

Proof. If the nonempty relative pieces were pairwise disjoint, one and the union of the others would disconnect X. Some pair therefore overlaps in a nonempty relatively open set, of positive measure. Integrating the covering multiplicity gives strict inequality. Closure only enlarges each piece. □

For $X = \partial H$ or S, use length; for $X = H$, use area. Each measure has the stated full-support property. The six V roles meet each target in positive measure; the C role does so on H and S, but need not meet ∂H. In particular,

$$\mathcal{H}^1(E) < L_E(T_C) + \sum_{i=0}^{5} L_E(T_i), \qquad E \in \{\partial H, S\}, \tag{7}$$

whenever these targets are covered. Empty center boundary traces are simply omitted from the collection. The local geometric caps remain separate from this common final budget argument.

FIGURE 4. The perimeter and full-skeleton targets. The skeleton is used only under the hypotheses of the corresponding trace budget.

3.1. Skeleton bounds.

Lemma 3.2 (Center skeleton cap). *If* $N_E(T_C) \in \{1, 2\}$ *and*

$$|T_C \cap \{M_0, \ldots, M_5\}| = 1,$$

then

$$L_S(T_C) < \frac{3}{2}.$$

Proof. This is the signed-center trace optimization lemma B.4. □

Lemma 3.3 (Positive-support skeleton cap). *Let* T_i *be the closure of an original open V triangle. If*

$$\exists j \in \{i-1, i+1\} \quad \mathcal{H}^1(T_i \cap r_j) > 0,$$

then

$$L_S(T_i) < \frac{3}{2}.$$

Proof. Normalize to $i = 0$ and translate the hexagon by V_0. The five components of S on which a unit triangle containing V_0 can have positive trace become the corresponding local components of the translated skeleton. The positive adjacent-radial trace becomes a positive boundary trace there. Proposition 2.15 and Lemma 3.2 give the bound; every nonlocal component is more than unit distance from V_0, apart from zero-measure endpoints. □

Lemma 3.4 (No-support skeleton cap)**.** *If*

$$A_i + B_i \leq 1,$$
$$\mathcal{H}^1(T_i \cap r_{i-1}) = 0, \qquad \mathcal{H}^1(T_i \cap r_{i+1}) = 0,$$

then

$$L_S(T_i) \leq 2.$$

Proof. For $s > 0$ and $j \notin \{i-1, i, i+1\}$, $\|V_i - sV_j\|^2 = 1 + s^2 - 2s\cos((j-i)\pi/3) > 1$. Thus only the incident boundary traces and own radial trace contribute; their total is $A_i + B_i + C_i \leq 2$. □

Lemma 3.5 (Supercritical skeleton cap)**.** *If $A_i + B_i > 1$, then*

$$L_S(T_i) < \frac{3}{2}.$$

Proof. The exact admissible-cell and polynomial-sign calculation is lemma B.5. □

Lemma 3.6 (Positive-support rescuer)**.** *Assume that the open roles cover S, that $N_E(T_C) \in \{1, 2\}$, and, after symmetry,*

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}.$$

If $N_+ \geq 2$, then for a supercritical index $s \neq 0$ there is $j \in \{s-1, s+1\}$ such that

$$M_s \in U_j, \qquad \mathcal{H}^1(T_j \cap r_s) > 0.$$

Proof. Choose a supercritical $s \neq 0$. Lemma 2.14 gives $M_s \notin T_s$, and the displayed midpoint identity gives $M_s \notin T_C$. The open cover therefore puts M_s in another U_j. Diameter locality restricts j to $s-1$ or $s+1$, and openness makes its trace on r_s have positive length. □

Theorem 3.7 (Direct CE1/CE2 skeleton count)**.** *If*

$$N_E(T_C) \in \{1, 2\},$$
$$|T_C \cap \{M_0, \dots, M_5\}| = 1,$$

and

$$N_+ + N_{\mathrm{sp}} \geq 3, \tag{8}$$

then the seven triangles do not cover S.

Proof. By proposition B.1 and lemma 2.11, every positive-support role is nonsupercritical. Hence the two index sets are disjoint. With $k = N_+ + N_{\mathrm{sp}}$, the preceding caps give

$$L_S(T_C) + \sum_i L_S(T_i) < \frac{3}{2} + \frac{3}{2}k + 2(6-k) = \frac{27-k}{2} \leq 12,$$

contrary to (7). □

3.2. **Boundary trace bounds.**

The following bounds suffice for the perimeter exit.

Theorem 3.8 (Boundary trace bounds)**.** *For closures of the original open roles,*

$$L_{\partial H}(T_C) = 0 \quad \text{(CE0)}, \qquad L_{\partial H}(T_C) < \tfrac{1}{2} \quad \text{(CE1 } \textit{or} \text{ CE2)},$$

and

$$L_{\partial H}(T_i) = A_i + B_i \leq \frac{2}{\sqrt{3}}. \tag{9}$$

A nonsupercritical role contributes at most 1*; a Vd1 or Vd2 role contributes less than* 1/2*.*

Proof. Boundary locality and the diameter bound give $A_i^2 + A_iB_i + B_i^2 \leq 1$, hence $3(A_i + B_i)^2/4 \leq 1$. The remaining caps are corollary A.16 and proposition B.1. □

Proposition 3.9 (Vd2 adjacent-midpoint branch)**.** *Suppose the C triangle is CE1 or CE2 and $N_+ = 1$. If a Vd2 role contains an adjacent midpoint, the perimeter is not covered.*

Proof. By lemma B.3, that role contributes strictly less than 1/3. The common boundary caps give

$$L_{\partial H}(T_C) + \sum_i L_{\partial H}(T_i) < \frac{1}{2} + \frac{2}{\sqrt{3}} + \frac{1}{3} + 4 < 6.$$

Apply lemma 3.1. □

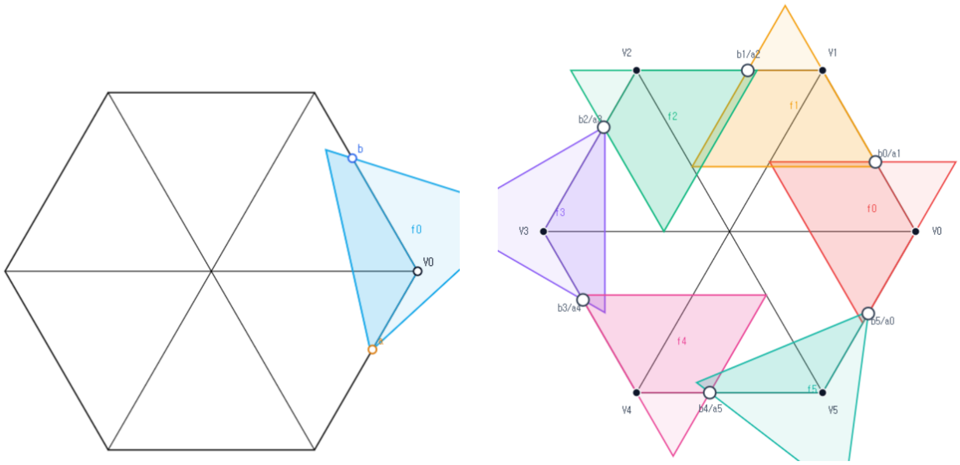
4. Area-Loss Estimate

The area-loss method records how much of each V triangle must lie outside H. The local estimate applies to every V triangle. A final cyclic argument adds these losses without imposing a center type.

Normalize area by $A_\triangle = \sqrt{3}/4$ and write

$$\mathcal{L}(T) := \frac{\operatorname{area}(T \setminus H)}{A_\triangle}.$$

The hexagon has normalized area six, and a unit equilateral triangle has normalized area one.



(a) local retained area (b) cyclic retained areas

FIGURE 5. The area-loss method. The dark regions indicate the portions retained in H; the loss $\mathcal{L}(T)$ is the area outside H.

4.1. **Local area loss.**

Theorem 4.1 (Local square loss)**.** *Let T be a closed unit equilateral triangle containing V_i, $V_i + a(V_{i-1} - V_i)$, and $V_i + b(V_{i+1} - V_i)$, where $a, b \in [0,1]$. Then*

$$\mathcal{L}(T) \geq \min\{a,b\}^2, \tag{10}$$

and, if $a + b > 1$,

$$\mathcal{L}(T) \geq \max\{a,b\}^2. \tag{11}$$

Proof. The two exhaustive orientation families give these lower bounds. Their polygon areas and the factorizations proving both inequalities are computed in lemma C.1. □

4.2. **Cyclic accumulation.** Assume here that the six V triangles cover ∂H, and choose strict handoffs ξ_i, a_i, b_i as in proposition 2.12. An *ascent* at index i means $\xi_{i-1} < \xi_i$, equivalently $a_i + b_i > 1$.

Lemma 4.2 (Cyclic area-loss certificate)**.** *The six actual V triangles satisfy*

$$|\{i \in \mathbb{Z}/6\mathbb{Z} : a_i + b_i > 1\}| \geq 2 \quad \Longrightarrow \quad \sum_{i=0}^{5} \mathcal{L}(T_i) > 1.$$

Proof. Put $m = \min_i \xi_i$ and $M = \max_i \xi_i$. The reflection $y_i = 1 - \xi_{-i-1}$ swaps each V triangle's two coordinates and preserves the number of ascents and the total loss. Hence assume $m \leq 1 - M$. Every selected V triangle coordinate is then at least m, so (10) gives loss at least m^2 for every V triangle.

Choose an ascent out of a minimum plateau, $\xi_{p-1} = m < \xi_p$. Its V triangle contains the coordinate $1 - m$ and loses at least $(1-m)^2$ by (11). A second ascent has one coordinate strictly larger than $1/2$ and hence loses strictly more than $1/4$.

The other four V triangles lose at least m^2 each, so the total loss is strictly larger than

$$4m^2 + (1-m)^2 + \frac{1}{4} = 5\left(m - \frac{1}{5}\right)^2 + \frac{21}{20} > 1.$$

□

Proposition 4.3 (Branch closed by area). *The branch $N_{\mathrm{gap}} = 0$, $N_+ \geq 2$ is impossible.*

Proof. The six V triangles cover ∂H. The at-least-two clause of proposition 2.12 supplies the selection for lemma 4.2. Thus the V triangles have total normalized inside area below five. The C triangle contributes at most one, whereas H has normalized area six. Apply lemma 3.1 with area. □

5. Finite-Enclosure Obstructions

The finite-enclosure method finds compact sets that the V triangles miss. Under a hypothetical cover those sets lie in the open C triangle, while their least equilateral enclosure has side at least one.

Put

$$h := \frac{\sqrt{3}}{2}.$$

The closed-disk notation and Λ are defined in Section 1. By lemma 2.1, a compact set contained in an open unit triangle has enclosure side below one. Conversely, $\Lambda(K) < 1$ gives such an open unit enclosure by slightly enlarging a smaller closed enclosure. Thus it is enough to exclude every open unit candidate containing the same fixed set K.

5.1. **Local capacities and radial witnesses.** Normalize at V_0 and define the four-anchor hull

$$K(a,b,c) := \operatorname{conv}\{V_0, X_5(1-a), X_0(b), (1-c)V_0\},$$
$$\mathcal{A} := \{(a,b,c) \in [0,1]^3 : \Lambda(K(a,b,c)) \leq 1\}.$$

For

$$0 \leq a, b \leq 1, \qquad a^2 + ab + b^2 \leq 1,$$

put

$$c_{\max}(a,b) := \max\{c \in [0,1] : (a,b,c) \in \mathcal{A}\}.$$

For $0 \leq a, c \leq 1$, also put

$$M_c(a) := \max\{b \in [0,1] : (a,b,c) \in \mathcal{A}\}.$$

In particular, the *diameter envelope* is

$$M_0(x) = \frac{-x + \sqrt{4 - 3x^2}}{2} \qquad (0 \leq x \leq 1). \tag{12}$$

Here M_0 without an argument denotes the radial midpoint, whereas $M_0(x)$ denotes this scalar function. For later rescuer estimates define

$$M_c^{\sup} := \frac{c + \sqrt{c^2 - 8c + 4}}{2} \qquad (0 \leq c \leq 1/2). \tag{13}$$

The value at $c = 1/2$ is a continuous endpoint convention; no supercritical triangle has that radial demand. The bound associated with this envelope is proved in lemma A.13. The exact admissible set and radial envelope are proved in proposition A.11.

The neighboring-arm capacities are the variational quantities

$$C_+(a,b) := \max\left\{ c \in [0,1] : \begin{array}{l} \exists T \ [T \text{ is a closed unit equilateral triangle,} \\ \{V_0, X_5(1-a), X_0(b), (1-c)V_1\} \subseteq T] \end{array} \right\},$$

$$C_-(a,b) := C_+(b,a), \qquad 0 \le a,b \le 1, \qquad a+b \le 1.$$

Only a bound, not the full piecewise neighboring capacity, is needed below. The direct four-caliper proof is lemma D.1.

Lemma 5.1 (Common-pair domination)**.** *If*

$$p,q \ge 0, \qquad p+q \le 1,$$

then

$$C_+(p,q), C_-(p,q) \le 1 - \min\{p,q\} \le c_{\max}(p,q).$$

Proof. This is the direct diagonal-capacity comparison in lemma D.2. □

Definition 5.2 (Fixed total radial endpoint)**.** Parametrize r_i by $Z_i(c) = (1-c)V_i$, measured from V_i toward O. Let $u_{j\to i}$ be the largest parameter in the actual positive trace of T_j on r_i for $j = i-1, i+1$, with value zero when no such positive trace exists. Put

$$\gamma_i := \max\{C_i, u_{i-1\to i}, u_{i+1\to i}\}, \qquad d_i := 1-\gamma_i, \qquad \widehat{P}_i := d_i V_i.$$

Thus $\widehat{P}_i$ is the innermost endpoint reached by any local V triangle on r_i, rather than just the endpoint of the own V triangle. This distinction is essential when a nonsupercritical V triangle supports an adjacent radial segment.

Lemma 5.3 (Fixed radial forcing and demand recovery)**.** *One has $0 < d_i < 1$ and $\widehat{P}_i \notin \bigcup_j U_j$. Under skeleton coverage, $\widehat{P}_i \in U_C$. More generally, for every $\gamma_i \le \Gamma \le 1$, the point $(1-\Gamma)V_i$ is missed by every open V triangle and is center-forced under skeleton coverage. If an arbitrary open unit candidate U' contains O and $\widehat{P}_i$, and its exit on r_i is d'_i measured from O, then $\gamma_i > 1 - d'_i$. If both neighboring endpoints are at most $1-d'_i$, it follows that*

$$C_i > 1 - d'_i.$$

It is sufficient that each neighboring V triangle be nonsupercritical with both actual boundary reaches greater than d'_i.

Proof. A local open triangle containing the total endpoint would extend its trace farther toward O. Nonlocal triangles are excluded by diameter. The last assertion follows from common-pair domination, which bounds a neighboring endpoint by $1 - \min\{A_j, B_j\}$. The full endpoint and strictness argument is in lemma D.8. □

Let selected lower bounds satisfy

$$0 \le a_j \le A_j, \qquad 0 \le b_j \le B_j \qquad (0 \le j \le 5),$$

$$a_j^2 + a_j b_j + b_j^2 \le 1 \qquad (0 \le j \le 5).$$

A neighboring term is used only when the actual V triangle has the corresponding positive radial trace:

$$\mathcal{H}^1(T_{i-1} \cap r_i) > 0 \Longrightarrow a_{i-1} + b_{i-1} \le 1,$$

$$\mathcal{H}^1(T_{i+1} \cap r_i) > 0 \Longrightarrow a_{i+1} + b_{i+1} \le 1.$$

Define

$$\Gamma_i := \max\Big(\{c_{\max}(a_i, b_i)\} \cup \{C_+(a_{i-1}, b_{i-1}) : \mathcal{H}^1(T_{i-1} \cap r_i) > 0\} \cup \{C_-(a_{i+1}, b_{i+1}) : \mathcal{H}^1(T_{i+1} \cap r_i) > 0\}\Big), \qquad D_i := (1-\Gamma_i)V_i.$$

The points D_i are the radial witnesses used below.

Lemma 5.4 (Type-aware radial forcing)**.** *For every i,*

$$D_i \in r_i \subseteq S, \qquad D_i \notin \bigcup_{j=0}^{5} U_j.$$

Consequently,

$$S \subseteq U_C \cup \bigcup_{j=0}^{5} U_j \Longrightarrow D_i \in U_C.$$

Proof. The variational capacities give $\gamma_i \le \Gamma_i \le 1$, including only actual positive adjacent traces. Apply lemma 5.3. The point O when $\Gamma_i = 1$ is handled by that lemma as well; no positive-distance claim is made there. $\square$

Corollary 5.5 (Uniform common-pair radial forcing)**.** *Suppose $p, q \ge 0$, $p + q \le 1$, and*

$$p \le a_i \le A_i, \qquad q \le b_i \le B_i \quad (0 \le i \le 5), \qquad 0 < c_* := c_{\max}(p, q) < 1.$$

Then $(1 - c_*)V_i \notin \bigcup_{j=0}^{5} U_j$ for every i, without a restriction on the normalized V types. Under a hypothetical cover of H,

$$(1 - c_*)V_i \in U_C \quad (0 \le i \le 5), \qquad \mathbb{B}(O, h(1 - c_*)) \subset U_C.$$

Proof. Reselect each boundary pair as (p, q), using convexity. Lemma 5.4 applies to these common bounds, including only the actual positive neighboring traces. By lemma 5.1, every included C_+ or C_- term is at most c_*, while the own-ray term equals c_*. Hence each maximum is exactly c_* and the forced points are the stated symmetric ones. Their convex hull contains the displayed disk. In particular, a Vd1, Vd2, or T3-like crossing does not change any witness coordinate. $\square$

5.2. **Fixed witnesses and candidate enclosures.** Each witness set is fixed from the original V triangles before an arbitrary enclosing candidate is chosen. The candidate's center parameters are recomputed, but its witness points are not. Gap endpoints are supplied by lemma 2.10; radial witnesses use the total endpoints, including all permitted neighboring traces.

Lemma 5.6 (The origin is implicit in the path witnesses)**.** *For $G = X_0(t)$, $0 < t < 1$, and $d_2, d_4 > 0$,*

$$G + \frac{1-t}{d_2}(d_2 V_2) + \frac{1}{d_4}(d_4 V_4) = 0.$$

Consequently $O \in \operatorname{conv}\{G, d_2V_2, d_4V_4\}$.

Proof. Use $V_2 = V_1 - V_0$ and $V_4 = -V_1$, and normalize the three positive coefficients to sum to one. $\square$

The three constructions are summarized after they have been defined. Normalize the C midpoint to M_0 in the nonzero-gap cases, and write σ for the unique supercritical index when $N_+ = 1$. Counts are upper bounds, allowing coincident gap endpoints.

Replacement is a separate skeleton-preserving operation, not a fourth witness construction. BC handles the center-aligned case. For an off-center supercritical role, its midpoint supplier leads to D, replacement, or the Vd2 perimeter obstruction.

5.3. **BC: one selected gap and one six-point construction.** Normalize the center midpoint to M_0 and choose any actual gap, reflected to J_0. The other incident edge $e_{5,0}$ may or may not contain a gap. Only the four middle edges $e_{1,2}, \dots, e_{4,5}$ must be gap-free. The following transfer supplies the tail bound needed by the path argument.

Lemma 5.7 (Shared-gap-anchor boundary transfer)**.** *If the original open roles cover ∂H and J_0 is an actual gap, then, independently of the gap status of $e_{5,0}$,*

$$B_5 > 1 - M_0(B_0) > B_0/2.$$

Here $M_0(z)$ is the scalar diameter envelope.

Proof. Put $G = X_0(B_0)$ and $L = X_5(B_5)$. Actual gap forcing gives $G \in U_C$, while maximality gives $G \in T_0$ and $L \notin U_5$. Boundary locality and the original perimeter cover give $L \in U_0 \cup U_C$. Thus one unit triangle contains L openly and G in its closure, so $\|G-L\| < 1$. Indeed, at equality one could move L slightly away from G inside that open triangle and exceed its diameter. The two incident-edge directions at V_0 meet at 120 degrees, hence

$$B_0^2 + B_0(1 - B_5) + (1 - B_5)^2 < 1.$$

Apply lemma A.17; its last inequality is strict because $B_0 > 0$. The proof includes singleton gaps and does not require L to be a witness in the final finite set. □

In the pure enclosure statement below the tail bound $B_5 \geq B_0/2$ is explicit. In the covering application, the preceding lemma establishes it for the original roles before an arbitrary enclosing candidate is chosen. We do not assume that the candidate also covers the discarded gap.

Define

$$K_{BC} := \{M_0, X_0(B_0), X_0(1 - A_1), \widehat{P}_2, \widehat{P}_3, \widehat{P}_4\}.$$

The midpoint is a structural anchor belonging to the original open C triangle. The other points are missed by the V triangles. The origin is implicit by lemma 5.6; the construction uses at most six points even though a candidate must also contain O.

Theorem 5.8 (Unified six-point selected-gap enclosure)**.** *Suppose J_0 is an actual gap, $A_i + B_i \leq 1$ for $1 \leq i \leq 5$, and*

$$B_i + A_{i+1} > 1 \quad (1 \leq i \leq 4), \qquad B_5 \geq B_0/2.$$

Then $\Lambda(K_{BC}) \geq 1$. Under skeleton coverage with distinguished C midpoint M_0, the fixed set K_{BC} lies in U_C. There is no condition on the gap status of $e_{5,0}$ or supercriticality of T_0, and no type restriction on the nonsupercritical path.

Proof. Forcing uses the two actual gap endpoints, the three total radial endpoints, and the C midpoint. The origin-hull identity then puts O in any open candidate containing K_{BC}. Its signed parameters turn the gap anchors and tail bound into a common pair of strict boundary lower bounds along the path. The CE2 threshold and CE1 return calculations in lemma D.9 exclude both candidate classes. □

Theorem 5.9 (Center-aligned path obstruction)**.** *Suppose a skeleton cover has at least one actual gap and its C triangle's unique midpoint is M_k. If every T_i with $i \neq k$ is nonsupercritical, the configuration is impossible.*

Proof. Normalize $k = 0$. The signed normal form confines possible gap edges to $e_{5,0}, e_{0,1}$. Select either actual gap and reflect it to J_0. The four middle edges are gap-free, and lemma 5.7 gives $B_5 > B_0/2$. Apply theorem 5.8. The midpoint belongs openly to U_C, and all other witnesses are center-forced. Compact-open containment contradicts $\Lambda(K_{BC}) \geq 1$ in either gap rank. □

Theorem 5.10 (Every skeleton cover has a supercritical V triangle)**.** *A skeleton cover by the seven original open roles satisfies $N_+ \geq 1$.*

Proof. If $N_+ = 0$ and there are no gaps, strict overlaps give $6 < \sum_i (A_i + B_i) \leq 6$. If a gap is present, all six roles are nonsupercritical, so theorem 5.9 applies. This proof uses only BC; replacement will invoke N0, not conversely. □

Lemma 5.11 (A unique midpoint supplier)**.** *In a nonzero-gap skeleton cover, the preceding N0 theorem and the skeleton budget leave only $N_+ = 1$, $N_{\mathrm{sp}} \leq 1$. Write M_k for the C midpoint and σ for the supercritical index. If $\sigma \neq k$, there is a unique positive-support role T_τ, where*

$$\tau \in \{\sigma - 1, \sigma + 1\}, \qquad M_\sigma \in U_\tau.$$

If T_τ is T3-like, then $\tau = k$. If $\tau \neq k$, the edge shared by T_τ, T_σ is center-free, in both CE1 and CE2.

Proof. N0 excludes $N_+ = 0$. Two supercritical roles would require a distinct positive-support rescuer by lemma 3.6, and theorem 3.7 would exclude them. The same count bound gives $N_{\mathrm{sp}} \leq 1$.

If $\sigma \neq k$, neither the C triangle nor T_σ contains M_σ. Diameter excludes nonlocal roles; a Vd0 role has no positive adjacent trace and cannot contain that point openly. Hence its unique supplier is an adjacent positive-support role. A T3-like supplier also misses its own midpoint by lemma 2.11. Unless $\tau = k$, neither the center nor the other roles could cover this second midpoint: all other roles are Vd0. Finally, when $k \notin \{\tau, \sigma\}$, the shared edge is not incident to V_k, whereas every open center boundary trace is incident to V_k by the signed normal form. □

5.4. **D: a four-point supported-rescuer enclosure.** Let $Y(t) = (1-t)V_0 + tV_5 = X_5(1-t)$. The geometric lemma below is stated independently of any V type or covering configuration.

Theorem 5.12 (Four-point rescuer geometry)**.** *Suppose $a \geq 0$, $\varepsilon > 0$, $\beta \geq 0$, and*

$$s = a + \varepsilon \leq 1, \qquad a \leq \varepsilon, \qquad \beta \leq \frac{\varepsilon}{s}.$$

Then

$$\Lambda\{O, \varepsilon V_1, Y(a), Y(1-\beta)\} \geq 1.$$

Proof. The convex combination

$$\frac{\varepsilon}{s}Y(a) + \frac{a}{s}(\varepsilon V_1) = \frac{\varepsilon}{s}V_0$$

forces M_0 into any open unit candidate containing the four points. The signed-center calculation then places the far endpoint of its left boundary trace strictly below a/s, whereas containment of $Y(1-\beta)$ requires that endpoint to exceed $1-\beta \geq a/s$. The signed parameters and endpoint cases are given in lemma D.10. □

Lemma 5.13 (Scalar rescuer ratio test)**.** *For* $0 \leq x, c \leq 1/2$ *and* $M = (c + \sqrt{c^2 - 8c + 4})/2$,

$$x \leq 1 - M \quad \iff \quad x^2 + (c-2)x + c \geq 0.$$

Proof. The smaller root is $1-M$, while the other root is at least one. The quadratic is nonnegative on $[0, 1/2]$ exactly before its smaller root. □

For the covering application, let T_0 supply M_1, with supported interval $[c, u]$ on r_1, measured from V_1 toward O, and put $\varepsilon = 1 - u$ and $P_T = \varepsilon V_1$. All interval endpoints here belong to the original triangle, not to its translated majorant. The local T3-like and Vd1 calculations establish

$$A_0 + \varepsilon \leq 1, \qquad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M_c^{\sup}, .$$

Together with $C_1 \geq c$, the boundary path, and the forced endpoint, lemma D.11 gives the single fixed witness

$$K_D = \{O, \varepsilon V_1, Y(A_0), Y(1 - B_5)\} \subset U_C, \qquad \Lambda(K_D) \geq 1.$$

The two local verifications are lemmas D.12 and D.13; the enclosure step does not distinguish one from two gaps.

Proposition 5.14 (T3-like supported-rescuer terminal)**.** *Suppose* $N_+ = 1$*, there is exactly one T3-like role and no Vd1/Vd2 role, and there is at least one actual gap. Then the skeleton is not covered.*

Proof. The center-aligned case is excluded by theorem 5.9. In the remaining case, lemma 5.11 puts the T3-like supplier at the center midpoint and, after reflection, the supercritical role at T_1. The actual-or-bounded frontier forces εV_1 into U_C, and radial coverage gives $C_1 \geq c$. The local calculation lemma D.12 gives the remaining ratio and size bounds. Apply the common adapter lemma D.11. □

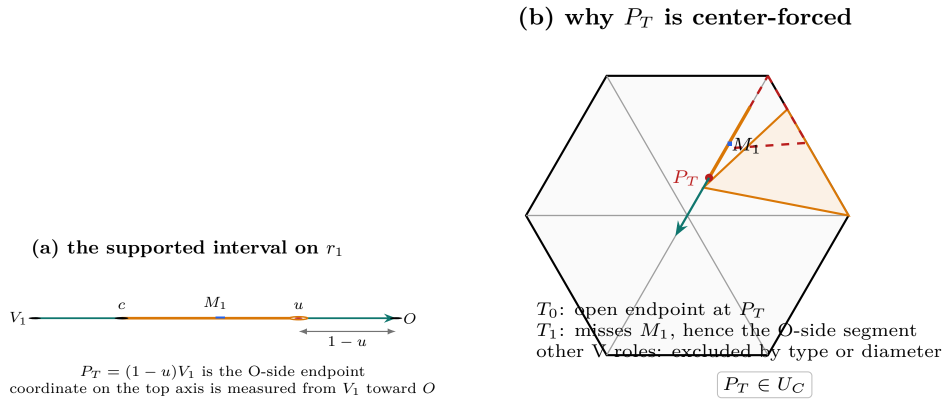


FIGURE 6. Construction of the T3-like rescuer endpoint. The interval $[c, u]$ is parametrized from V_1 toward O, so its O-side endpoint is $P_T = (1-u)V_1$. The T3-like open role omits its endpoint, the adjacent supercritical role contains V_1 but misses M_1 and therefore misses the O-side segment by convexity, and all remaining V roles are excluded. Thus P_T is forced into the C triangle.

5.5. **The remaining placement router.**

Proposition 5.15 (One-Vd placement assembly)**.** *Suppose the C triangle is CE1 or CE2, $N_+ = 1$, exactly one V triangle is Vd1 or Vd2, and there is an actual gap. The skeleton is not covered.*

Proof. An additional positive-support role closes by the skeleton budget. Otherwise lemma 5.11 gives the unique supplier or the center-aligned placement. The latter closes by theorem 5.9. A Vd2 supplier closes by proposition 3.9. A center-based Vd1 supplier has its endpoint forced by the common frontier and $C_1 \geq c$ by radial coverage; lemmas D.11 and D.13 then give D. An away Vd1 supplier has a center-free shared edge, so proposition D.16 applies and contradicts N0. The latter preserves only the skeleton, not all of H or the input gap rank. ☐

5.6. **The zero-gap nine-point obstruction.** Assume

$$N_{\mathrm{gap}} = 0, \qquad N_+ = 1.$$

Rotate the unique supercritical role to index 4. The exact-one handoff calculation lemma E.2 gives

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4. \tag{14}$$

Define

$$a := 1 - \xi_3, \qquad b := \xi_4, \qquad p := 1 - b, \qquad q := 1 - a. \tag{15}$$

Then

$$\begin{gathered} 0 < a, b < 1, \qquad a = a_4 < A_4, \qquad b = b_4 < B_4, \\ a + b > 1, \qquad a^2 + ab + b^2 < 1, \\ a_i \geq p, \qquad b_i \geq q \quad (0 \leq i \leq 5). \end{gathered} \tag{16}$$

Define the two boundary anchors

$$P_A := V_4 + a(V_3 - V_4), \qquad P_B := V_4 + b(V_5 - V_4),$$

and the feasible union

$$\mathcal{R}_{AB}(a,b) := \bigcup \left\{ T \cap H : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ V_4, P_A, P_B \in T \end{array} \right\}.$$

Put

$$c_* := c_{\max}(p,q), \qquad r_* := h(1-c_*), \qquad P_i^{\mathrm{rad}} := (1-c_*)V_i \quad (0 \le i \le 5).$$

The bounds in lemma E.3 give $0 < c_* < 1$. Applying corollary 5.5 to the common pair gives

$$\mathbb{B}(O, r_*) \subseteq \operatorname{conv}\{P_0^{\mathrm{rad}}, \dots, P_5^{\mathrm{rad}}\} \subset U_C. \tag{17}$$

The two circles adjacent to the supercritical frontier are

$$\mathcal{C}_- := \partial\mathbb{B}(X_2(b), 1), \qquad \mathcal{C}_+ := \partial\mathbb{B}(X_5(1-a), 1).$$

Let ℓ_-, ℓ_+ be the two straight pieces of the interior frontier of $\mathcal{R}_{AB}(a,b)$, in the order of theorem E.1. Define

$$\{Q_0\} = \ell_- \cap \ell_+, \qquad \{Q_-\} = \ell_- \cap \mathcal{C}_-, \qquad \{Q_+\} = \ell_+ \cap \mathcal{C}_+.$$

The coordinate formulas and selected first roots are given in Appendix E, equations (89) and (93). Lemma E.4 proves that these intersections are singletons on the indicated pieces. The nine-point set is

$$K_F := \{P_i^{\mathrm{rad}} : 0 \le i \le 5\} \cup \{Q_-, Q_0, Q_+\}.$$

Their exact construction and exclusions in theorem E.1 and lemmas E.4 and E.5 use contained vertices and actual handoffs, not Vd0 locality, and give

$$Q_-, Q_0, Q_+ \in U_C. \tag{18}$$

For the enclosure calculation use the compact comparison set

$$K_{\mathrm{wit}}(a,b) := \mathbb{B}(O, r_*) \cup \{Q_-, Q_0, Q_+\} \subset U_C. \tag{19}$$

Theorem 5.16 (Witness enclosure)**.** *For $0 < a, b < 1$, $a+b > 1$, and $a^2+ab+b^2 < 1$, the comparison set $K_{\mathrm{wit}}(a,b)$ satisfies*

$$\Lambda(K_{\mathrm{wit}}(a,b)) \ge 1.$$

Proof. The disk closes the range $c_* \le 2/3$. For $c_* > 2/3$, the Newton inner points reduce the enclosure to four supporting contacts: the lines through two consecutive points and the two exposed disk tangencies. The line bounds are analytic; the paired tangent residuals use the exact certificate and simultaneous radius transfer. The complete calculation is lemma E.10. $\square$

Theorem 5.17 (Zero-gap obstruction)**.** *There is no hypothetical cover with*

$$N_{\mathrm{gap}} = 0, \qquad N_+ = 1.$$

Proof. The construction gives the compact containment $K_{\mathrm{wit}}(a,b) \subset U_C$, while theorem 5.16 gives $\Lambda(K_{\mathrm{wit}}(a,b)) \ge 1$. This contradicts lemma 2.1. $\square$

5.7. **Summary of the fixed witnesses.**

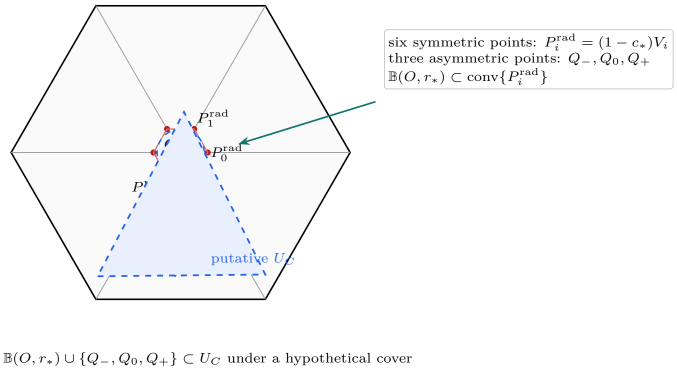


Figure 7. The type-independent zero-gap nine-point witness. The six symmetric radial points force a centered disk into the convex C triangle, while the three asymmetric frontier points supply the missing directional support. Marker shapes separate the two constructions: circles are radial witnesses and triangles are strict-supercritical AB-frontier witnesses. The displayed placement is illustrative; the point definitions are the exact ones in the preceding lemmas.

Table 2: Fixed finite-enclosure recipes. Ordinary nonsupercritical V triangles are not subdivided by type.

Conditions	Fixed witness set	Disk and count
BC: at least one incident gap; $T_1, \ldots, T_5$ nonsupercritical	$\{M_0, X_0(B_0), X_0(1-A_1), \widehat{P}_2, \widehat{P}_3, \widehat{P}_4\}$	No disk; at most 6
D: center-based T3-like or Vd1 rescuer; one or two gaps	$\{O, \varepsilon V_1, Y(A_0), Y(1-B_5)\}$	No disk; at most 4
F: zero gaps, $N_+ = 1$	The nine-point set K_F	Disk; at most 9

6. Completion of the Proof

Proof of Theorem 1.1. Suppose seven open unit equilateral triangles cover H. By theorem 5.10, $N_+ \geq 1$. With no gaps, the nine-point obstruction theorem 5.17 excludes $N_+ = 1$, and proposition 4.3 excludes $N_+ \geq 2$.

With a gap, lemma 5.11 gives $N_+ = 1, N_{\mathrm{sp}} \leq 1$. Let M_k be the C midpoint and σ the supercritical index. The case $\sigma = k$ is excluded by theorem 5.9. Otherwise M_σ has a unique adjacent positive-support supplier. A T3-like supplier is center-based and is excluded by proposition 5.14; a Vd2 supplier is excluded by proposition 3.9. For Vd1, a center-based supplier uses D, whereas an away supplier admits replacement followed by N0, as proved in proposition 5.15. Thus no cover exists. The scaling equivalence proposition 2.2 gives corollary 1.2. $\square$

Appendix A. Structural, Shared-Local, and Signed-Center Optimization

This appendix solves the coordinate problems used by the geometric interfaces in Section 2. It also proves the shared equilateral-enclosure calculus used by the later optimization appendices. Each calculation is attached to the body statement it supplies.

A.1. **Center and V triangle incidence feasibility.**

Lemma A.1 (Separation of nonadjacent boundary edges)**.** *Relative-interior points on two nonadjacent edges of H are more than unit distance apart.*

Proof. Up to symmetry take

$$x = (1-t)V_0 + tV_1, \qquad 0 < t < 1.$$

For $y = (1-s)V_2 + sV_3$, $0 < s < 1$, direct expansion gives

$$\|x-y\|^2 - 1 = (1-t)(2-t) + s(s+t) > 0.$$

For the other orbit, $y = (1-s)V_3 + sV_4$, put $u = t+s$ and obtain

$$\|x-y\|^2 - 1 = (u-1)^2 + 2 > 0.$$

These are the two nonadjacent-edge configurations. Since the diameter of a unit equilateral triangle is 1, the result supplies Proposition 2.4. □

Lemma A.2 (Shared corner chart and wedge reduction)**.** *After rotating indices to $i = 0$, write*

$$X = V_0 + x(V_1 - V_0) + y(V_5 - V_0), \qquad W = \{(x,y) : x \geq 0,\ y \geq 0\}.$$

If a closed unit equilateral triangle T contains V_0, then

$$T \cap H = T \cap W.$$

Normalized Euclidean area is twice the corresponding (x,y)-coordinate area.

Proof. The two basis vectors have length one and angle 120°, so

$$\|(x,y)\|^2 = x^2 + y^2 - xy, \qquad |\det(V_1 - V_0, V_5 - V_0)| = \frac{\sqrt{3}}{2}.$$

The hexagon is

$$H = \{0 \leq x, y \leq 2,\ |x-y| \leq 1\} \subset W.$$

Conversely, if $(x,y) \in T \cap W$, then the diameter-one condition gives $x^2+y^2-xy \leq 1$. Hence

$$|x-y|^2 \leq x^2 + y^2 - xy \leq 1, \qquad \frac{3}{4}x^2 \leq x^2 + y^2 - xy,$$

and the same estimate holds for y. Thus $|x-y| \leq 1$ and $0 \leq x, y \leq 2/\sqrt{3} < 2$, proving $(x,y) \in H$. The determinant identity proves the area assertion. □

Lemma A.3 (Local wedge calculation)**.** *The incidence conclusions of Lemma 2.5 hold.*

Proof. Use lemma A.2 at V_0. In the unit ball,

$$x^2 + y^2 - xy = (y - x/2)^2 + 3x^2/4 = (x - y)^2 + xy.$$

The adjacent radial segments are

$$r_1 = \{(1, y) : 0 \le y \le 1\}, \qquad r_5 = \{(x, 1) : 0 \le x \le 1\}.$$

Suppose $T_0 \cap r_1$ has positive length and let m be the largest x-coordinate of a vertex of T_0. If $m > 1$, a maximizing vertex (m, y) lies in the unit ball and has $0 < y < 1$, so it lies in H. If $m = 1$, positive-length contact with the support line $x = 1$ forces a side there, and both endpoints satisfy $1 + y^2 - y \le 1$ and lie in H. The argument for r_5 is symmetric. If both arms are met but only one vertex lies in H, that vertex would have $x > 1$ and $y > 1$, contradicting $(x - y)^2 + xy > 1$.

Finally, if all three vertices lay in convex H, then $T_0 \subset H$, contrary to the extreme point V_0 lying in $\mathrm{int}(T_0)$. Hence $o(T_0) \ge 1$, completing the calculation. $\square$

A.2. **Shared orientation charts and exact-trace normalization.** Set

$$D_t := 1 - t + t^2, \qquad z := \sqrt{D_t}, \qquad 0 \le t \le 1.$$

In the shared corner chart, physical rotation through $60°$ is $(x, y) \mapsto (x - y, x)$. After relabelling the vertices, every orientation belongs to one of the two families

	first side vector	second side vector
I	$z^{-1}(1, t)$	$z^{-1}(1 - t, 1)$
II	$z^{-1}(t, -(1 - t))$	$z^{-1}(1, t)$.

They cover successive $60°$ direction sectors, including their common endpoints. The following half-plane descriptions allow closed containment of the corner; strict interior containment is recorded separately. Here U, V are affine slack functions, not the open triangles U_i. In Type I write

$$U = \alpha + y - tx, \qquad V = \beta + x - (1 - t)y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}, \tag{20}$$

and in Type II write

$$U = \alpha + (1 - t)x + ty, \qquad V = \beta + tx - y, \qquad T = \{U \ge 0, V \ge 0, U + V \le z\}. \tag{21}$$

In both charts

$$V_0 \in \mathrm{int}(T) \iff \alpha > 0, \quad \beta > 0, \quad \alpha + \beta < z. \tag{22}$$

The Type-II vertex in the wedge is

$$Q = \left(\frac{z - \alpha - t\beta}{D_t}, \frac{t(z - \alpha) + (1 - t)\beta}{D_t} \right). \tag{23}$$

Lemma A.4 (Raw $(3, 0)$ trace-normalization calculation)**.** *The translation asserted in Proposition 2.6 exists and preserves both the open and closed traces in H.*

Proof. Normalize to $i = 0$. A Type-II triangle cannot have $o = 3$, because the point Q in (23) lies in the wedge; together with the diameter-one condition, Lemma 2.5 puts it in H. Hence T has Type-I form. Its vertices are

$$Q_0 = \left(-\frac{(1 - t)\alpha + \beta}{D_t}, -\frac{\alpha + t\beta}{D_t} \right),$$

$$Q_1 = \left(\frac{z-(1-t)\alpha-\beta}{D_t}, \frac{tz-\alpha-t\beta}{D_t}\right),$$
$$Q_2 = \left(\frac{(1-t)z-(1-t)\alpha-\beta}{D_t}, \frac{z-\alpha-t\beta}{D_t}\right). \tag{24}$$

Thus $o = 3$ is equivalent to

$$\alpha + t\beta > tz, \qquad (1-t)\alpha + \beta > (1-t)z, \tag{25}$$

which forces $0 < t < 1$. Put $s = \alpha + \beta$ and $L = z - s > 0$. Then

$$\alpha > \frac{tL}{1-t}, \qquad \beta > \frac{(1-t)L}{t}. \tag{26}$$

For $(x, y) \in W$ satisfying

$$(1-t)x + ty \le L, \tag{27}$$

the minimum of U is attained at $(L/(1-t), 0)$ and that of V at $(0, L/t)$. Hence

$$T \cap W = \{(x, y) \in W : (1-t)x + ty \le L\}. \tag{28}$$

Keep t, s fixed and set

$$\widehat{\alpha} := \frac{tL}{1-t}, \qquad \widehat{\beta} := s - \widehat{\alpha}. \tag{29}$$

Then $0 < \widehat{\alpha} < \alpha$, $\widehat{\beta} > \beta > 0$, and $\widehat{\alpha} + \widehat{\beta} = s < z$. Changing these affine constants is a translation and preserves $V_0 \in \mathrm{int}(\widehat{T})$. Its third inequality remains (27); $\widehat{U} = 0$ only at the endpoint $(L/(1-t), 0)$ of that common boundary and $\widehat{V} > 0$. Therefore

$$\widehat{T} \cap W = T \cap W,$$

and their interiors have the same trace on W. The local wedge calculation identifies these with the closed and open traces in H.

Equation (29) places Q_1 on the boundary of H; Q_0 remains outside and the second strict inequality in (26) keeps Q_2 outside. Hence $(o(\widehat{T}), n(\widehat{T})) = (2, 0)$. Since the three reach segments lie in H, their open and closed traces, and therefore A_i, B_i, C_i and $A_i + B_i > 1$, are unchanged. □

A.3. **T3-like trace-majorant feasibility.** The calculation uses the local wedge and the two orientation charts just introduced. Its output is a closed majorant, not a replacement open triangle.

Lemma A.5 (T3-like translation calculation)**.** *The closed-trace majorant in Proposition 2.8 exists. In the calculation chart, every original* T3-like *triangle also satisfies $A_i + B_i \le 1$ and $M_i \notin T_i$.*

Proof. Use the wedge chart of lemma A.3 and the orientation forms (20)– (21) in Subsection A.2. In Type I, direct inversion shows that when exactly two vertices lie outside W, the sole wedge vertex has both coordinates strictly below 1: the relevant bounds are $(1-t)/z$ and t/z, with strictness from $\alpha > 0$ or $\beta > 0$. The support argument in lemma A.3 then excludes positive-length contact with either adjacent radial line. Thus a T3-like triangle has Type II.

Reflect in $x = y$ if necessary so that $T_0 \cap r_1$ has positive length. The endpoint cases $t = 0, 1$ give no such interval, hence $0 < t < 1$. The exact axis reaches are

$$A_0 = \beta, \qquad B_0 = z - \alpha - \beta,$$

and the support condition is

$$Q_x > 1, \qquad Q_y \le 1,$$

where Q is (23). In particular

$$A_0 + B_0 = z - \alpha < z \le 1.$$

The same support inequality gives

$$\beta < \frac{z - D_t}{t} < \frac{1-t}{2},$$

where the second comparison is equivalent to $(1-z)^2 > 0$. At $M_0 = (1/2, 1/2)$ the affine coordinate V equals $\beta - (1-t)/2 < 0$, so $M_0 \notin T_0$.

Define $\widehat{T}$ by replacing α with 0 and retaining t, β. This changes only the translation. At the origin, $\widehat{U} = 0$ and $0 < \widehat{V} = \beta < z$, so V_0 lies in the relative interior of a side. For $(x, y) \in W$,

$$\widehat{U} = (1-t)x + ty \ge 0, \qquad \widehat{V} = V, \qquad \widehat{U} + \widehat{V} = U + V - \alpha.$$

Thus $T \cap W \subset \widehat{T} \cap W$. The old x-axis reach is B_0 and the new reach is $B_0 + \alpha$, so the inclusion is strict in H.

The two outside vertices remain outside, while V_0 and the new wedge vertex lie in H. Moreover $\widehat{Q}_x > Q_x > 1$, and $Q_x > 1$ gives

$$tD_t(1 - \widehat{Q}_y) > D_t(1-z) + (1-t)\alpha > 0.$$

Hence $\widehat{Q}_y < 1$ and $(o(\widehat{T}), n(\widehat{T})) = (2, 1)$. At every nonzero point of W, $\widehat{U} > 0$; the other two displayed slack relations show that old interior points other than V_0 remain interior. Reflection proves the other supported-arm case. □

A.4. **Strict-handoff feasibility.**

Lemma A.6 (Cyclic strict-handoff selection)**.** *Under the hypotheses of Proposition 2.12, all six overlap intervals are nonempty, and the selectors can preserve the stated one- or two-supercritical-index alternatives.*

Proof. Since V_i is interior to a diameter-one triangle, $0 < A_i < 1$ and $0 < B_i < 1$. No nonincident V triangle contains a relative-interior point of $e_{i,i+1}$. Thus the relatively open incident traces cover the edge and overlap, so

$$(1 - A_{i+1}, B_i) \ne \varnothing.$$

Choosing ξ_i in this interval places both selected anchors in U_i. Their squared distance is $a_i^2 + a_i b_i + b_i^2$, which is strictly below 1 because both are interior points of the same open unit triangle.

Put $\ell_i := 1 - A_{i+1}$ and $u_i := B_i$. Then

$$A_i + B_i > 1 \iff \ell_{i-1} < u_i, \qquad a_i + b_i > 1 \iff \xi_{i-1} < \xi_i. \tag{30}$$

If i is actually nonsupercritical, then

$$\xi_i < u_i \le \ell_{i-1} < \xi_{i-1},$$

so no selection makes it supercritical. Also

$$\sum_{i=0}^{5} (a_i + b_i) = \sum_{i=0}^{5} (1 - \xi_{i-1} + \xi_i) = 6. \tag{31}$$

If p is the sole actual supercritical index, the other five selected sums are below 1, and (31) forces the pth sum above 1.

For two nonadjacent supercritical indices p, q, choose

$$\ell_{r-1} < z_r < u_r \qquad (r \in \{p, q\}),$$

and then

$$\xi_{r-1} \in (\ell_{r-1}, \min\{u_{r-1}, z_r\}), \qquad \xi_r \in (\max\{\ell_r, z_r\}, u_r).$$

The pairs are disjoint and create ascents at p, q. Any three indices on a six-cycle contain two nonadjacent indices. If the only two supercritical indices are adjacent, say $p, p+1$, the remaining inequalities give

$$\ell_{p-1} < \ell_{p+1}, \qquad \max\{\ell_{p-1}, \ell_p\} < \min\{u_p, u_{p+1}\}.$$

Choose ξ_p between the latter bounds and choose $\xi_{p-1} < \xi_p < \xi_{p+1}$ in their overlap intervals. By (30), both selected indices remain supercritical. $\square$

A.5. **Shared equilateral-enclosure optimization.** Let R denote counterclockwise rotation through $2\pi/3$. For a nonempty compact $K \subset \mathbb{R}^2$, put

$$h_K(n) := \max_{x \in K} \langle x, n \rangle.$$

Proposition A.7 (Equilateral enclosure gauge)**.** *The least side length $\Lambda(K)$ of a closed equilateral triangle containing K is*

$$\Lambda(K) = \frac{2}{\sqrt{3}} \min_{\|n\|=1} \sum_{j=0}^{2} h_K(\mathsf{R}^j n). \tag{32}$$

It is monotone under inclusion, translation invariant, and positively homogeneous.

Proof. For fixed outward normals $n, \mathsf{R}n, \mathsf{R}^2 n$, the least containing triangle has the corresponding three support numbers, and its side is $2/\sqrt{3}$ times their sum. Translation invariance follows from

$$n + \mathsf{R}n + \mathsf{R}^2 n = 0,$$

while monotonicity and homogeneity are immediate. $\square$

Lemma A.8 (Support-cell rotation)**.** *A least equilateral enclosure of a nondegenerate polygon can be chosen with one side containing a polygon edge. For a centered disk together with finitely many points, it suffices to test point–point and point–disk ties, or a disk-only orientation. Here a* support cell *is an open angular interval on which the maximizing point or disk is fixed in each of the three support directions; a* tie *is an endpoint where a maximizer changes.*

Proof. By proposition A.7, the side is the three-term support sum divided by $h = \sqrt{3}/2$. Parametrize unit normals by $n(\theta) = (\cos\theta, \sin\theta)$. For a disk of radius $\rho > 0$, its support value is ρ in every unit direction. On a polygon cell with fixed support vertices the sum is $g(\theta) = a\cos\theta + b\sin\theta > 0$, so $g'' = -g < 0$ and no interior minimum occurs. In the disk case a cell with q disk supports has sum $q\rho + \langle v, n(\theta)\rangle$, where v is the sum of the active point vectors rotated back by their respective multiples of $120°$. If a point is active its projection exceeds $\rho \geq 0$, and the inner product is positive, again giving a negative second derivative. A disk-only cell has constant support sum 3ρ. Each application must still identify its exposed ties. $\square$

A.5.1. *Finite caliper criterion.* Let J denote counterclockwise rotation through $\pi/2$. For a nonempty finite set F define

$$W_F(N) := \sum_{k=0}^{2} \max_{z\in F} \langle z, \mathsf{R}^k N\rangle. \tag{33}$$

Theorem A.9 (Finite caliper certificate)**.** *If F has at least two distinct points, then F is contained in a closed unit equilateral triangle if and only if there are distinct $p, q \in F$ and $\varepsilon \in \{-1, 1\}$ such that*

$$W_F\big(\varepsilon \mathsf{J}(q-p)\big) \le \frac{\sqrt{3}}{2}\|p-q\|. \tag{34}$$

A singleton is contained in equilateral triangles of arbitrarily small positive side length.

Proof. Apply proposition A.7 and lemma A.8 to conv$F$. A minimizing normal is perpendicular to a hull edge $[p, q]$ and, after cyclic rotation, equals $\varepsilon\mathsf{J}(q-p)/\|p-q\|$. Homogeneity gives (34); a nondegenerate segment follows by checking its two normals.

□

Proposition A.10 (Disk–finite-set calipers)**.** *Let $\rho > 0$, $P = \{p_1, \ldots, p_m\} \subset \mathbb{R}^2$, and $K = \mathrm{conv}(\mathbb{B}(O, \rho) \cup P)$. Use the rotation J defined above. Unless the disk is the active support in all three directions at a minimizing orientation, a minimizing normal belongs to the finite list*

$$\pm\frac{\mathsf{J}(p_k - p_i)}{\|p_k - p_i\|} \quad (p_i \ne p_k)$$

or

$$\frac{\rho p_i \pm \sqrt{\|p_i\|^2 - \rho^2}\,\mathsf{J}p_i}{\|p_i\|^2} \quad (\|p_i\| \ge \rho).$$

Thus one side of a least enclosing equilateral triangle either contains two points, or is tangent to the disk and contains one point.

Proof. Apply lemma A.8. A point–point tie has normal perpendicular to $p_k - p_i$; solving $\langle p_i, n\rangle = \rho$ with $\|n\| = 1$ gives the displayed point–disk normals. Rotation through 120 degrees identifies the three supporting directions. A disk-only cell contributes 3ρ and is the only additional alternative. □

A.6. **The local admissible set and radial envelope.** Let u, v be unit vectors meeting at 120° and define

$$K(a, b, c) := \mathrm{conv}\{0, au, bv, c(u + v)\}.$$

This is the translate of the body normalization at V_0, with $u = V_5 - V_0$ and $v = V_1 - V_0$. The exact local admissible set is

$$\mathcal{A} := \{(a, b, c) \in [0, 1]^3 : \Lambda(K(a, b, c)) \le 1\}. \tag{35}$$

This set records the three reach lower bounds that a closed unit V triangle can realize.

Proposition A.11 (Exact local admissible set)**.** *Put*

$$s := a+b, \qquad d := a^2+ab+b^2, \qquad m := \min\{a,b\}, \qquad M := \max\{a,b\},$$
$$q := s^4 - s^2 + ab,$$

and

$$\begin{aligned} F_L &:= c^4 - c^2 + mc - m^2, \\ F_T &:= (s^2-1)c^2 + Mc - M^2, \\ F_S &:= (m^2-1)c^2 + (2mM^2+M)c + M^4 - M^2. \end{aligned}$$

Then

$$\mathcal{A} = \mathcal{A}_L \cup \mathcal{A}_T \cup \mathcal{A}_S, \tag{36}$$

where all variables lie in $[0,1]$ *and*

$$\begin{aligned} \mathcal{A}_L &:= \{d \le 1,\ s \le 1,\ q \le 0,\ F_L \le 0\}, \\ \mathcal{A}_T &:= \{d \le 1,\ s \le 1,\ q \ge 0,\ F_T \le 0,\ c \le 2M\}, \\ \mathcal{A}_S &:= \{d \le 1,\ s > 1,\ F_S \le 0,\ c \le 1/2\}. \end{aligned}$$

The set is coordinatewise down-closed. Define its maximal radial demand by

$$c_{\max}(a,b) := \max\{c \in [0,1] : (a,b,c) \in \mathcal{A}\}.$$

Then

$$c_{\max}(a,b) = \begin{cases} 1, & a=b=0, \\ C_L(m), & s \le 1,\ q \le 0,\ (a,b) \ne (0,0), \\ \dfrac{2M}{1+\sqrt{4s^2-3}}, & s \le 1,\ q > 0, \\ \dfrac{M(2mM+1) - M\sqrt{4d-3}}{2(1-m^2)}, & s > 1, \end{cases} \tag{37}$$

where $C_L(m)$ *is the unique root in* $[\sqrt{3}/2, 1]$ *of*

$$z^4 - z^2 + mz - m^2 = 0.$$

Proof. First assume $s > 0$ and $cs \le ab$. When $a,b > 0$,

$$c(u+v) = \frac{c}{a}(au) + \frac{c}{b}(bv) + \left(1 - \frac{c}{a} - \frac{c}{b}\right) 0 \in \operatorname{conv}\{0, au, bv\};$$

the zero-coordinate cases follow by continuity. The distance from au to bv is $\sqrt{d}$. With $T = -bu - av$, the triangle $\operatorname{conv}\{au, bv, T\}$ is equilateral of side $\sqrt{d}$ and

$$0 = \frac{b^2}{d}(au) + \frac{a^2}{d}(bv) + \frac{ab}{d}T.$$

Thus the least enclosure side is $\sqrt{d}$. When $a=b=0$, it is c.

Now assume $cs > ab$. The hull has cyclic vertices $0, bv, c(u+v), au$. Lemma A.8 reduces its least enclosure side to its four edge normals. For the edges through 0 one obtains

$$L_{0A} = b + \max\{a,c\}, \qquad L_{B0} = a + \max\{b,c\}.$$

For the edge from au to $c(u+v)$ put

$$R_a := \sqrt{a^2 - ac + c^2}, \qquad n_0 := \frac{1}{2R_a}(\sqrt{3}a, 2c-a).$$

The three support values in the rotated directions are

$$\frac{\sqrt{3}ac}{2R_a}, \qquad \frac{\sqrt{3}a(a-c)_+}{2R_a}, \qquad \frac{\sqrt{3}c\max\{b,(c-a)_+\}}{2R_a}.$$

Consequently

$$L_{AC} = \begin{cases} (a^2+bc)/R_a, & a \geq c, \\ c(a+b)/R_a, & a < c \leq a+b, \\ c^2/R_a, & c > a+b, \end{cases}$$

and L_{CB} is obtained by interchanging a, b. Hence

$$\Lambda(K(a,b,c)) = \min\{L_{0A}, L_{B0}, L_{AC}, L_{CB}\}.$$

Assume by symmetry that $a = m \leq b = M$. Reading the active support points in these four formulas gives

range s	range q	active supports in cyclic order: first	second	third	unit frontier equation
$s<1$	$q<0$	$\{0\}$	$\{c(u+v)\}$	$\{au, c(u+v)\}$	$F_L = 0$
$s<1$	$q>0$	$\{0\}$	$\{bv, c(u+v)\}$	$\{au\}$	$F_T = 0$
$s=1$		$\{0, bv\}$	$\{bv, c(u+v)\}$	$\{au\}$	$c = M$
$s>1$		$\{bv\}$	$\{bv, c(u+v)\}$	$\{au\}$	$F_S = 0$

The inactive-support inequalities reduce to $q \leq 0$, $q \geq 0$, and $s > 1$. The positive unit-frontier equations square to $F_L = 0, F_T = 0, F_S = 0$.

It remains to select the geometric components introduced by squaring. In Cell T the roots are

$$c_- := \frac{2M}{1+\sqrt{4s^2-3}}, \qquad c_+ := \frac{2M}{1-\sqrt{4s^2-3}}.$$

The polynomial inequality also has a formal high component, while the geometric fiber is the component joined to $c = 0$. Since

$$c_- \leq 2M \leq c_+,$$

the exact selector is $c \leq 2M$.

In Cell S,

$$F_S(a,b,0) = M^2(M^2-1) < 0, \qquad F_S(a,b,M) = M^2(s^2-1) > 0.$$

Moreover $4F_S(a,b,1/2)$ increases with m, and at $m = 1-M$ equals $M^2(2M-1)^2$. Because $s > 1$, one has $F_S(a,b,1/2) > 0$. The quadratic opens downward, so $c \leq 1/2$ selects exactly the smaller-root component containing 0.

In Cell L, the fiber from $c = 0$ reaches one root in $[\sqrt{3}/2, 1]$; uniqueness there follows from $4c^3 - 2c + m > 0$. At $q = 0$ the L and T formulas meet at $c = s$. This proves (36), including all equality and degenerate cases by continuity. Convexity of any containing triangle proves coordinatewise down-closedness. Solving the three selected frontier equations gives (37). □

For later variational use, the raw forward envelope retains its established notation

$$M_c(a) := \max\{b \in [0,1] : (a,b,c) \in \mathcal{A}\}.$$

No closed-form evaluation of M_c is needed in the geometric body.

A.7. **Self-midpoint corollary.**

Lemma A.12 (Self-midpoint admissibility corollary)**.** *If $a+b>1$, then*

$$\Lambda\left(\operatorname{conv}\left\{0, au, \frac{u+v}{2}, bv\right\}\right) > 1.$$

This supplies Lemma 2.14.

Proof. Put $m=\min\{a,b\}$ and $M=\max\{a,b\}$. If the four-point hull were admissible with $a+b>1$, Cell S of proposition A.11 would require $F_S(m,M,1/2)\le 0$. However,

$$4F_S(1-M,M,1/2) = M^2(2M-1)^2 \ge 0,$$
$$\frac{\partial}{\partial m}\big(4F_S(m,M,1/2)\big) = 2m+4M^2 > 0.$$

Since $m>1-M$, this is impossible. If $a+b=1$ and the four points have positive margin, increase the two boundary demands slightly to obtain the same contradiction. ☐

Lemma A.13 (Strict-supercritical outgoing envelope)**.** *If a V triangle is supercritical and has own-radial reach at least $c\in[0,1/2)$, then both boundary reaches are strictly below $M_c^{\sup}$ from* (13)*. No supercritical triangle has own-radial reach at least* $1/2$*.*

Proof. Let m and M be the smaller and larger actual boundary reaches. For $c>0$, the selected supercritical cell gives $F_S(m,M,c)\le 0$, whereas $m>1-M$ and

$$\partial_m F_S = 2c(M^2+cm) > 0,$$
$$F_S(1-M,M,c) = M(c-M)(1-2c+cM-M^2).$$

Since $M>1/2>c$, these identities imply $1-2c+cM-M^2>0$. Solving the quadratic gives $M<(c+\sqrt{c^2-8c+4})/2$. For $c=0$, each boundary reach is strictly less than one because its opposite distinguished vertex is excluded from the closed V triangle. The endpoint $c=1/2$ is excluded by the self-midpoint obstruction. ☐

A.8. **Signed-center optimization.** The CE1 and CE2 cases share one signed cell. The parameters below are used only in the appendices; the geometric body uses the exact C triangle type and midpoint interfaces.

Proposition A.14 (Signed CE1/CE2 center normal form)**.** *Let T_C be a closed unit equilateral triangle with $O\in\operatorname{int}(T_C)$ and a positive-length boundary trace. Normalize such a trace to $e_{0,1}$ and use affine coordinates*

$$X = V_0 + b(V_1-V_0) + a(V_5-V_0).$$

There are parameters

$$0<R<1, \qquad W := 1-R, \qquad E := \sqrt{1-RW},$$
$$\eta := 1-E, \qquad P := E(1-E),$$

and nonnegative slacks α,δ, with $k:=\eta+\alpha+\delta$, such that

$$\begin{aligned} F_0 &= R+\alpha-a+Wb, \\ F_1 &= Rb+Wa-k, \qquad\qquad T_C = \{F_0,F_1,F_2\ge 0\}. \\ F_2 &= W+\delta-b+Ra, \end{aligned} \tag{38}$$

Define

$$\Delta_R := P - \alpha - W\delta, \qquad \Delta_L := P - R\alpha - \delta. \tag{39}$$

Then

$$T_C \cap e_{0,1} = \left[\frac{k}{R}, W + \delta\right], \qquad \mathcal{H}^1(T_C \cap e_{0,1}) = \frac{\Delta_R}{R}, \tag{40}$$

and the possible companion trace is

$$T_C \cap e_{5,0} = \left[\frac{k}{W}, R + \alpha\right] \quad \textit{when } \Delta_L > 0, \qquad \mathcal{H}^1(T_C \cap e_{5,0}) = \frac{[\Delta_L]_+}{W}. \tag{41}$$

The normalized trace requires $\Delta_R > 0$, and

$$T_C \textit{ is } \mathrm{CE1} \iff \Delta_L \le 0, \qquad T_C \textit{ is } \mathrm{CE2} \iff \Delta_L > 0. \tag{42}$$

The six radial exits from O are

$$\begin{aligned} d_0^C &= E - \alpha - \delta, & d_1^C &= \frac{\delta}{R}, \quad d_2^C = \delta, \\ d_3^C &= \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\}, & d_4^C &= \alpha, \quad d_5^C = \frac{\alpha}{W}. \end{aligned} \tag{43}$$

Moreover,

$$T_C \cap \{M_0, \dots, M_5\} = \{M_0\}. \tag{44}$$

For the closure of an original open C triangle, $\alpha, \delta > 0$.

Proof. Write $T_C \cap e_{0,1} = [s, t]$ with $0 \le s < t \le 1$. The two sides through the endpoints have a positive slope ratio. Reflect across the perpendicular bisector if necessary, call the ratio $R \le 1$, and put $W = 1 - R$. The three side equations are

$$F_1 = Rb + Wa - Rs, \qquad F_2 = -b + Ra + t, \qquad F_0 = Wb - a + E + Rs - t,$$

where $E = \sqrt{1 - RW}$ is the unit-side relation. If $R = 1$, then $F_0(O) = s - t < 0$, so $0 < R < 1$.

Set $\alpha := F_0(O)$ and $\delta := F_2(O)$. Then

$$t = W + \delta, \qquad Rs = \eta + \alpha + \delta = k,$$

which gives (38). The gradients are the three equilateral side-normal directions and $F_0 + F_1 + F_2 = E$.

On $e_{0,1}$, $a = 0$ and the active inequalities are $Rb \ge k$ and $b \le W + \delta$. Thus its length is

$$W + \delta - \frac{k}{R} = \frac{P - \alpha - W\delta}{R}.$$

On $e_{5,0}$, $b = 0$ and the active inequalities are $Wa \ge k$ and $a \le R + \alpha$, giving signed length

$$R + \alpha - \frac{k}{W} = \frac{P - R\alpha - \delta}{W}.$$

This proves the two trace formulas.

The positive right trace implies

$$\delta < \frac{P}{W} = \frac{ER}{1 + E} < \frac{R}{2}.$$

On $e_{1,2}$, parametrized by $b = 1 + q, a = q$, one has $F_2 = \delta - R - Wq < 0$. The center classification leaves $e_{5,0}$ as the only possible companion edge, and its signed length proves (42); $\Delta_L = 0$ is only point contact.

Substituting the six ray parametrizations into (38) gives (43). Finally,

$$W(\alpha+\delta) \le \alpha + W\delta < P$$

implies

$$E-\alpha-\delta > E-\frac{P}{W} = \frac{E(E-R)}{W} > \frac{1}{2},$$

where the last inequality is equivalent to $(E-R)^2 > 0$. Also

$$\alpha < P < \min\left\{\frac{R}{2}, \frac{W}{2}\right\}, \qquad \delta < \frac{R}{2}.$$

Therefore $d_0^C > 1/2$ and $d_i^C < 1/2$ for $i = 1, \ldots, 5$, proving the midpoint formula. Interior containment of O in the original open triangle makes all three side slacks positive, so $\alpha, \delta > 0$. □

Lemma A.15 (CE2 total slack)**.** *In the CE2 sign range, with $T := \alpha + \delta$,*

$$T < \frac{\eta}{E}. \tag{45}$$

Proof. The CE2 inequalities are $\alpha + W\delta < P$ and $R\alpha + \delta < P$. Multiply them by W and R and add. Since $W + R^2 = W^2 + R = E^2$, this gives $E^2 T < P = E\eta$. □

Corollary A.16 (Common center-boundary formula)**.** *The full center-boundary contribution is*

$$L_{\partial H}(T_C) = \frac{\Delta_R}{R} + \frac{[\Delta_L]_+}{W}. \tag{46}$$

In CE1,

$$L_{\partial H}(T_C) \le P \le \frac{\sqrt{3}}{2} - \frac{3}{4},$$

and in CE2,

$$L_{\partial H}(T_C) = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

Proof. The first formula sums (40) and (41). In CE1, $R\alpha + \delta \ge P$; multiplication by W, together with $\alpha \ge RW\alpha$, gives $\alpha + W\delta \ge WP$. Thus $\Delta_R \le RP$ and the contribution is at most P. Since $E \ge \sqrt{3}/2$ and $E(1-E)$ decreases there, the stated cap follows.

In CE2, direct addition gives

$$L_{\partial H}(T_C) = \frac{P - E^2(\alpha+\delta)}{RW} = \frac{E}{1+E} - \frac{E^2(\alpha+\delta)}{RW} < \frac{1}{2}.$$

□

Lemma A.17 (Adjacent-edge diameter transfer)**.** *For $0 \le q \le 1$, the zero-radial forward envelope is*

$$M_0(q) = \frac{-q + \sqrt{4-3q^2}}{2}.$$

Here M_0 without an argument remains the radial midpoint. If a unit equilateral triangle reaches parameters q, b on adjacent boundary edges, then $b \le M_0(q)$. The function is strictly decreasing and, for $q > 0$,

$$M_0(q) < 1 - \frac{q}{2}, \qquad 1 - M_0(q) > \frac{q}{2}.$$

Proof. The squared distance between the boundary points is $q^2 + qb + b^2 \le 1$. Solving the quadratic in b gives the formula. Differentiation gives

$$M_0'(q) = \frac{-1 - 3q/\sqrt{4-3q^2}}{2} < 0.$$

For $q > 0$, $\sqrt{4-3q^2} < 2$, which gives both strict comparisons. $\square$

Appendix B. Trace-Length Optimization

This appendix evaluates the trace quantities whose exact interfaces appear in Section 3. The center and V triangle coordinates are used only here; the body retains the geometric hypotheses and final bounds.

B.1. **Vd1 and Vd2 corner optimization.** Use temporary coordinates

$$X = V_0 + x(V_5 - V_0) + y(V_1 - V_0),$$

so $V_0 = (0,0)$, $V_5 = (1,0)$, $V_1 = (0,1)$, $O = (1,1)$, and $\|(x,y)\|^2 = x^2 + y^2 - xy$.

Proposition B.1 (Vd1/Vd2 corner normal form)**.** *Let T be the closure of an original* Vd1 *or* Vd2 *triangle at V_0, and let A_0, B_0 be its exact reaches toward V_5, V_1. There is a unique $t > 0$ such that, with $d = \sqrt{t^2+t+1}$,*

$$T = \left\{ (x,y) : \begin{aligned} x - (t+1)y &\le A_0, \\ ty - (t+1)x &\le tB_0, \\ tx + y &\le d - A_0 - tB_0 \end{aligned} \right\}. \tag{47}$$

The raw traces, parametrized from the indicated outer vertex, are

$$\begin{array}{c|c} r_0 & 0 \le s \le \dfrac{d - A_0 - tB_0}{t+1} \\ r_1 & \dfrac{t(1-B_0)}{t+1} \le s \le \dfrac{d - A_0 - tB_0 - 1}{t} \\ r_5 & \dfrac{1-A_0}{t+1} \le s \le d - A_0 - tB_0 - t. \end{array} \tag{48}$$

Intersecting with $[0,1]$ gives the actual traces, and

$$A_0 + B_0 < \frac{1}{2}. \tag{49}$$

Proof. The triangle has one vertex W outside H and two vertices U, Z in H. Because V_0 is an interior convex combination, both coordinates of W are negative. The axis endpoints $(A_0, 0), (0, B_0)$ lie on the two sides through W. Put $p = U - W$ and $q = Z - W$. Their order selects

$$q = (p_x - p_y, p_x), \qquad p_x > p_y > 0.$$

Writing $t = (p_x - p_y)/p_y$ gives uniquely

$$p = \frac{(t+1, 1)}{d}, \qquad q = \frac{(t, t+1)}{d},$$

which yields (47); substitution of $(s,s), (s,1), (1,s)$ yields (48).

If r_1 has positive trace, clearing its endpoint inequality gives

$$(t+1)A_0 + tB_0 < (t+1)(d-1) - t^2.$$

Since the left side is at least $t(A_0 + B_0)$,

$$A_0 + B_0 < \frac{(t+1)(d-1) - t^2}{t} < \frac{1}{2}.$$

The last inequality follows from

$$\left(t^2 + \frac{3t}{2} + 1\right)^2 - (t+1)^2(t^2 + t + 1) = \frac{t^2}{4} > 0.$$

Reflection treats support on r_5. $\square$

Lemma B.2 (Orientation of a forward Vd1 supplier)**.** *In proposition B.1, if a Vd1 role contains M_1 openly, then $t \geq 1$.*

Proof. Suppose $t < 1$. Open containment of $M_1 = (1/2, 1)$ gives $A_0 + tB_0 < d - 1 - t/2$. The backward radial interval has endpoints $\ell_5 = (1 - A_0)/(t+1)$ and $u_5 = d - A_0 - tB_0 - t$, so

$$u_5 > 1 - t/2 > \frac{1}{t+1} > \ell_5, \qquad 0 < \ell_5 < 1.$$

Here $1 - t/2 - 1/(t+1) = t(1-t)/(2(t+1)) > 0$. Thus r_5 has positive trace as well as r_1, contradicting Vd1. $\square$

Lemma B.3 (Vd2 adjacent-midpoint cap)**.** *If an original open V triangle has* Vd2 *closure T_i and*

$$\{M_{i-1}, M_{i+1}\} \cap T_i \neq \varnothing,$$

then its exact boundary reaches satisfy

$$A_i + B_i < \frac{1}{3}.$$

Proof. In Proposition B.1, reflection sends

$$(t, A_0, B_0, M_1, M_5) \longmapsto (1/t, B_0, A_0, M_5, M_1).$$

Thus assume $t \geq 1$ and use $A_0 + B_0 \leq A_0 + tB_0$.

If $M_5 = (1, 1/2)$ lies in T, the third half-plane gives

$$A_0 + B_0 \leq A_0 + tB_0 \leq d - t - \frac{1}{2} \leq \sqrt{3} - \frac{3}{2} < \frac{1}{3}.$$

If $M_1 = (1/2, 1)$ lies in T, the second and third half-planes give

$$B_0 \geq \frac{t-1}{2t}, \qquad A_0 + B_0 \leq S(t) := d - t - \frac{1}{2t}.$$

Positive-length contact with r_5 gives

$$A_0 + B_0 < R(t) := \frac{2(t+1)d - 3t^2 - t - 2}{2t}.$$

Direct differentiation shows S increasing and R decreasing on $[1, \infty)$. For S', the sign reduces to

$$(2t+1)^2 t^4 - (t^2 + t + 1)(2t^2 - 1)^2 = (t+1)(t^3 + 3t^2 - 1) > 0,$$

while $R' < 0$ follows from $d > t + 1/2$. Splitting at $t = 4/3$ gives

$$1 \leq t \leq \frac{4}{3}: \quad A_0 + B_0 \leq \frac{8\sqrt{37} - 41}{24} < \frac{1}{3},$$

$$t \geq \frac{4}{3}: \quad A_0 + B_0 < \frac{7\sqrt{37} - 39}{12} < \frac{1}{3}.$$

The final comparisons are $8\sqrt{37} < 49$ and $7\sqrt{37} < 43$. $\square$

B.2. **Conditional skeleton caps.**

Lemma B.4 (Center-skeleton calculation). *Under the hypotheses of Lemma 3.2, one has $L_S(T_C) < 3/2$.*

Proof. Normalize the unique midpoint as M_0 and a positive boundary trace to $e_{0,1}$. Use the signed parameters and six radial exits of proposition A.14. The positive trace condition is

$$\alpha + W\delta < P. \tag{50}$$

Summing the boundary and radial contributions gives

$$L_S(T_C) \le K_R + \mathcal{E}(\alpha, \delta), \qquad K_R := W + E - \frac{\eta}{R},$$

where

$$\mathcal{E}(\alpha, \delta) := -\frac{\alpha}{R} + \frac{\alpha}{W} + \delta + \min\left\{\frac{\alpha}{R}, \frac{\delta}{W}\right\} + \frac{[P - (R\alpha + \delta)]_+}{W}.$$

Splitting according to the minimum and positive part gives

$$\mathcal{E}(\alpha, \delta) \le \frac{P}{W}. \tag{51}$$

If $\alpha/R \le \delta/W$, the expression without the positive part is $\alpha/W + \delta$; when that part is active, the excess over P/W is $\alpha - R\delta/W \le 0$. The reflected order gives $\delta - W\alpha/R \le 0$. If the positive part vanishes, (50) gives the same bound strictly.

Put $A := 1 + R - R^2$. Then

$$\frac{3}{2} - K_R - \frac{P}{W} = \frac{1 + A - 2EA}{2RW}.$$

Both sides of $1 + A > 2EA$ are positive, and

$$(1 + A)^2 - 4A^2E^2 = R^2W^2(5 + 4R - 4R^2) > 0.$$

Therefore $K_R + P/W < 3/2$, as required. $\square$

For the remaining cap, reflect the local directions if necessary so that $0 \le A_i \le B_i \le 1$. The actual reach triple of a supercritical V triangle lies in the selected cell

$$\begin{aligned} A_i^2 + A_iB_i + B_i^2 &\le 1, \\ C_i &\le \frac{1}{2}, \\ (A_i^2 - 1)C_i^2 + (2A_iB_i^2 + B_i)C_i + (B_i^4 - B_i^2) &\le 0, \end{aligned} \tag{52}$$

where C_i belongs to the connected component containing 0 in the corresponding one-dimensional fiber. This is Cell S of Proposition A.11. The selector is essential: the quadratic is negative at $C_i = 0$, positive at $C_i = 1/2$, and opens downward.

Lemma B.5 (Supercritical skeleton calculation). *Every V triangle with $A_i + B_i > 1$ has $L_S(T_i) < 3/2$.*

Proof. The corner normal form, Proposition B.1, makes Vd1 and Vd2 triangles nonsupercritical. Lemma 2.11 gives the same conclusion for T3-like triangles. Hence a supercritical triangle is Vd0 and has no positive adjacent-radial support. Reflect so that $A_i \le B_i$. Then

$$L_S(T_i) = A_i + B_i + C_i.$$

Set

$$s := A_i + B_i, \qquad \delta := B_i - A_i, \qquad \gamma_0 := \frac{3}{2} - s.$$

The diameter constraint gives

$$1 < s \le \frac{2}{\sqrt{3}}, \qquad 0 \le \delta \le \sqrt{4 - 3s^2}.$$

Let $P(\gamma)$ be the quadratic in (52). Substitution and multiplication by 16 yield

$$\begin{aligned} 16P(\gamma_0) = Q(s,\delta) = {} & \delta^4 + 8\delta^3 s - 6\delta^3 + 14\delta^2 s^2 - 18\delta^2 s + 5\delta^2 \\ & - 8\delta s^3 + 30\delta s^2 - 34\delta s + 12\delta \\ & + s^4 - 6s^3 - 19s^2 + 60s - 36. \end{aligned}$$

Its derivative factors as

$$\frac{\partial Q}{\partial \delta} = 2(2\delta + 4s - 3)\big(\delta^2 + \delta(4s-3) + (s-1)(2-s)\big) \ge 0.$$

Therefore

$$Q(s,\delta) \ge Q(s,0) = (s-1)(s^3 - 5s^2 - 24s + 36) > 0.$$

The cubic decreases on $[1, 2/\sqrt{3}]$ and at the right endpoint equals $88/3 - 136\sqrt{3}/9 > 0$. Hence $P(\gamma_0) > 0$. The selected smaller-root component in (52) forces $C_i < \gamma_0$, so $A_i + B_i + C_i < 3/2$. □

Appendix C. Area-Loss Optimization

This appendix proves the pointwise local loss bound of theorem 4.1. The cyclic accumulation is proved in the body.

C.1. **Local square-loss optimization.**

Lemma C.1 (Solved local square-loss optimization)**.** *Every triangle satisfying the hypotheses of theorem 4.1 obeys equations (10) and (11).*

Proof. Use the shared corner chart of lemma A.2 and the orientation forms equations (20) and (21). Write $D = 1 - t + t^2$, $z = \sqrt{D}$, with $0 \le t \le 1$. In Type I, containment of $(0,0), (0,a), (b,0)$ gives

$$\alpha \ge tb, \quad \beta \ge (1-t)a, \quad \alpha + \beta + (1-t)b \le z, \quad \alpha + \beta + ta \le z. \tag{53}$$

The two inequalities defining W become

$$(1-t)U + V \ge (1-t)\alpha + \beta, \qquad U + tV \ge \alpha + t\beta.$$

For fixed t, lowering α or β therefore enlarges $T \cap W$. The outside loss is minimized at $\alpha = tb$, $\beta = (1-t)a$. At that placement the portion outside W has vertices, in the (U,V)-plane,

$$(0,0), \quad (a + tb, 0), \quad (tb, (1-t)a), \quad (0, b + (1-t)a).$$

Since the Jacobian of $(x,y) \mapsto (U,V)$ has absolute value D,

$$\mathcal{L}(T) \ge \mathcal{L}_{\mathrm{I}}(a,b;t) = \frac{(1-t)a^2 + tb^2 + 2t(1-t)ab}{D}. \tag{54}$$

If $a \le b$, then

$$D(\mathcal{L}_{\mathrm{I}} - a^2) = t\{b^2 - a^2 + (1-t)(a^2 + 2ab)\} \ge 0;$$

the reflected factorization proves $\mathcal{L}_{\mathrm{I}} \ge b^2$ when $b \le a$. This proves (10) in Type I.

Suppose now that $a+b>1$ and $a\le b$. Set
$$r=\frac{a}{b},\qquad t_0=\frac{1-r^2}{1+2r}.$$
Since $b>1/(1+r)$, feasibility in (54) and (53) excludes $0\le t<t_0$: for $0<t<t_0$,
$$\left(\frac{1+r(1-t)}{1+r}\right)^2-D=\frac{t(1+2r)(t_0-t)}{(1+r)^2}>0,$$
so $b+(1-t)a>z$, while $t=0$ would give $a+b\le 1$. Hence $t\ge t_0$, and
$$D(\mathcal{L}_{\mathrm{I}}-b^2)=(1-t)b^2\{r^2-1+t(2r+1)\}\ge 0.$$
If $b\le a$, put $r=b/a$ and $t_1=(r^2+2r)/(1+2r)$. The identical reflected calculation excludes $t>t_1$ and gives
$$D(\mathcal{L}_{\mathrm{I}}-a^2)=ta^2\{r^2+2r-t(1+2r)\}\ge 0.$$
Thus (11) holds in Type I.

In Type II, substituting the axes in equation (21) gives the exact positive y- and x-axis reaches
$$A=\beta,\qquad B=z-\alpha-\beta,\qquad A+B\le z\le 1. \tag{55}$$
In particular Type II is impossible when $a+b>1$. Its wedge intersection is the quadrilateral with vertices
$$(0,0),\quad (B,0),\quad \left(\frac{B+(1-t)A}{D},\frac{A+tB}{D}\right),\quad (0,A).$$
Hence
$$\mathcal{L}(T)=\mathcal{L}_{\mathrm{II}}(A,B;t)=1-\frac{(1-t)A^2+tB^2+2AB}{D}. \tag{56}$$
Let $S=A+B$. If $A\le B$, write $A=rS$, $B=(1-r)S$ with $0\le r\le 1/2$. Since $S^2\le D$, and with
$$C=(1-t)r^2+t(1-r)^2+2r(1-r),$$
we have $\mathcal{L}_{\mathrm{II}}\ge 1-C$. The exact factorization
$$1-C-r^2D=(1-t)(1-2r+tr^2)\ge 0$$
gives $\mathcal{L}_{\mathrm{II}}\ge r^2D\ge A^2$. Reflection gives $\mathcal{L}_{\mathrm{II}}\ge B^2$ when $B\le A$. Since $A\ge a$ and $B\ge b$, this proves (10). The two orientation types are exhaustive, so the theorem follows. $\square$

Appendix D. Nonzero-Gap Finite-Enclosure Optimization

This appendix proves the selected-gap and supported-rescuer enclosure statements and the two-vertex replacement. The witnesses are fixed in Section 5; candidate parameters vary only in their enclosure proofs.

D.1. **Direct neighboring-ray domination.** The capacities keep their variational definitions. The following proof establishes the bound needed by all active witnesses without evaluating the cubic plateau and radical branches of the full function.

Lemma D.1 (Sharp diagonal neighboring capacity)**.** *For* $0 \le m \le 1/2$*, one has* $C_+(m,m) = C_-(m,m) = 1-m$*.*

Proof. Use oblique coordinates with metric x^2+y^2-xy. The four points are $P=(0,0)$, $A=(m,0)$, $B=(0,m)$, and $Z=(c,1)$. The case $m=0$ is immediate. If $0<m\le 1/2$ and $c>1-m$, their hull has cyclic order P,A,Z,B. lemma A.8 reduces the enclosure to its four outward edge normals. At PA and BP, the side lengths are $c+m$ and $1+m$, both greater than one.

Embed (x,y) as $(x-y/2,hy)$. At the other two edges take unnormalized normals $n_A=(h,1/2-c+m)$ and $n_B=(h(m-1),m/2+c-1/2)$. Put

$$D_A=(c-m)^2-(c-m)+1, \qquad N_A=m+c^2-cm-c+1,$$
$$D_B=c^2+cm-c+m^2-2m+1, \quad N_B=cm+c^2-c-m+1.$$

Selecting the contained points (A,Z,P) in the three normal directions at n_A, and (B,P,Z) at n_B, gives support lower bounds $L_{AZ}\ge N_A/\sqrt{D_A}$ and $L_{ZB}\ge N_B/\sqrt{D_B}$. No active-support selector is assumed. With $q=c+m-1>0$,

$$\begin{aligned} N_A^2-D_A &= q^2\big((q+1-3m)^2+4m^2-2m+1\big)\\ &\quad + q(1-2m)(6m^2-2m+1)+m^2(1-2m)^2>0,\\ N_B^2-D_B &= q^4+(2-2m)q^3+(m^2-4m+2)q^2\\ &\quad +(1-m)(1-2m)q>0. \end{aligned}$$

Here $4m^2-2m+1\ge 3/4$, $6m^2-2m+1>0$, and $m^2-4m+2\ge 1/4$. The numerators are positive: $N_A\ge(1+m)(3-m)/4$ and $N_B=1-m+(1-m)q+q^2$. Thus every caliper requires side greater than one. Equality is realized by the unit triangle $\operatorname{conv}\{(-m,0),(1-m,0),(1-m,1)\}$. Reflection gives C_-. □

Lemma D.2 (Solved common-pair comparison)**.** *The inequalities in lemma 5.1 hold.*

Proof. Put $m=\min(p,q)$. By convexity, any triangle containing the p,q boundary anchors also contains the shorter m,m anchors. Hence lemma D.1 gives $C_\pm(p,q)\le 1-m$. If $q=m$, the unit triangle used for equality contains $(p,0)$, $(0,q)$, and $(1-q,1-q)$ because $p\le 1-q$. The last point is on the own ray, so $c_{\max}(p,q)\ge 1-q$. Reflect for $p=m$. Increasing an anchor shrinks the feasible family, giving antitonicity. □

D.2. **Exact direct output formulas.** The following outgoing capacities follow from Proposition A.11 and supply the CE1 return.

For a nonsupercritical triangle with incoming reach at least a and radial reach at least c, let

$$\mathcal{B}_c(a)=\max\{b:(a,b,c)\text{ is admissible and } a+b\le 1\}.$$

This is the outgoing capacity of a single triangle.

For $1/2<c\le\sqrt{3}/2$, put

$$h_c=\frac{2c}{1+\sqrt{16c^2-3}}.$$

For $\sqrt{3}/2 < c < 1$, put

$$\ell(c) = \frac{c}{2}\left(1 - \sqrt{4c^2 - 3}\right), \qquad r(c) = c - \ell(c).$$

Also set

$$\mathcal{Q}_-(a,c) = \frac{\sqrt{a^2 - ac + c^2}}{c} - a,$$
$$\mathcal{Q}_+(a,c) = \frac{2(1-a^2)c^2}{2ac^2 + c + \sqrt{(2ac^2 + c)^2 - 4(1-c^2)(1-a^2)c^2}}.$$

Proposition D.3 (Four direct high-radial outputs)**.** *For $1/2 < c \le \sqrt{3}/2$,*

$$\mathcal{B}_c(a) = \begin{cases} 1-a, & 0 \le a \le 1-c, \\ \mathcal{Q}_+(a,c), & 1-c < a < h_c, \\ h_c, & a = h_c, \\ \mathcal{Q}_-(a,c), & h_c < a < c, \\ 1-a, & c \le a \le 1. \end{cases}$$

For $\sqrt{3}/2 < c < 1$,

$$\mathcal{B}_c(a) = \begin{cases} 1-a, & 0 \le a \le 1-c, \\ \mathcal{Q}_+(a,c), & 1-c < a \le \ell(c), \\ \ell(c), & \ell(c) < a < r(c), \\ \mathcal{Q}_-(a,c), & r(c) \le a < c, \\ 1-a, & c \le a \le 1. \end{cases}$$

At $a = \ell(c)$, the selected value is $r(c)$; immediately to the right it is $\ell(c)$.

Proof. Since $c > 1/2$, any containing V triangle contains the radial midpoint. Therefore $a + b \le 1$, and the supercritical cell of Proposition A.11 is absent. On the cap $b = 1 - a$, the T-cell inequality becomes

$$M(c - M) \le 0,$$

where $M = \max\{a, b\}$. Thus the linear cap is feasible exactly when $a \le 1 - c$ or $a \ge c$.

Off the linear cap, a maximal point lies on $F_L = 0$ or $F_T = 0$. The selected L-cell equation has the two ordered roots $\ell(c), r(c)$; the selector $q \le 0$ retains the smaller coordinate. The ordered halves of $F_T = 0$ give $\mathcal{Q}_+$ and $\mathcal{Q}_-$. Their equality boundary $a = b$ is h_c in the low-radial range, while their $q = 0$ boundaries are $(\ell(c), r(c))$ and $(r(c), \ell(c))$ in the high range. These conditions produce the two displayed formulas.

The formal other root in the $\mathcal{Q}_+$ quadratic violates the geometric selector $c \le 2\max\{a, b\}$ and is discarded. At the high-range junction, the selected T-cell point has value $r(c)$, whereas the adjacent selected L-cell point has value $\ell(c)$; this is the genuine downward jump stated above. □

D.3. **High-radial and selected-chord calculations.** For

$$0 < e < 1 - \frac{\sqrt{3}}{2},$$

define the selected low root

$$\epsilon(e) = \frac{1-e}{2}\left(1 - \sqrt{4(1-e)^2 - 3}\right). \tag{57}$$

It satisfies

$$e < \epsilon(e), \qquad \epsilon(e) < \frac{2e}{1-2e}, \tag{58}$$

and, for $e \le 1/8$,

$$\epsilon(e) \le 2e + 5e^2. \tag{59}$$

These follow by substituting the proposed upper bounds in the selected quadratic and checking its sign.

Lemma D.4 (Direct high-radial threshold)**.** *Let $0 < e < 1 - \sqrt{3}/2$. A nonsupercritical V triangle has incoming boundary reach A and outgoing boundary reach B. If*

$$C \ge 1 - e, \qquad A > \epsilon(e),$$

then

$$\boxed{B \le \epsilon(e).}$$

Proof. The condition $C > 1/2$ forces $A + B \le 1$. In the exact cells of Proposition A.11, the only selected component with $A > \epsilon(e)$ has $B \le \epsilon(e)$; equivalently the two roots of the selected T-cell quadratic separate the feasible fiber. The selector $c \le 2\max\{A, B\}$ removes the formal high root. □

Lemma D.5 (Selected-arc chord bounds)**.** *Put $c = 1 - e$, with $0 < e < 1/4$, and define $q_*(p) = 1 - \mathcal{Q}_+(p, c)$ on the selected branch. Its right endpoint is*

$$(p_e, q_e) = \begin{cases} (\epsilon(e), e + \epsilon(e)), & c > h, \\ (h_c, 1 - h_c), & c \le h, \end{cases} \qquad h = \sqrt{3}/2.$$

The function q_* is increasing and concave on $[e, p_e]$, with $q_*(e) = e$. Thus, for $e < p \le p_e$,

$$q_*(p) \ge p + \lambda_e(p - e), \qquad \lambda_e = \frac{q_e - p_e}{p_e - e}.$$

One has $\lambda_e > 1 - 4e$ when $0 < e < 1 - h$, and $\lambda_e > 1 - 5e$ throughout $0 < e < 1/4$. A following actual incoming reach q, forced by a weak boundary handoff, satisfies $q \ge q_*(p)$; equality is not assumed for actual reaches.

Proof. The selected boundary is parametrized by

$$q_* = e + cx, \qquad p = e + cx - 1 + \sqrt{1 - x + x^2}.$$

Indeed its selected quadratic gives $x(1-x) = (q_* - p)(2 - q_* + p)$. If $p = p(x)$, then

$$p'(x) = c + \frac{2x - 1}{2\sqrt{1 - x + x^2}} \ge c - \frac{1}{2} > 0, \qquad p''(x) = \frac{3}{4(1 - x + x^2)^{3/2}} > 0$$

on the selected interval $0 \le x \le 1$. Hence $dq_*/dp > 0$ and $d^2q_*/dp^2 = -cp''/(p')^3 < 0$. The endpoint formulas follow from proposition D.3, and concavity places the graph above its endpoint chord. For $c > h$,

$$\lambda_e = \frac{e}{\epsilon(e) - e} > \frac{1-2e}{1+2e} > 1 - 4e,$$

using (58). For $3/4 < c \le h$, the formula for h_c gives

$$h_c \le \frac{3}{2(1+\sqrt{6})} < \frac{7}{16}, \qquad \lambda_e > \frac{2}{7-16e} > 1 - 5e.$$

For the first bound, h_c decreases with c on this interval and $17^2 < 49 \cdot 6$. For the second, $e \ge 1 - h > 1/8$, and $80e^2 - 51e + 5 < 0$ on $[1/8, 1/4]$: it decreases there and has value $-1/8$ at $1/8$. Finally the preceding outgoing reach is at most $\mathcal{Q}_+(p, c)$; a weak handoff therefore gives $q \ge 1 - \mathcal{Q}_+(p, c)$. □

D.4. **CE1 reverse-path calculation.** The first lemma uses only the T_4 radial demand. The full return adds the T_3 and T_2 demands; BC establishes them before using the latter.

Use the signed center parameters of proposition A.14, with $w = 1 - R$, $A = \alpha$, and $D = \delta$. Assume

$$A > 0, \qquad D > 0, \qquad A + wD < P, \qquad D + RA \ge P. \tag{60}$$

Put

$$X = R - D, \qquad m = \frac{A}{R}, \qquad h_0 = \frac{\eta + A + D}{2R}. \tag{61}$$

Lemma D.6 (First CE1 return step)**.** *Assume* (60)*, and let* $T_1, \dots, T_4$ *be nonsupercritical V triangles of arbitrary types, with actual reaches satisfying*

$$A_1 \ge X, \qquad B_i + A_{i+1} \ge 1 \ (1 \le i \le 3), \qquad B_4 \ge h_0. \tag{62}$$

If $C_4 \ge 1 - A$*, then the reflected* T_4 *state lies on the selected* $\mathcal{Q}_+$ *component and*

$$B_3 > L_1, \qquad L_1 = (2 - 4A)h_0 - (1 - 4A)A. \tag{63}$$

No own-radial demand at T_2 *or* T_3 *is assumed.*

Proof. The sign difference in equation (60) gives

$$RD > wA,$$

so the center exit on r_3 is $m = A/R$. Moreover

$$0 < A < P < \frac{1}{8}, \qquad 0 < D < \frac{R}{2}, \qquad 0 < m < \frac{w}{2} < \frac{1}{2}. \tag{64}$$

We first record

$$\epsilon(A) < X, \qquad h_0 > A. \tag{65}$$

Indeed, $A + wD < P$ and $D = R - X$ give $A < wX - \eta$. Convexity of

$$wX - \frac{X}{2(1+X)}$$

on $R/2 < X < R$ gives

$$A < \frac{X}{2(1+X)}.$$

The selected-root bound equation (58) yields $\epsilon(A) < X$. Also

$$2R(h_0 - A) = \eta + D + (1 - 2R)A > 0;$$

when $R > 1/2$, use $A < P = \eta E$ for the last sign.

The assumed transverse radial reach at T_4 is at least $1 - A$. Apply the exact local catalogue to the reflected boundary pair (B_4, A_4). If the local point is not on the selected $\mathcal{Q}_+$ component, the constant, $\mathcal{Q}_-$, and linear components together with equation (65) give

$$1 - A_4 \geq 1 - \epsilon(A) > 1 - X.$$

The center-free edges then carry this lower bound backward to $B_1 > 1 - X$, contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$.

Hence the T_4 state is selected $\mathcal{Q}_+$. The selected-chord estimate lemma D.5 and the supplied floor $B_4 \geq h_0$ give equation (63). □

Proposition D.7 (CE1 tail-input three-transverse obstruction). *Under the domain, path, and tail hypotheses of lemma D.6, the additional own-radial demands*

$$C_4 \geq 1 - A, \qquad C_3 \geq 1 - m, \qquad C_2 \geq 1 - D \tag{66}$$

are incompatible.

Proof. Apply lemma D.6. The selected-state condition forces

$$\boxed{X > \frac{1}{2}.} \tag{67}$$

To see this, selectedness gives

$$h_0 \leq \epsilon(A) < \frac{2A}{1 - 2A}.$$

If $X \leq 1/2$, the lower CE1 inequality gives

$$2Rh_0 = \eta + A + D \geq w(R + A).$$

Consequently

$$f_R(A) < 0, \qquad f_R(z) = w(R + z)(1 - 2z) - 4Rz.$$

The quadratic is strictly concave and $f_R(0) > 0$. If $R \leq 1/2$, then

$$f_R(P) \geq \frac{1 - E}{2}(6E^3 - 2E^2 - 5E + 2) > 0.$$

If $R > 1/2$, then $A < A_* = w/2 - \eta$ and

$$f_R(A_*) = \frac{1 - E}{2}(E + 11R - 3ER - 5) > 0.$$

Concavity contradicts $f_R(A) < 0$ in both cases, proving equation (67). Thus

$$m < \frac{w}{2} < \frac{1}{4}. \tag{68}$$

If $B_3 \geq 1/2$, then $B_3 > 1 - X$ and the preceding immediate contradiction applies. Assume $B_3 < 1/2$. At T_3 the reflected incoming reach is $B_3 > h_0 > m$, while its assumed transverse radial reach is at least $1 - m$. If this state is not selected $\mathcal{Q}_+$, the exact local catalogue gives

$$B_2 \geq 1 - B_3 > \frac{1}{2} > 1 - X,$$

and the remaining center-free edge again contradicts T_1.

Thus both T_4 and T_3 are selected $\mathcal{Q}_+$. A second use of lemma D.5 gives

$$B_2 > (2-5m)B_3 - (1-5m)m.$$

Since $2-5m>0$, equation (63) yields

$$\boxed{B_2 > L_2, \qquad L_2 = (2-5m)L_1 - (1-5m)m.} \tag{69}$$

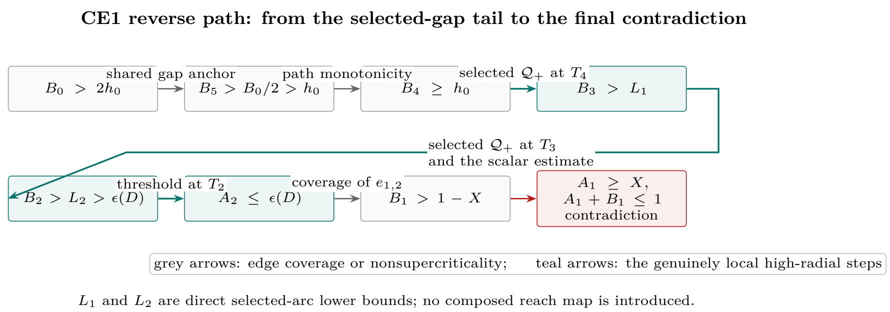


FIGURE 8. Logical structure of the CE1 direct reverse-path certificate. The proof starts from the BC tail input, moves through the actual boundary reaches, uses two selected-arc estimates and one high-radial threshold, and returns to T_1 with incompatible incoming and outgoing demands.

We now prove the scalar estimate

$$\boxed{D < \frac{1}{10}, \qquad L_2 > 2D + 3D^2 > \epsilon(D), \qquad \epsilon(D) < X.} \tag{70}$$

The T_4 selected-state inequality and equation (59) give

$$D \leq D_h := A(4R-1) + 10RA^2 - \eta. \tag{71}$$

If $D \geq 1/10$, then $A + wD < P$ gives

$$A < \overline{A} = \frac{w(5R-1)}{10}.$$

The right side of equation (71) is increasing in A, while $\eta > Rw/2$, so

$$D < U(R) := \overline{A}(4R-1) + 10R\overline{A}^2 - \frac{Rw}{2}.$$

Direct expansion gives

$$\frac{1}{10} - U(R) = -\frac{R}{10}P_4(R),$$

where

$$P_4(R) = 25R^4 - 60R^3 + 26R^2 + 22R - 14.$$

For $y = 2R-1 \in (0,1)$,

$$16P_4(R) = 25y^4 - 20y^3 - 106y^2 + 124y - 39 \leq -101y^2 + 124y - 39 < 0.$$

Hence $U(R) < 1/10$, a contradiction. This proves $D < 1/10$.

Put

$$\Psi(D) = L_2 - 2D - 3D^2, \qquad J(D) = L_2 - \frac{23}{10}D.$$

Since $D < 1/10$,

$$\Psi(D) - J(D) = 3D\left(\frac{1}{10} - D\right) > 0.$$

For fixed R, A, differentiation of equation (69) gives

$$J'(D) = \frac{S(R,A)}{10R^2},$$

where

$$S(R,A) = 100A^2 - (40R+50)A + R(20-23R).$$

The nonempty interval $P - RA \le D \le D_h$ implies

$$A > \frac{4Rw}{25R-4} =: A_*.$$

On the relevant interval $\partial S/\partial A < -45$, and

$$S(R,A) < S(R,A_*) = -\frac{Rq_3(R)}{(25R-4)^2},$$

where

$$q_3(R) = 8775R^3 - 14260R^2 + 7928R - 1120.$$

For $y = 2R - 1$,

$$8q_3(R) = 8775y^3 - 2195y^2 + 997y + 3007 > 0.$$

Thus J decreases with D, so $J(D) \ge J(D_h)$.

At $D = D_h$, one has $h_0 = 2A + 5A^2$. If $L_{2,h}$ is the corresponding value, then

$$L_{2,h} - A(1+4R) = \frac{A}{R^2}Q(R,A),$$

where

$$\begin{aligned} Q(R,A) = {} & 100A^3R - 40A^2R^2 - 30A^2R + 12AR^2 - 15AR + 5A \\ & - 4R^3 + 5R^2 - R. \end{aligned}$$

This polynomial is concave in A on $0 \le A \le Rw/2$. Its endpoint values are

$$Q(R,0) = Rw(4R-1) > 0$$

and

$$Q\left(R, \frac{Rw}{2}\right) = \frac{Rw}{2}(25R^5 - 30R^4 + 20R^3 - 3R^2 - 7R + 3) > 0.$$

The last positivity follows after putting $y = 2R - 1$:

$$32p(R) = 25y^5 + 65y^4 + 90y^3 + 106y^2 - 35y + 5 > 0.$$

Hence $L_{2,h} > A(1+4R)$.

Finally $A < P = \eta E$, $A < Rw/2$, and $1/E > 1 + Rw/2$ give

$$\frac{D_h}{A} < C_* := 4R - 2 + 5R^2w - \frac{Rw}{2}.$$

Moreover

$$1 + 4R - \frac{23}{10}C_* = \frac{230R^3 - 253R^2 - 81R + 112}{20} > 0.$$

Thus $J(D_h) > 0$, so

$$L_2 > 2D + 3D^2.$$

Substitution in the selected-root quadratic gives $\epsilon(D) < 2D + 3D^2$, and the argument proving $\epsilon(A) < X$ also gives $\epsilon(D) < X$. This proves equation (70).

The assumed transverse radial reach at T_2 is at least $1 - D$. Since $B_2 > \epsilon(D)$, lemma D.4, applied after reflection, gives

$$A_2 \leq \epsilon(D).$$

Coverage of $e_{1,2}$ therefore forces

$$B_1 \geq 1 - A_2 \geq 1 - \epsilon(D) > 1 - X,$$

contradicting $A_1 \geq X$ and $A_1 + B_1 \leq 1$. Thus the tail-input boundary and radial hypotheses are incompatible. □

D.5. **Fixed endpoints and arbitrary candidates.**

Lemma D.8 (Total radial endpoint and maximum-to-own-demand)**.** *The assertions of lemma 5.3 hold for arbitrary V types.*

Proof. No closed V triangle contains O: its distinguished vertex is interior and at distance one from O, so moving that vertex slightly away from O inside the open triangle would contradict diameter one. Each own trace has positive length. Hence $0 < \gamma_i < 1$ and $0 < d_i < 1$.

If a local open role contained $Z_i(\gamma_i)$, openness would give a point $Z_i(c)$ with $c > \gamma_i$ in that role. A role with no positive neighboring trace would acquire one; otherwise its maximal endpoint would increase. Both contradict the definition of γ_i. The nonlocal roles are excluded by

$$\|d_i V_i - V_{i\pm 2}\|^2 = 1 + d_i + d_i^2 > 1, \qquad \|d_i V_i - V_{i+3}\| = 1 + d_i > 1.$$

Thus every total endpoint is missed by all open V roles. For $\gamma_i \leq \Gamma < 1$, the same proof with $d = 1 - \Gamma$ excludes $(1 - \Gamma)V_i$: openness would give a local endpoint strictly above $\Gamma \geq \gamma_i$. For $\Gamma = 1$, the point is O, which was already excluded even from the closed V triangles.

Now let an arbitrary open unit candidate contain O and $\widehat{P}_i$. If its radial exit is d_i', open containment gives $d_i < d_i'$ and hence $\gamma_i > 1 - d_i'$. If both neighboring endpoints are at most $1 - d_i'$, the own term must realize this strict demand. For a nonsupercritical neighbor with both actual boundary reaches above d_i', the bound $C_\pm(A_j, B_j) \leq 1 - \min\{A_j, B_j\} < 1 - d_i'$ excludes its contribution. No skeleton-cover hypothesis for this new candidate is used. □

Lemma D.9 (Fixed six-point transverse enclosure)**.** *The set K_{BC} in theorem 5.8 has $\Lambda(K_{BC}) \geq 1$ for arbitrary V types on the nonsupercritical path.*

Proof. Suppose an open unit candidate U' contains K_{BC}. Its origin is implicit in the witness hull, and M_0 fixes the unique-midpoint normalization. Write its signed parameters as $R, W, E, \eta, P, \alpha, \delta$ and put

$$X = R - \delta, \qquad Q = \frac{\eta + \alpha + \delta}{2R}.$$

The two selected gap endpoints give $B_0 > 2Q$ and $A_1 > X$. The explicit tail premise gives $B_5 \geq B_0/2 > Q$. The four middle gap-free edges and nonsupercriticality imply

$$A_1 < A_2 < A_3 < A_4 < A_5, \qquad A_i > X, \quad B_i > Q \quad (1 \leq i \leq 5). \tag{72}$$

The pair (X, Q) is positive and has sum below one because it is strictly dominated by a nonsupercritical actual pair.

CE2 candidates. Use the selected high-radial root

$$e(d) = \frac{1-d}{2}\left(1 - \sqrt{4(1-d)^2 - 3}\right).$$

The strict CE2 center inequalities give

$$X > e(\alpha), \qquad \min\{e(\alpha), e(\delta)\} < Q. \tag{73}$$

For completeness, $WX > \eta + \alpha$, and testing the low-root equation at this bound reduces the first assertion to

$$\pi(t) = (\eta + t)(1 - 2t) - 2Wt > 0 \qquad (0 \le t \le P).$$

The polynomial is concave, $\pi(0) = \eta > 0$, and $\pi(P) = \eta(\eta + 2ER^2) > 0$. For the second assertion put $T = \alpha + \delta$. The weighted sum of the two strict center inequalities gives $T < \eta/E$. With $d = \min\{\alpha, \delta\}$,

$$\min\{e(\alpha), e(\delta)\} < \frac{2d}{1-2d} \le \frac{T}{1-T} < Q.$$

The final comparison follows from concavity of $\chi(T) = (\eta + T)(1 - T) - 2RT$, positive at zero and at η/E, where it has value $\eta(E - R)^2/E^2$.

If $e(\alpha) < Q$, both coordinates of (X, Q) exceed $e(\alpha)$. The high-radial threshold therefore gives $c_{\max}(X, Q) < 1 - \alpha$. Otherwise $e(\delta) < Q \le e(\alpha) < X$ and it gives $c_{\max}(X, Q) < 1 - \delta$. Common-pair domination and equation (72) bound all local contributors on r_4 in the first case and r_2 in the second. Thus the candidate misses the corresponding fixed total endpoint.

CE1 candidates. Now $R\alpha + \delta \ge P$ and $\alpha + W\delta < P$. Set $m = \alpha/R$; the transverse exits are α, m, δ on r_4, r_3, r_2. Combining the two signed inequalities gives $E^2\alpha < RP$, so $m < \eta/E$. Also

$$X > \frac{R}{1+E} > \frac{\eta}{E} > m, \qquad 2R(Q - m) = \eta + \delta - \alpha > 0.$$

Here $E > W$ proves the middle inequality, and $\alpha < P = E\eta < \eta$ proves the last sign. Consequently

$$\min\{X, Q\} > m > \alpha. \tag{74}$$

The neighboring capacities on r_4, r_3 are therefore below the candidate's required demands. Fixed-endpoint containment and lemma D.8 recover

$$C_4 > 1 - \alpha, \qquad C_3 > 1 - m. \tag{75}$$

Apply lemma D.6 with $h_0 = Q$. The path inequalities give $B_4 > Q$, and only the recovered demand $C_4 > 1 - \alpha$ is needed. Thus

$$B_3 > L_1, \qquad L_1 = (2 - 4\alpha)Q - (1 - 4\alpha)\alpha.$$

The identity

$$R(L_1 - \delta) = (1 - 2\alpha)\eta + (W - 2\alpha)(\alpha + \delta) + 4R\alpha^2 > 0 \tag{76}$$

follows by expansion, with every sign justified by $0 < \alpha < P < \min\{1/2, W/2\}$. Hence $B_3 > \delta$. Backward path monotonicity gives $B_1 > B_3 > \delta$, and $A_1, A_3 > X > \delta$. Both neighbors of r_2 have capacities below $1 - \delta$. Candidate containment now recovers

$$C_2 > 1 - \delta. \tag{77}$$

All three own-radial hypotheses of proposition D.7 now hold. Its conclusion contradicts $A_1 > X$ and nonsupercriticality, excluding CE1 as well. □

Lemma D.10 (Fixed four-point supported-rescuer proof)**.** *The geometric statement of theorem 5.12 holds.*

Proof. If $a = 0$ or $\beta = 0$, the set contains O and a hexagon vertex at distance one, so it cannot lie in an open unit equilateral triangle. Otherwise suppose a candidate contains the four points. The convex combination in the theorem's body proof puts $(\varepsilon/s)V_0$ in it, and $\varepsilon/s \geq 1/2$ puts M_0 in it by convexity with O. The candidate has a positive boundary trace and unique midpoint M_0. Use the left-active reflected signed normal form when needed; a positive right-edge trace is not an extra hypothesis. The left trace has endpoints $k/W, R + \alpha$, and the exit on r_1 is δ/R, with $k = \eta + \alpha + \delta$ and $\eta, \alpha, \delta > 0$. Containment of $Y(a)$ and εV_1 gives

$$Wa > k, \qquad \delta > R\varepsilon,$$

so $\alpha < a - Rs$ and $R < a/s$. Since $s \leq 1$,

$$R + \alpha < a + R(1 - s) \leq a/s.$$

Containment of $Y(1 - \beta)$ would require $1 - \beta < R + \alpha$, whereas $\beta \leq \varepsilon/s$ gives $1 - \beta \geq a/s$. This is impossible. □

D.6. **The supported-rescuer adapter.**

Lemma D.11 (Supported endpoint to four-point enclosure)**.** *Suppose T_0 has actual boundary endpoint $a = A_0$ toward V_5 and supported interval $[c, u]$ on r_1, measured from V_1 toward O. Put $\varepsilon = 1 - u > 0$ and $M = M_c^{\sup}$. Assume*

$$\varepsilon V_1 \in U_C, \qquad A_0 + \varepsilon \leq 1, \qquad \frac{A_0}{A_0 + \varepsilon} \leq 1 - M.$$

If T_1 is uniquely supercritical with $C_1 \geq c$, and the four center-free edges from $e_{1,2}$ through $e_{4,5}$ join nonsupercritical T_2, T_3, T_4, T_5, then the skeleton is not covered. The conclusion applies to both possible nonzero gap ranks.

Proof. The strict-supercritical envelope gives $B_1 < M$, and the four handoffs give $B_5 \leq B_4 \leq B_3 \leq B_2 \leq B_1$. Thus $B_5 < 1 - A_0$, so the left-edge actual gap has endpoints $Y(A_0), Y(1 - B_5)$, with $Y(t) = (1 - t)V_0 + tV_5$. Under coverage these endpoints and $O, \varepsilon V_1$ belong to U_C. Since $M \geq 1/2$, the hypotheses imply

$$A_0 \leq \varepsilon, \qquad B_5 < M \leq \frac{\varepsilon}{A_0 + \varepsilon}.$$

Apply theorem 5.12 with geometric parameters $a = A_0$, $\beta = B_5$. The fixed set has enclosure number at least one, contrary to compact containment in U_C. □

D.7. **T3-like supported-tail adapter.**

Lemma D.12 (Solved T3-like supported-tail adapter)**.** *Under the off-center supplier placement of proposition 5.14, the actual supported interval satisfies the hypotheses of lemma D.11.*

Proof. Normalize T_0 as the T3-like supplier of M_1, with uniquely supercritical T_1 and all remaining roles nonsupercritical Vd0. Use the original Type-II chart (21), writing its positive first slack as α_T and its actual reach β as $a = A_0$. Thus $0 < t < 1$

and $z = \sqrt{1 - t + t^2}$. On r_1, whose coordinates are $(1, s)$, direct substitution gives the actual interval $c \le s \le u$, where

$$c = \frac{1 + \alpha_T + a - z}{1 - t}, \qquad u = a + t, \qquad \varepsilon = 1 - u. \tag{78}$$

Open midpoint containment gives $0 < c < 1/2 < u < 1$. The supercritical own role ends before the midpoint and the other roles have no positive trace on r_1. Hence u is its total radial frontier: $P_T = \varepsilon V_1 \in U_C$ by lemma 5.3. Coverage of the near endpoint c gives $C_1 \ge c$, since the center exits before M_1.

Put

$$x = \frac{a}{1 - t}, \qquad \theta = \frac{t}{1 + z}.$$

Then $0 < \theta < 1/2$, because $z > t$, and

$$a + \varepsilon = 1 - t < 1, \qquad c = x + \theta + \frac{\alpha_T}{1 - t} > x + \theta, \qquad \frac{a}{a + \varepsilon} = x. \tag{79}$$

The midpoint bounds imply

$$\frac{1 - 4\theta + \theta^2}{2(1 - 2\theta)} < x < \frac{1}{2} - \theta,$$

using $t = \theta(2 - \theta)/(1 - \theta^2)$ for the lower endpoint. For $0 < \theta \le 1/5$, this interval lies to the right of the vertex of $Q_\theta(x) = 2x^2 + (\theta - 1)x + \theta$, and

$$Q_\theta(x) \ge \frac{\theta(1 - 5\theta + 11\theta^2 - \theta^3)}{2(1 - 2\theta)^2} \ge 0.$$

For $1/5 \le \theta < 1/2$, its unrestricted minimum is $(10\theta - 1 - \theta^2)/8 > 0$. Consequently

$$x^2 + (c - 2)x + c = Q_\theta(x) + \frac{\alpha_T}{1 - t}(x + 1) > 0.$$

Here $0 < x, c < 1/2$, so lemma 5.13 gives $x \le 1 - M_c^{\sup}$. Together with (79) and the forced endpoint, these are the common D hypotheses. Setting $\alpha_T = 0$ would give only a lower bound on the actual near endpoint; that translation is not made in this argument. □

Lemma D.13 (Solved Vd1 supported-tail adapter)**.** *In the center-based Vd1 supplier placement, with $M_1 \in U_0$ and T_1 uniquely supercritical, the actual corner data imply the size and ratio inequalities required by lemma D.11.*

Proof. Set $a = A_0$, $b = B_0$. By lemma B.2, the original Vd1 corner parameter satisfies $t \ge 1$. Let $d = \sqrt{t^2 + t + 1}$ and write the actual supported interval as $[c, u]$. Open midpoint containment gives $0 < c < 1/2 < u < 1$. Then

$$c = \frac{t(1 - b)}{t + 1}, \qquad u = \frac{d - a - tb - 1}{t}.$$

The midpoint condition gives

$$a + tb \le d - 1 - \frac{t}{2}.$$

Eliminating $tb = t - c(t + 1)$ gives

$$a \le F(t, c) := d - 1 - \frac{3t}{2} + c(t + 1).$$

For fixed $c \le 1/2$, the function F decreases with t, so

$$a \le F(1,c) = \sqrt{3} - \frac{5}{2} + 2c =: L(c).$$

The inequality $2L(c) \le 1 - M_c^{\sup}$ reduces to

$$20c^2 + (18\sqrt{3} - 52)c + 47 - 24\sqrt{3} \ge 0,$$

whose derivative is negative on $[0, 1/2]$ and whose value at $1/2$ is $26 - 15\sqrt{3} > 0$. The unsquared right side $12 - 4\sqrt{3} - 9c$ is positive there, so this squaring preserves the inequality. Hence

$$a \le 1 - M_c^{\sup}.$$

Put $\varepsilon = 1 - u$. Direct substitution gives

$$\varepsilon = \varepsilon_0 + \frac{a}{t}, \qquad \varepsilon_0 = \frac{2t + 1 - c(t+1) - d}{t} > 0.$$

The map

$$z \longmapsto \frac{z}{z + \varepsilon_0 + z/t}$$

is increasing, while $\varepsilon_0 + F(t,c)/t = 1/2$. Therefore

$$\frac{a}{a + \varepsilon} \le \frac{F(t,c)}{F(t,c) + 1/2} \le 2L(c) \le 1 - M_c^{\sup}.$$

Positive forward support also gives

$$(t+1)a + tb < (t+1)(d-1) - t^2 < d - 1,$$

because $d < t+1$. Thus $u > a$ and $a + \varepsilon = 1 + a - u < 1$. As in the T3-like placement, u is the total frontier and the near endpoint is supplied by T_1, so $\varepsilon V_1 \in U_C$ and $C_1 \ge c$. The size and ratio requirements of the common rescuer lemma have now all been verified. □

D.8. **Vd1 two-chart replacement.** The replacement below uses two different vertex charts. It preserves the five skeleton pieces that can change, then invokes the all-nonsupercritical obstruction without assuming preservation of the input gap rank.

Lemma D.14 (Half-square admissibility). *If $(x, y, z) \in \mathcal{A}$ satisfies*

$$x \ge \frac{1}{2}, \qquad 0 \le z \le \frac{1}{2},$$

then

$$y \le 1 - z.$$

Proof. Suppose $y > 1 - z$. Then $x + y > 1$, so the selected supercritical cell of proposition A.11 applies. In the ordered half $x \le y$ its necessary polynomial is

$$F(x,y,z) = (x^2 - 1)z^2 + (2xy^2 + y)z + y^4 - y^2 \le 0.$$

It is nondecreasing in x, its derivative in y is positive for $y \ge 1 - z$, and

$$F\left(\frac{1}{2}, 1 - z, z\right) = \frac{z^2(2z-5)(2z-1)}{4} \ge 0.$$

This is a contradiction. In the reflected half $y < x$, apply the same argument to $F(y, x, z)$. At the joining corner,

$$F(1 - z, 1 - z, z) = z(1 - 2z) \ge 0,$$

so no selected feasible point is lost at equality. □

Lemma D.15 (Two-vertex replacement from scalar margins)**.** *Consider two adjacent V roles. Only the five pieces*

$$e_{5,0}, \quad e_{0,1}, \quad e_{1,2}, \quad r_0, \quad r_1$$

may be affected by their replacement, and the shared edge is center-free. Let $a, c, B, r \in [0, 1)$ *be the boundary reach on* $e_{5,0}$*, the own-radial reach on* r_0*, the boundary reach on* $e_{1,2}$*, and the radial demand on* r_1 *needed to overlap the unchanged center interval. If*

$$a < \tfrac{1}{2}, \qquad a + c < 1, \qquad a + B < 1, \qquad r < \max\{1 - a, 1 - B\},$$

there are two open unit equilateral replacements preserving these five skeleton pieces, with actual boundary sums below one and no positive adjacent support. No center-class formula is assumed.

Proof. Choose first $p_2 \in (a, 1 - B)$ with $\max\{p_2, 1 - p_2\} > r$. This is possible because the supremum on that interval is $\max\{1-a, 1-B\}$. Next choose $a < p_1 < \min\{p_2, 1/2, 1 - c\}$ and

$$0 < \varepsilon < \min\{p_1 - a, p_2 - p_1, 1 - B - p_2, 1 - p_1 - c, \max(p_2, 1 - p_2) - r\}.$$

All five margins are positive. Use the separate vertex charts

$$X_0(x, y) = V_0 + x(V_5 - V_0) + y(V_1 - V_0), \qquad X_1(x, y) = V_1 + x(V_0 - V_1) + y(V_2 - V_1).$$

Both have metric $x^2 + y^2 - xy$. Define

$$\Delta_p^- = \mathrm{conv}\{(0, 1-p), (1, 1-p), (0, -p)\}, \quad D_{p,\varepsilon}^- = \mathrm{int}(\Delta_p^-) + (-\varepsilon, 0) \quad (p \le 1/2),$$

$$\Delta_p^+ = \mathrm{conv}\{(p, 0), (p, 1), (p-1, 0)\}, \quad D_{p,\varepsilon}^+ = \mathrm{int}(\Delta_p^+) + (0, -\varepsilon) \quad (p > 1/2).$$

Their edges have unit length. The minus half-planes are $x > -\varepsilon$, $y < 1 - p$, $y > x + \varepsilon - p$; the plus half-planes are $y > -\varepsilon$, $x < p$, $x > y + \varepsilon + p - 1$. Substitution on the axes and diagonal gives their actual reaches:

	A	B	C
$D_{p,\varepsilon}^-$	$p - \varepsilon$	$1 - p$	$1 - p$
$D_{p,\varepsilon}^+$	p	$1 - p - \varepsilon$	p

Both contain the chart origin. On the adjacent support lines $(z, 1)$ and $(1, z)$ the half-planes are incompatible for $0 \le z \le 1$, so neither has positive adjacent support. Set $U_0' = X_0(D_{p_1,\varepsilon}^-)$ and use $U_1' = X_1(D_{p_2,\varepsilon}^-)$ for $p_2 \le 1/2$, or $U_1' = X_1(D_{p_2,\varepsilon}^+)$ otherwise. The two charts and this split cannot be identified. Both actual boundary sums are $1 - \varepsilon$.

The first replacement reaches $p_1 - \varepsilon > a$ on $e_{5,0}$ and $1 - p_1 > c$ on r_0. On the shared edge the reaches sum to at least

$$(1 - p_1) + (p_2 - \varepsilon) > 1,$$

so the open traces overlap. The second reaches more than B on $e_{1,2}$ and $\max(p_2, 1 - p_2) > r$ on r_1. Thus all five pieces remain covered; all other pieces are unchanged. □

Proposition D.16 (Vd1 input to the two-vertex replacement)**.** *Assume a skeleton cover has a unique Vd1 role T_0 supplying M_1, a unique supercritical role T_1, and all other V roles are nonsupercritical Vd0. Suppose $\mathrm{e}_{0,1}$ is center-free and no other V role has positive trace on r_0 or r_1. The scalar replacement applies in CE1 and CE2 and yields a skeleton cover with $N'_+ = 0$.*

Proof. Let A_0, B_0, C_0 be the actual incoming, outgoing, and own-radial reaches of the Vd1 role. Its corner normal form and lemma B.2 supply $\vartheta \geq 1$ and

$$\omega := \sqrt{\vartheta^2 + \vartheta + 1}$$

such that

$$C_0 = \frac{\omega - A_0 - \vartheta B_0}{\vartheta + 1}, \qquad \lambda = \frac{\vartheta(1 - B_0)}{\vartheta + 1},$$

$$u_{\mathrm{adj}} = \frac{\omega - A_0 - \vartheta B_0 - 1}{\vartheta}.$$

The strict midpoint and one-sided-support conditions give

$$A_0 + C_0 < 1, \qquad A_0 < \lambda \leq \frac{1}{2},$$

$$B_0 < \frac{1}{2}, \qquad u_{\mathrm{adj}} < 1 - A_0.$$

For the first inequality, evaluation on the closed midpoint face reduces strictness to

$$(2\vartheta^2 + 4\vartheta + 1)^2 - 4(\vartheta + 1)^2(\vartheta^2 + \vartheta + 1) = 4\vartheta^3 + 4\vartheta^2 - 4\vartheta - 3 > 0.$$

The remaining inequalities follow directly from the displayed endpoint formulas and the strict midpoint inequalities.

The center-free shared edge and the supported interval give

$$A_1 \geq 1 - B_0 > \frac{1}{2}, \qquad C_1 \geq \lambda$$

for the actual reaches of T_1. The selected supercritical component has $C_1 \leq 1/2$, so lemma D.14 yields

$$B_1 \leq 1 - C_1 \leq 1 - \lambda.$$

Since $A_0 < \lambda$,

$$A_0 + B_1 < 1.$$

Define the C-triangle reach on r_1, measured from O, by

$$d_1^C := \max\{x \in [0,1] : xV_1 \in T_C\}, \qquad c_1^{\mathrm{req}} := 1 - d_1^C.$$

Only U_C, U_1, and the adjacent trace of U_0 can contribute a positive interval to r_1. Open skeleton coverage gives

$$c_1^{\mathrm{req}} < \max\{C_1, u_{\mathrm{adj}}\} \leq \max\{1 - B_1, 1 - A_0\}.$$

Apply lemma D.15 with

$$(a, c, B, r) = (A_0, C_0, B_1, c_1^{\mathrm{req}}).$$

All four scalar conditions were proved above. The four untouched roles remain nonsupercritical Vd0, while the two new actual boundary sums are $1 - \varepsilon < 1$. The resulting skeleton cover contradicts N0. The original C triangle is fixed; its class and the input gap rank were not used in the construction. This operation does not claim to preserve coverage of the interior of H. $\square$

Appendix E. Zero-Gap Nine-Point Optimization

This appendix proves the frontier construction, witness exclusions, and four-contact enclosure bound for the zero-gap obstruction. It also introduces inner comparison points for the exact certificate, without changing the three frontier witnesses Q_-, Q_0, Q_+.

E.1. **Corner-clipped AB-unions.** In this subsection the corner is the local origin $O_{\rm loc}$, not the hexagon center O. The local anchors A, B used here are also distinct from the Newton comparison points introduced later. The corner chart

$$\Psi_4(u,v) = V_4 + u(V_5 - V_4) + v(V_3 - V_4)$$

has coordinate axes corresponding to the forward and backward edge directions. We identify a local vector $ue_A + ve_B$ with its pair (u,v) when applying this chart.

Let e_A, e_B be unit vectors with angle $120°$, put

$$O_{\rm loc} = 0, \qquad A = ae_A, \qquad B = be_B, \qquad \mathcal{C}_{\rm loc} = \{ue_A + ve_B : u, v \geq 0\},$$

and define

$$\mathcal{R}^{\rm loc}_{AB}(a,b) = \mathcal{C}_{\rm loc} \cap \bigcup \left\{ T : \begin{array}{l} T \text{ is a closed unit equilateral triangle,} \\ O_{\rm loc}, A, B \in T \end{array} \right\}. \tag{80}$$

By the wedge reduction, the global union from the body is $\Psi_4(\mathcal{R}^{\rm loc}_{AB}(b,a))$; its two anchor parameters are exchanged.

The strict supercritical branch admits a simpler closed form. In the next statement write $x = ue_A + ve_B$ and $\|x\|^2 = u^2 + v^2 - uv$.

Theorem E.1 (Strict supercritical AB-union)**.** *Assume $a, b > 0$ and*

$$a + b > 1, \qquad \rho = a^2 + ab + b^2 < 1,$$

put $h = \sqrt{3}/2$, $D = \sqrt{4\rho - 3}$, and define

$$\alpha = \frac{h(a + 2b - aD)}{2\rho}, \qquad \beta = \frac{h(a - b + (a+b)D)}{2\rho},$$

$$\gamma = \frac{h(-a + b + (a+b)D)}{2\rho}, \qquad \delta = \frac{h(2a + b - bD)}{2\rho}.$$

Then all four coefficients are positive and

$$\mathcal{R}^{\rm loc}_{AB}(a,b) = (\mathcal{C}_{\rm loc} \cap \mathcal{D}_A \cap \mathcal{H}_A) \cup (\mathcal{C}_{\rm loc} \cap \mathcal{D}_B \cap \mathcal{H}_B), \tag{81}$$

where

$$\mathcal{D}_A = \{(u,v) : (u-a)^2 + v^2 - (u-a)v \leq 1\},$$
$$\mathcal{D}_B = \{(u,v) : u^2 + (v-b)^2 - u(v-b) \leq 1\},$$
$$\mathcal{H}_A = \{(u,v) : \alpha(u-a) + \beta v \leq 0\},$$
$$\mathcal{H}_B = \{(u,v) : \gamma u + \delta(v-b) \leq 0\}.$$

Its non-axis frontier consists, in counterclockwise order, of the A-circle, the A-line, the B-line, and the B-circle.

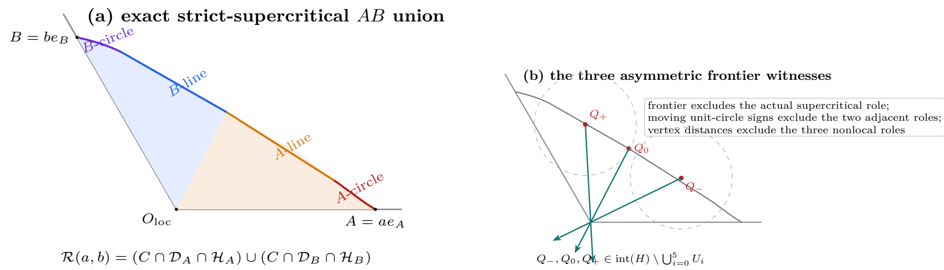


FIGURE 9. The strict-supercritical AB frontier and the three asymmetric witnesses. The source-conditioned union has, in counterclockwise order, an A-circle arc, an A-line segment, a B-line segment, and a B-circle arc. The points Q_-, Q_0, Q_+ are selected on this exact frontier and are then excluded from the six actual V roles by frontier, moving-circle, and vertex-distance arguments. The drawing is schematic; the labelled frontier order and witness roles are exact.

Proof. The identities

$$4\rho - 3(a+b)^2 = (a-b)^2, \qquad (a+b)^2 D^2 - (a-b)^2 = 4\rho((a+b)^2 - 1)$$

give $0 < D < 1$ and positivity of the four coefficients.

We apply the caliper criterion of Theorem A.9. For a point $x = (u, v)$ in the relative interior of $\mathcal{C}_{\mathrm{loc}}$ but outside $\triangle O_{\mathrm{loc}} AB$, the four hull vertices occur in the order $O_{\mathrm{loc}}, B, x, A$. The two cone-axis hull edges cannot certify a unit triangle: their support sums, divided by h, are respectively

$$\max\{a, u\} + \max\{b, v-u\}, \qquad \max\{b, v\} + \max\{a, u-v\},$$

and are at least $a + b > 1$. It remains to test the Ax and Bx calipers.

For the Ax caliper put $p = u - a$ and $r^2 = p^2 + v^2 - pv$. Exact evaluation of the three supports gives

$$\frac{G_A}{h} = \frac{\max\{r^2, -ap + (a+b)v\} + N_A}{r}, \qquad N_A = \begin{cases} 0, & p \le 0, \\ ap, & 0 \le p \le v, \\ (a+b)p - bv, & p \ge v. \end{cases} \tag{82}$$

No support label is omitted in these three sectors. If $p \le 0$, $G_A \le h$ is equivalent to

$$r \le 1, \qquad -ap + (a+b)v \le r.$$

The second inequality is the half-plane condition in (81), because

$$r^2 - (-ap + (a+b)v)^2 = \frac{(cp + dv)(c'p + d'v)}{4\rho},$$

where

$$c = a + 2b - aD, \quad d = a - b + (a+b)D, \quad c' = a + 2b + aD, \quad d' = a - b - (a+b)D,$$

and $c, c', d > 0 > d'$. Thus the low sector is exactly $\mathcal{D}_A \cap \mathcal{H}_A$. In the middle sector $0 < p \le v$ one has $r \le v$, whereas the numerator in (82) is at least $(a+b)v > r$, so no fit is possible.

In the high sector $p \geq v$, the inequalities $G_A \leq h$ imply

$$r^2 + (a+b)p - bv \leq r, \qquad bp + av \leq r. \tag{83}$$

Write $x - A = ry(\theta)$, $0 \leq \theta \leq 60°$, with cone coordinates

$$y(\theta) = te_A + we_B, \quad t = \frac{2}{\sqrt{3}}\cos(\theta - 30°), \quad w = \frac{2}{\sqrt{3}}\sin\theta.$$

Then (83) gives

$$r \leq f(\theta) = 1 - (a+b)t + bw, \qquad S(\theta) = bt + aw \leq 1.$$

Let n_B be the unit vector with dual coordinates (γ, δ). The identities

$$b\gamma + (a+b)\delta = h, \quad (a+b)\gamma + a\delta = \frac{h}{2}(1+D), \quad b\delta - a\gamma = \frac{h}{2}(1-D)$$

show that the unique angle θ_B of this normal satisfies $S(\theta_B) = 1$ and $f(\theta_B) = (1-D)/2$. The function S has at most one critical point, a maximum, so $S(\theta) \leq 1$ implies $\theta \leq \theta_B$. On the feasible part $f, f' \geq 0$; hence $f(\theta)\cos(\theta_B - \theta)$ is increasing. Consequently

$$\langle n_B, x - B\rangle \leq a\gamma - b\delta + \frac{h}{2}(1-D) = 0.$$

The diameter bound also gives $\|x - B\| \leq 1$. Thus every high sector fit belongs to $\mathcal{D}_B \cap \mathcal{H}_B$. Reflection exchanges a, u with b, v and supplies the corresponding result for the Bx caliper. This proves (81) in the cone interior; the axes follow by taking limits, since both sides are closed.

For completeness, the line–circle junctions are obtained by solving the two displayed line and circle equations. The line normals satisfy

$$\alpha^2 + \alpha\beta + \beta^2 = h^2, \qquad \gamma^2 + \gamma\delta + \delta^2 = h^2,$$

and the line determinant is

$$\alpha\delta - \gamma\beta = \frac{3(1-D)(D+3)}{16\rho} > 0.$$

The successive cone slopes are

$$+\infty, \quad \frac{2}{1-D}, \quad \frac{a(1-D)+2b}{2a+b(1-D)}, \quad \frac{1-D}{2}, \quad 0;$$

the two middle comparisons reduce to $a(4-(1-D)^2) > 0$ and $b(4-(1-D)^2) > 0$. This proves the asserted four-piece order. ☐

E.2. **Exact-one handoffs and common reach bounds.**

Lemma E.2 (Exact-one handoff chain)**.** *Assume the six V triangles cover ∂H and exactly one actual V triangle is supercritical. Rotate its index to 4. The strict handoffs supplied by Proposition 2.12 satisfy*

$$\xi_3 < \xi_2 < \xi_1 < \xi_0 < \xi_5 < \xi_4.$$

With

$$a = 1 - \xi_3, \qquad b = \xi_4, \qquad p = 1 - b, \qquad q = 1 - a,$$

one has

$$0 < a, b < 1, \qquad a + b > 1, \qquad a^2 + ab + b^2 < 1, \tag{84}$$

and every selected V triangle obeys

$$a_i \geq p, \qquad b_i \geq q.$$

Proof. Every selected V triangle other than 4 is strictly nonsupercritical, so $\xi_i < \xi_{i-1}$ for $i \neq 4$, which is precisely (E.2). In particular ξ_3 is the global minimum and ξ_4 the global maximum. The two anchors of V triangle 4 lie in its open triangle; their mutual squared distance is $a^2 + ab + b^2$, strictly below the diameter squared of the closed unit triangle. This proves (84). Finally every $\xi_j \in [\xi_3, \xi_4]$, so $1 - \xi_{i-1} \geq 1 - \xi_4 = p$ and $\xi_i \geq \xi_3 = q$. □

E.3. **Six directly forced radial points.** Put $h = \sqrt{3}/2$ and

$$s = p + q, \qquad m = \min(p, q), \qquad M = \max(p, q), \qquad \chi = s^4 - s^2 + pq.$$

The strict domain (84) gives

$$pq > (1 - s)(2 - s), \qquad m > 1 - s, \qquad s > 1 - m. \tag{85}$$

Let $c_* = c_{\max}(p, q)$ be the exact radial envelope of Proposition A.11. On the present subcritical domain, equation (37) becomes

$$c_* = \begin{cases} C_L(m), & \chi \leq 0, \\ \dfrac{2M}{1 + \sqrt{4s^2 - 3}}, & \chi > 0, \end{cases} \tag{86}$$

where $C_L(m)$ is the unique root in $[\sqrt{3}/2, 1]$ of

$$z^4 - z^2 + mz - m^2 = 0.$$

At $\chi = 0$ the two expressions agree and equal s.

Lemma E.3 (Direct radial forcing)**.** *One has*

$$0 < c_* < 1, \qquad c_* > 1 - m. \tag{87}$$

Assume that $U_C, U_0, \ldots, U_5$ cover H, with no restriction on the normalized V types. Then, for every i,

$$D_i = (1 - c_*)V_i \in U_C.$$

Consequently U_C contains the centered closed disk

$$\mathcal{D}_\eta = \{x : \|x\| \leq \eta\}, \qquad \eta = h(1 - c_*). \tag{88}$$

Proof. On the C_L branch, $C_L(m) < 1$ because the defining polynomial is positive at 1; if $s < \sqrt{3}/2$, then $c_* \geq \sqrt{3}/2 > s > 1 - m$, while if $s \geq \sqrt{3}/2$ its value at s is $\chi \leq 0$, so $c_* \geq s > 1 - m$. On the other branch put $E = \sqrt{4s^2 - 3}$. The selector gives $sE > M - m$, whence $c_* < s < 1$. Moreover, with

$$A_* = \frac{2M}{1 - m} - 1,$$

one has

$$A_*^2 - E^2 = \frac{4(1 - s)\{(1 - m)(1 - 2m) + m(2 - m)(1 - s)\}}{(1 - m)^2} > 0.$$

Thus $A_* > E$ and $c_* > 1 - m$. This proves (87).

Every selected pair dominates (p, q) by lemma E.2. Apply corollary 5.5 to obtain the six forced points and their inscribed disk. □

We next define the three asymmetric witnesses. At V_4 use local coordinates

$$P = \Psi_4(u,v), \qquad \mathsf{q}(u,v) = u^2 + v^2 - uv.$$

The forward-reach lower bound is b and the backward-reach lower bound is a. Accordingly, set

$$\alpha = \frac{h(b+2a-bD)}{2\rho}, \qquad \beta = \frac{h(b-a+(a+b)D)}{2\rho},$$
$$\gamma = \frac{h(-b+a+(a+b)D)}{2\rho}, \qquad \delta = \frac{h(2b+a-aD)}{2\rho},$$

where now $\rho = a^2 + ab + b^2$ and $D = \sqrt{4\rho - 3}$. The frontier coefficients here are not the signed-center slacks. Applying Theorem E.1 with the exchanged pair (b,a), the two local line pieces are

$$L_-(\lambda) = (b - \beta\lambda, \alpha\lambda), \qquad L_+(\mu) = (\delta\mu, a - \gamma\mu).$$

They meet at $\lambda_* = \mu_* = 8h\rho/(3(D+3))$ in

$$J = \left(\frac{2b+a-aD}{D+3}, \frac{b+2a-bD}{D+3}\right). \tag{89}$$

Put

$$E_- = (1-b, 2-b), \qquad E_+ = (2-a, 1-a).$$

The actual adjacent handoffs have local centers

$$Z_5(t) = (1+t, t), \qquad 1-a \le t \le b, \quad \text{and} \quad Z_3(y) = (y, 1+y), \qquad 1-b \le y \le a.$$

The restrictions of the two unit-circle equations are

$$g_-(\lambda) = \frac{3}{4}\lambda^2 - 3(\alpha + b\beta)\lambda + 3b^2 - 3b + 2,$$
$$g_+(\mu) = \frac{3}{4}\mu^2 - 3(\delta + a\gamma)\mu + 3a^2 - 3a + 2. \tag{90}$$

Let λ_- and μ_+ be their first roots; explicitly,

$$\lambda_- = 2(\alpha + b\beta) - \frac{2}{3}\sqrt{9(\alpha + b\beta)^2 - 9b^2 + 9b - 6},$$
$$\mu_+ = 2(\delta + a\gamma) - \frac{2}{3}\sqrt{9(\delta + a\gamma)^2 - 9a^2 + 9a - 6}.$$

Lemma E.4 (Moving adjacent-triangle signs)**.** *Throughout* (84),

$$\lambda_* < \lambda_- < h^{-1}, \qquad \mu_* < \mu_+ < h^{-1}. \tag{91}$$

Moreover $J_u, J_v < 1/2$. The point $L_+(\mu_+)$ satisfies

$$(L_+(\mu_+))_u + (L_+(\mu_+))_v < 3 - 2a,$$

and the reflected bound holds for $L_-(\lambda_-)$. Writing $F(P,t) = \mathsf{q}(P - Z_5(t)) - 1$, one has, for every $t \in [1-a, b]$,

$$F(L_+(\mu_+), t) \ge 0, \qquad F(J,t) > 0, \qquad F(L_-(\lambda_-), t) > 0. \tag{92}$$

Under the reflection $(u,v,a,b) \leftrightarrow (v,u,b,a)$, the analogous signs hold for every $y \in [1-b, a]$: the first-root point $L_-(\lambda_-)$ has nonnegative sign against $Z_3(y)$, and the other two witnesses have strictly positive sign.

Proof. Put

$$U = a + b - 1, \qquad z = a - b, \qquad D^2 = z^2 + 6U + 3U^2.$$

Then $0 < U < 1/6$ and $z^2 < \Omega := 1 - 6U - 3U^2$. We give the fixed-line calculation for the circle centered at $E_+ = (2-a, 1-a)$; reflection gives the calculation for E_-. Put

$$F(P,t) = \mathsf{q}(P - (1+t,t)) - 1, \qquad t_0 = 1 - a,$$

so this circle is $F(P, t_0) = 0$. At the junction, exact expansion gives

$$(D+3)^2 F(J, t_0) = 3D(z^2 + Uz + 1 - 3U) + P_J,$$

where

$$P_J = z^4 + 5z^2 + 6Uz^2 + 3U^2z^2 + 9Uz + 3U + 15U^2 > 0;$$

indeed $z^2 + Uz + 1 - 3U \geq 1 - 3U - U^2/4 > 0$, and $5z^2 + 9Uz + 15U^2$ is positive definite. Thus $F(J, t_0) > 0$. Also

$$J_u + J_v = \frac{(1+U)(3-D)}{3+D} < 1.$$

Indeed this inequality is $3U < D(2+U)$, which follows from $D^2 \geq 3U(2+U)$ and $(2+U)^3 > 3U$. Since

$$\frac{\partial F}{\partial t}(P,t) = 1 + 2t - P_u - P_v,$$

$F(J,t)$ is strictly increasing for $t \geq 0$, and therefore is positive throughout $[1-a, b]$.

On the opposite line L_-, the derivative at the junction, after multiplication by a positive factor, has at $t = 1 - a$ the numerator

$$N_0 = D(9 + 3z^2 + 6Uz - 9U^2) + P_0,$$

where

$$\begin{aligned} P_0 = {} & 18U^3 + 6U^2z^2 + 63U^2 + 18Uz^2 + 18Uz + 54U \\ & + 2z^4 + 9z^2 + 9. \end{aligned}$$

The coefficient of D is at least $9 - 6U - 9U^2 > 0$, while $P_0 \geq 9 + 54U - 18U > 0$. Hence $N_0 > 0$. At the other endpoint $t = b$, the corresponding numerator is

$$N_1 = D(9 + 3z^2 + 6U - 3U^2) + P_1,$$

where

$$\begin{aligned} P_1 = {} & 9U^4 - 3U^3z + 45U^3 + 9U^2z^2 - 6U^2z + 72U^2 \\ & - Uz^3 + 21Uz^2 + 9Uz + 45U + 2z^4 + 9z^2 + 9. \end{aligned}$$

Its D-coefficient is positive and, using $|z| < 1$,

$$P_1 \geq 9 + 45U - 9U - U - 6U^2 - 3U^3 > 0.$$

The junction derivative is affine in t, so it is positive throughout $[1-a, b]$. Since the restriction of F to L_- is a convex quadratic, is positive at the junction, and is increasing there, it has no intersection on the opposite line side for any actual adjacent handoff.

On the correct line L_+, the value at its outer endpoint h^{-1} has, after multiplication by a positive factor, numerator

$$A_* = P_2 - 4DU(1 + U + z),$$

where

$$P_2 = 3U^4 + 6U^3 z + 6U^3 + 4U^2 z^2 + 12U^2 z + 12U^2 \\ + 2Uz^3 + 6Uz^2 + 2Uz + 6U + z^4 + 2z^2 - 3.$$

For fixed U, replace the odd powers of z by $x = |z|$. The resulting polynomial $P_2^+(U, x)$ satisfies

$$\partial_x P_2^+ = 2(3U^3 + 4U^2 x + 6U^2 + 3Ux^2 + 6Ux + U + 2x^3 + 2x) > 0,$$

and therefore its supremum for $z^2 < \Omega$ occurs at $x = \sqrt{\Omega}$. There

$$P_2^+(U, \sqrt{\Omega}) = 4U(U + \sqrt{\Omega} - 3) < 0.$$

As $1 + U + z = 2a > 0$, this gives $A_* < 0$. The positive junction value and negative endpoint value force the first root strictly between them, proving the L_+ half of (91); reflection proves the other half.

For the coordinate bounds, the exact identities

$$D^2 - (3U - z)^2 = 6U(1 + z - U), \qquad D^2 - (3U + z)^2 = 6U(1 - z - U)$$

give $J_u, J_v < 1/2$. It remains to control the coordinate sum along L_+. For $E = L_+(h^{-1})$, the sign of $3 - 2a - E_u - E_v$ is the sign of

$$D(6U + 2z + 6) + B(U, z),$$

where

$$B(U, z) = -9U^3 - 9U^2 z - 9U^2 - 3Uz^2 - 18Uz + 3U \\ - 3z^3 + 3z^2 - 3z + 3.$$

Here

$$B_z = -3(3z^2 + 2Uz + 6U - 2z + 3U^2 + 1) < 0,$$

because the quadratic in parentheses has discriminant $-32U^2 - 80U - 8 < 0$. Hence its minimum on $z^2 < \Omega$ is at $z = \sqrt{\Omega}$, where $B(U, \sqrt{\Omega}) = 6(1 - 3U - \sqrt{\Omega}) > 0$ because $(1 - 3U)^2 - \Omega = 12U^2$. Thus $E_u + E_v < 3 - 2a$. Also $J_u + J_v < 1 < 3 - 2a$, and the coordinate sum is affine on the line segment, so the same strict bound holds at the first root. Now

$$\frac{\partial F}{\partial t}(P, t) = 1 + 2t - P_u - P_v.$$

Since $F(L_+(\mu_+), 1-a) = 0$, the coordinate-sum bound makes its t-derivative strictly positive for $t \geq 1 - a$. Together with the no-opposite-line result, this proves (92). Reflection gives both the bound on L_- and the complete $Z_3(y)$ statement. $\square$

Define the local and global witnesses by

$$Q_-^{\text{loc}} = L_-(\lambda_-), \qquad Q_0^{\text{loc}} = J, \qquad Q_+^{\text{loc}} = L_+(\mu_+), \tag{93}$$

and put $Q_j = \Psi_4(Q_j^{\text{loc}})$ for $j \in \{-, 0, +\}$.

Lemma E.5 (Direct asymmetric forcing)**.** *Assume that $U_C, U_0, \ldots, U_5$ cover H. Then*

$$Q_-, Q_0, Q_+ \in \text{int}(H) \setminus \bigcup_{i=0}^{5} U_i,$$

and consequently all three witnesses belong to U_C.

Proof. The root order and positivity of the line coordinates give $0 < u, v < 1$ for all three points. Each point has local coordinates in $\mathcal{R}^{\mathrm{loc}}_{AB}(b, a)$ and hence lies in a closed unit triangle containing V_4, so $\mathsf{q}(u, v) \le 1$; hence $|u - v| < 1$ and all three points lie in $\mathrm{int}(H)$.

In local coordinates the closure of U_4 is represented in $\mathcal{R}^{\mathrm{loc}}_{AB}(b, a)$, since U_4 contains the actual handoffs $X_3(\xi_3), X_4(\xi_4)$ and its distinguished vertex V_4. All three witnesses lie on its exact non-axis frontier. If one belonged to the open triangle U_4, a sufficiently small plane ball about it would lie both in U_4 and in the interior of the corner cone, making the witness an interior point of $\mathcal{R}^{\mathrm{loc}}_{AB}(b, a)$. This contradiction excludes U_4.

For the triangles U_0, U_1, U_2, their contained vertices have local coordinates

i	0	1	2
V_i	$(2, 1)$	$(2, 2)$	$(1, 2)$.

For $P = (u, v)$,

$$\|P - (2, 1)\|^2 - 1 = \mathsf{q}(u, v) + 2 - 3u, \quad \|P - (1, 2)\|^2 - 1 = \mathsf{q}(u, v) + 2 - 3v.$$

The inequalities $J_u, J_v < 1/2$ and the line order give distance greater than one from V_0 for Q_-, Q_0 and from V_2 for Q_0, Q_+. For the remaining pair, the first-root circle and the sum bound in Lemma E.4 give

$$\|Q_+ - V_0\|^2 - 1 = a(3 - a - (Q_+)_u - (Q_+)_v) > a^2,$$

and, by reflection, $\|Q_- - V_2\|^2 - 1 > b^2$. All three points are farther than one from $V_1 = (2, 2)$, because writing $u = 1 - \xi$, $v = 1 - \eta$ gives

$$\|P - V_1\|^2 = 1 + \xi + \eta + \xi^2 + \eta^2 - \xi\eta > 1.$$

Since two points in one open unit equilateral triangle have distance strictly less than one, these inequalities exclude U_0, U_1, U_2.

The actual handoff $X_5(\xi_5) \in U_5$ has local coordinates $Z_5(\xi_5)$, and $1 - a < \xi_5 < b$ by (E.2). The three signs in (92) therefore put every witness at distance at least one from $X_5(\xi_5)$, excluding U_5. Similarly, if $y = 1 - \xi_2$, then $1 - b < y < a$ and $X_2(\xi_2) \in U_3$ has local coordinates $Z_3(y)$; the reflected signs exclude U_3. Thus no V triangle contains any witness. The assumed cover now forces all three interior witnesses into U_C. □

Together with lemma E.3, this proves the containment $K_{\mathrm{wit}}(a, b) \subset U_C$ used in the body.

E.4. **Newton inner reduction.** For the nontrivial regime $c_* > 2/3$, we first remove the square radicals in the first-root witnesses. Normalize $\bar{\alpha} = \alpha/h$, and similarly for the other three coefficients, and put

$$\xi = \lambda/h, \quad \zeta = \mu/h, \quad \xi_* = \zeta_* = \frac{8\rho}{3(D + 3)}.$$

Define $\widetilde{g}_-(x) = g_-(hx)$ and $\widetilde{g}_+(x) = g_+(hx)$ from (90). Take one Newton step at the junction:

$$\widehat{\xi}_- = \xi_* - \frac{\widetilde{g}_-(\xi_*)}{\widetilde{g}'_-(\xi_*)}, \qquad \widehat{\zeta}_+ = \zeta_* - \frac{\widetilde{g}_+(\zeta_*)}{\widetilde{g}'_+(\zeta_*)}.$$

Define the global Newton vectors by

$$\begin{aligned} A &= \Psi_4\left(b - \frac{3}{4}\bar{\beta}\widehat{\xi}_-, \frac{3}{4}\bar{\alpha}\widehat{\xi}_-\right), \\ B &= \Psi_4(J), \\ C &= \Psi_4\left(\frac{3}{4}\bar{\delta}\widehat{\zeta}_+, a - \frac{3}{4}\bar{\gamma}\widehat{\zeta}_+\right), \end{aligned}$$

All norms and cross products below are taken in these global vectors.

Lemma E.6 (Newton inner reduction)**.** *Their local coordinates are rational in a, b, D, and their global coordinates are rational over $\mathbb{Q}(\sqrt{3})$ in those variables. They satisfy*

$$A \in (Q_0, Q_-), \qquad C \in (Q_0, Q_+).$$

Consequently

$$\widehat{K} = \operatorname{conv}(\mathcal{D}_\eta \cup \{A, B, C\}),$$

satisfies

$$\widehat{K} \subseteq \operatorname{conv}(K_{\mathrm{wit}}(a,b)), \qquad \Lambda(K_{\mathrm{wit}}(a,b)) \geq \Lambda(\widehat{K}). \tag{94}$$

Proof. Each rescaled quadratic $\widetilde{g}_-, \widetilde{g}_+$ is strictly convex, positive, and decreasing at the junction. Its tangent zero therefore lies strictly between the junction and its first root. This gives the two open-segment inclusions. The normalized coefficients and Newton quotients are rational in a, b, D; applying Ψ_4 gives the stated global field. Convexity proves the containment. $\square$

E.5. **Ordered convex chain and supporting-line bounds.**

Proposition E.7 (Four-contact witness geometry)**.** *Assume* (84) *and $c_* > 2/3$. The rays A, B, C occur in that order in one open $120°$ cone, and*

$$\operatorname{cross}(B - A, C - B) > 0, \qquad \|A\|, \|C\| > \eta.$$

Moreover the outward distances of the lines AB, BC satisfy

$$d_{AB} := \frac{\operatorname{cross}(A,B)}{\|B - A\|} \geq h - 2\eta, \qquad d_{BC} := \frac{\operatorname{cross}(B,C)}{\|C - B\|} \geq h - 2\eta. \tag{95}$$

Proof. The local coordinates of all three Newton points lie in $(0,1)^2$. Centering at O gives coefficient vectors $(u-1, v-1)$ in the two incident edge directions, so their rays lie in one open $120°$ cone. The two line identities

$$d_{AB} = \alpha(1-b) + \beta > 0, \qquad d_{BC} = \gamma + \delta(1-a) > 0$$

give their order A, B, C. Put $s_A = h\widehat{\xi}_- - \lambda_* > 0$ and $s_C = h\widehat{\zeta}_+ - \mu_* > 0$. The frontier directions give

$$\operatorname{cross}(B - A, C - B) = h s_A s_C(\alpha\delta - \beta\gamma) > 0,$$

since $\alpha\delta - \beta\gamma = 3(1-D)(D+3)/(16\rho) > 0$. For $A = (-X, -Y)$ in the centered cone basis, $X > 1/2$, and

$$\|A\|^2 = (Y - X/2)^2 + 3X^2/4 > 3/16.$$

The reflected calculation gives

$$\|A\|, \|C\| > h/2 > \eta. \tag{96}$$

It remains to prove the two supporting-line bounds. Put $\tau = h - 2\eta = h(2c_* - 1)$. We now give the exact analytic verification. By reflection take $a \le b$ and set $m = 1 - b$, $M = 1 - a$, $U = a + b - 1$, $s = m + M$, and $w = M - m$. The smaller of the two normalized left sides in (95) is

$$d = \frac{\mathcal{A} + U(2m-1) + D(\mathcal{A}+U)}{2\rho}, \qquad \mathcal{A} = 1 - m + m^2. \tag{97}$$

On the C_L sheet, write $F_L(z) = z^4 - z^2 + mz - m^2$. This defining polynomial evaluated at $r = 1 - m/2 + m^2/2$ is

$$F_L(r) = \frac{m^2(m-1)}{16}(m^5 - 3m^4 + 11m^3 - 17m^2 + 28m - 12) > 0.$$

Indeed the polynomial in parentheses is increasing on $(0, 1/2)$ and has value $-33/32$ at $1/2$; its derivative is $28 - 34m + m^2(5m^2 - 12m + 33) > 0$. Since $F_L(h) = -(m - h/2)^2 \le 0$ and the selected root is unique, $2c_* - 1 \le \mathcal{A}$. Put

$$X_0 = \mathcal{A} + U, \qquad Y_0 = 2\rho\mathcal{A} - \mathcal{A} - U(2m-1).$$

Then

$$d - \mathcal{A} = \frac{D(\mathcal{A}+U) - Y_0}{2\rho},$$

and exact expansion gives

$$\begin{aligned} Y_0 = {} & 2(m^2 - m + 1)U^2 + (2m^3 - 2m + 3)U \\ & + (m^2 - m + 1)(2m^2 - 2m + 1) > 0. \end{aligned}$$

Furthermore

$$D^2 X_0^2 - Y_0^2 = 4m\rho\Phi(U, m),$$

where

$$\begin{aligned} \Phi(U, m) = {} & (1 - m)(m^2 - m + 2)U^2 \\ & + (2 - m^2)(m^2 - m + 1)U \\ & - m(1 - m)^2(m^2 - m + 1). \end{aligned}$$

In these variables

$$\chi = U^4 - 4U^3 + 5U^2 - (m + 2)U + m(1 - m).$$

Since $U^2(U^2 - 4U + 5) > 0$, the selector $\chi \le 0$ implies $U > m(1 - m)/(m + 2)$; Φ is increasing in U and at that lower endpoint is

$$\frac{m^2(1 - m)(m^4 - 2m^3 + 6m^2 - 6m + 4)}{(m + 2)^2} > 0.$$

The last quartic equals $(m^2 - m)^2 + 5m^2 - 6m + 4 > 0$. Thus $d > \mathcal{A} \ge 2c_* - 1$.

On the other radial sheet let $E = \sqrt{4s^2 - 3}$. Then $0 \le w < sE$ and $c_* = (s + w)/(1 + E)$. At fixed U, differentiation of (97) can be checked without an implicit estimate. Namely,

$$d = \frac{1 + D}{2} - UB(D, w), \qquad B(D, w) = \frac{(1 + U)(3 + D) + w(1 - D)}{D^2 + 3}.$$

Here $D^2 = w^2 + 6U + 3U^2$, so $0 \le D' = w/D \le 1$, and

$$B_w = \frac{1 - D}{D^2 + 3} \ge 0.$$

Writing $N = (1+U)(3+D) + w(1-D)$, direct differentiation gives

$$B_D = \frac{(1+U-w)(D^2+3) - 2DN}{(D^2+3)^2} > -\frac{34}{27}.$$

Indeed the first term in the numerator is positive, while $1+U-w = 2a > 0$, $N < (7/6)4+1 = 17/3$, $D < 1$, and $D^2+3 \geq 3$. If $B_D < 0$, then $B_D D' \geq B_D$; otherwise $B_D D' \geq 0$. Hence $B' = B_w + B_D D' > -34/27$, and $U < 1/6$ yields

$$d' < \frac{1}{2} + \frac{1}{6}\frac{34}{27} = \frac{115}{162} < 1,$$

whereas $(2c_* - 1)' = 2/(1+E) \geq 1$. Hence their difference decreases with w. At the boundary $w = sE$, where $\chi = 0$ and $c_* = s$, the preceding sheet applies: this is an admissible point because $h < s < 1$ and $D^2 = 4s^4 - 12s + 9 < 1$. Indeed $s^4 - 3s + 2 = (s-1)(s^3+s^2+s-2) < 0$ on this interval; the second factor is increasing and already equals $(7h-5)/4 > 0$ at $s = h$. The preceding sheet gives $d \geq 2s-1$. Therefore (95) holds on both sheets, and reflection supplies the other line.

This proves both supporting-line bounds. Since $0 < \eta < h/3$, their common lower bound $h - 2\eta$ is strictly greater than η, as required for the exposed-edge geometry used below. □

E.6. **Four supporting contacts and paired radius transfer.** Let R and J denote counterclockwise rotations through $2\pi/3$ and $\pi/2$, respectively. For the inner witness define

$$H(n) = \max\{\eta, \langle A, n\rangle, \langle B, n\rangle, \langle C, n\rangle\}, \qquad \Psi(n) = \sum_{j=0}^{2} H(\mathsf{R}^j n).$$

The support formula gives $\Lambda(\widehat{K}) = h^{-1} \min_{\|n\|=1} \Psi(n)$. The minimum can be located without introducing another convex body.

Lemma E.8 (Four-contact enclosure formula)**.** *For the Newton inner witness in the range $c_* > 2/3$, put*

$$n_{AB} = -\frac{\mathsf{J}(B-A)}{\|B-A\|}, \qquad n_{BC} = -\frac{\mathsf{J}(C-B)}{\|C-B\|},$$

$$t_A = \frac{\eta A - \sqrt{\|A\|^2 - \eta^2}\,\mathsf{J}A}{\|A\|^2}, \qquad t_C = \frac{\eta C + \sqrt{\|C\|^2 - \eta^2}\,\mathsf{J}C}{\|C\|^2}.$$

Then

$$\Lambda(\widehat{K}) = \frac{1}{h} \min\{\Psi(n_{AB}), \Psi(n_{BC}), \Psi(t_A), \Psi(t_C)\}. \tag{98}$$

Proof. By proposition E.7, the rays A, B, C are ordered in one open 120° cone, the chain turns strictly counterclockwise, and the supporting-line distances exceed η. The third point and the disk are strictly on the inward side of each of AB, BC, so both segments are exposed edges. Moreover

$$\operatorname{cross}(C-A, B-A) < 0, \qquad \operatorname{cross}(C-A, -A) > 0.$$

Thus B and the origin are on opposite sides of AC, which is not an exposed edge. The other two straight boundary pieces are the exposed disk tangents through A and C. There is no exposed disk tangent through B. The indicated signs select the tangent before A and the tangent after C in the counterclockwise boundary traversal.

The support-cell rotation lemma lemma A.8 places a minimum at these exposed ties or on a disk-only cell. In the latter case the same value occurs at a boundary tie, since $\|A\| > \eta$. Rotation through 120° leaves just the four displayed normals. The exposed-hull argument, rather than the general caliper theorem alone, is what discards the other point and disk ties. $\square$

At the first two contacts the disk supplies the other two supports, so

$$\Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h, \qquad \Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h. \tag{99}$$

For the two tangencies, put

$$\Delta = \mathrm{cross}(C, \mathsf{R}A) > 0, \quad \nu = \langle C, \mathsf{R}A\rangle, \quad N_X = \|X\|^2 \quad (X = A, C).$$

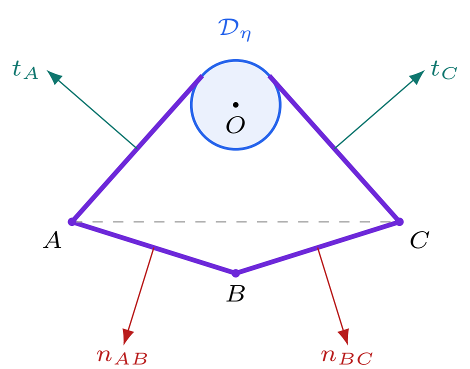
Choosing C as a support source at $\mathsf{R}t_A$ and A at $\mathsf{R}^2 t_C$, with the disk in the other directions, gives

$$\Psi(t_A) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_A - \eta^2}}{N_A}, \qquad \Psi(t_C) \geq 2\eta + \frac{\eta\nu + \Delta\sqrt{N_C - \eta^2}}{N_C}. \tag{100}$$

Consequently the only remaining targets are the paired residual signs

$$P_X(\eta) := (N_X - \eta^2)\Delta^2 - ((h - 2\eta)N_X - \eta\nu)^2 \geq 0 \quad (X = A, C). \tag{101}$$

(a) the four exposed contacts



(b) four support sums

Two consecutive-point contacts:

$\Psi(n_{AB}) \geq d_{AB} + 2\eta \geq h,$

$\Psi(n_{BC}) \geq d_{BC} + 2\eta \geq h.$

Two exposed disk tangencies:

$P_A(\eta), P_C(\eta) \geq 0$

$\implies \quad \Psi(t_A), \Psi(t_C) \geq h.$

Hence $\Lambda(\widehat{K}) \geq 1$.

FIGURE 10. The four-contact terminal for the zero-gap witness. The supporting lines through AB and BC and the two outer disk tangencies exhaust the active changes of support source. The dashed segment AC is not exposed: B and the disk center lie on opposite sides. The drawing is schematic; the exact witness geometry and the paired tangent residuals are proved in the text. No auxiliary Minkowski sum or cap covering is needed.

Indeed $P_X(\eta) \geq 0$ gives $\Delta\sqrt{N_X - \eta^2} \geq |(h - 2\eta)N_X - \eta\nu|$, which implies the required lower bound even when the expression inside the absolute value is negative. No sign is lost by squaring.

The exact certificate uses a smaller rationally specified radius. The following elementary transfer is the reason those existing signs suffice for the actual-radius tangencies.

Lemma E.9 (Simultaneous radius transfer). *Let X, Y be vectors, $d \geq 0$, and $0 \leq e_0 \leq 1/3$. If*

$$\|(1-2e_0)X - e_0 Y\| \leq d, \qquad \|(1-2e_0)Y - e_0 X\| \leq d,$$

then both inequalities hold at every $e \in [e_0, 1/3]$.

Proof. The difference of the two vectors at e_0 is $(1-e_0)(X-Y)$, so

$$\|X - Y\| \leq \frac{2d}{1-e_0} \leq 3d.$$

At $e = 1/3$ the vectors become $(X-Y)/3$ and $(Y-X)/3$, both in the closed radius-d ball. Each vector depends affinely on e, and the ball is convex. Interpolation proves both inequalities, including the endpoint case. This transfers the pair together; it does not assert monotonicity of either squared residual separately. ☐

Lemma E.10 (Solved witness-enclosure optimization). *For every parameter pair in equation (16),*

$$\Lambda(K_{\mathrm{wit}}(a,b)) \geq 1.$$

Proof. If $c_* \leq 2/3$, the centered disk gives $\Lambda(K_{\mathrm{wit}}) \geq 3(1-c_*) \geq 1$. Otherwise $0 < \eta < h/3$, and it suffices to enclose $\widehat{K}$ by equation (94). The line contacts satisfy equation (99).

theorem F.3 proves the unchanged eight integer-polynomial signs at the rational radius $0 < \bar{\eta} = h(1-\bar{c}) \leq \eta$. The Gram factors and paired transfer in lemma F.2 turn those signs into $P_A(\eta), P_C(\eta) \geq 0$. Thus both tangent sums in equation (100) are at least h as well. All four contacts in equation (98) have support sum at least h. Therefore

$$\Lambda(K_{\mathrm{wit}}) \geq \Lambda(\widehat{K}) \geq 1.$$

☐

Appendix F. Exact Disk-Tangent Residual Certificate

The two disk-tangent contact bounds in the zero-gap nine-point theorem are certified by exact symbolic arithmetic. This section gives the mathematical reduction and verification algorithms. The finite sparse data and the two exact programs form the electronic appendix at an immutable revision; they are included in the accompanying source package. The PDF prints the certificate construction and checking rules, not every machine coefficient. No floating-point arithmetic, interval arithmetic, adaptive subdivision, branch-and-bound, or unrecorded numerical tolerance is used.

F.1. **Certificate manifest.** The machine-readable provenance manifest identifies the six sparse-data shards and both exact verifiers by full Git blob and records their roles. They decode to one canonical sparse-polynomial transcript with SHA–256 digest

`dc46aaf263655d5159ecd3a81db72ee82477951d06172f4743b248df37209485`

The derivation verifier reconstructs the transcript from the witness formulas and compares every coefficient with the authenticated object. The positivity verifier derives the Bernstein coefficients from that object and checks their signs exactly.

F.2. **Rational radial upper envelopes.** Retain the notation of Section E. Thus

$$p = 1-b, \qquad q = 1-a, \qquad s = p+q, \qquad m = \min\{p,q\}, \qquad M = \max\{p,q\},$$
$$\chi = s^4 - s^2 + mM, \qquad h = \frac{\sqrt{3}}{2}, \qquad c_* = c_{\max}(p,q).$$

The strict witness domain gives

$$mM > (1-s)(2-s), \qquad m > 1-s, \qquad s > \frac{2}{3}. \tag{102}$$

Put

$$F_m(x) = x^4 - x^2 + mx - m^2, \qquad \kappa = F'_m(s) = 4s^3 - 2s + m.$$

Then

$$\kappa > 4s^3 - 3s + 1 = (s+1)(2s-1)^2 > 0.$$

Define

$$\bar{c}_L = s - \frac{\chi}{\kappa} \quad (\chi \le 0), \qquad \bar{c}_T = s - \frac{\chi}{1-m} \quad (\chi \ge 0). \tag{103}$$

Lemma F.1 (Branchwise radial upper bounds)**.** *On the corresponding radial cells,*

$$c_* \le \bar{c}_L < 1, \qquad c_* \le \bar{c}_T < 1.$$

At $\chi = 0$ both upper bounds equal $s = c_*$.

Proof. On the L cell, $F_m(s) = \chi \le 0$ and $F_m(c_*) = 0$. For $x \ge s > 2/3$, $F''_m(x) > 0$, so convexity gives

$$0 \ge F_m(s) + F'_m(s)(c_* - s),$$

and hence $c_* \le \bar{c}_L$. If $U = 1-s$, then

$$\chi + U\kappa > U\{m + 1 - 4U + 8U^2 - 3U^3\} > 0$$

by (102); therefore $\bar{c}_L < 1$.

On the T cell put

$$E = \sqrt{4s^2 - 3}, \qquad m_- = \frac{s(1-E)}{2}, \qquad m_+ = \frac{s(1+E)}{2}.$$

Then

$$\chi = (m - m_-)(m_+ - m), \qquad s - c_* = \frac{2(m - m_-)}{1+E}.$$

Consequently

$$\frac{\chi}{s - c_*} = \frac{1+E}{2}(m_+ - m) \le 1 - m,$$

which gives $c_* \le \bar{c}_T$. The strict inequality $\chi > 0$ gives $\bar{c}_T < s < 1$. $\square$

Reflect so that $z = a - b \ge 0$ and put $U = a + b - 1$. Then

$$s = 1 - U, \qquad m = \frac{1 - U - z}{2}, \qquad D^2 = z^2 + 6U + 3U^2 =: K(U,z),$$

and

$$\Xi(U,z) = 4\chi = 1 - 10U + 21U^2 - 16U^3 + 4U^4 - z^2,$$
$$G(U,z) = 2\kappa = 5 - 21U + 24U^2 - 8U^3 - z.$$

Thus

$$\bar{c}_L = 1 - U - \frac{\Xi(U,z)}{2G(U,z)}, \qquad \bar{c}_T = 1 - U - \frac{\Xi(U,z)}{2(1+U+z)}. \tag{104}$$

F.3. **Gram reduction and the geometric implication.** On the active cell let $\bar{c}$ denote the corresponding expression in (104), and put

$$\bar{\eta} = h(1-\bar{c}), \qquad e = 1-\bar{c}, \qquad t = 2\bar{c} - 1.$$

Lemma F.1 gives $0 < \bar{\eta} \le \eta = h(1 - c_*)$. The algebra below is evaluated at $\bar{\eta}$; lemma E.9 will transfer both residual signs to the actual radius η.

Represent a global vector as (x, hy) and put $C' = \mathsf{R}^{-1}C$. Define

$$N_A = \|A\|^2, \qquad N_C = \|C\|^2, \qquad \nu = \langle A, C' \rangle,$$
$$\Delta_0 = x_{C'}y_A - y_{C'}x_A > 0.$$

Thus $h\Delta_0 = \mathrm{cross}(C', A) = \mathrm{cross}(C, \mathsf{R}A) > 0$. The authenticated code uses the opposite signed determinant, but every occurrence is squared; its integer polynomials and digest are unchanged. The Gram identity is

$$h^2\Delta_0^2 = N_A N_C - \nu^2. \tag{105}$$

Define

$$Q_A = \Delta_0^2 - \|tA - eC'\|^2, \qquad Q_C = \Delta_0^2 - \|tC' - eA\|^2. \tag{106}$$

Lemma F.2 (Residual signs imply the actual-radius tangent bounds)**.** *If $Q_A, Q_C \ge 0$, then the two actual-radius residuals $P_A(\eta), P_C(\eta)$ in equation (101) are nonnegative. Consequently both tangent contact sums in equation (100) are at least h.*

Proof. Let $\bar{\tau} = h - 2\bar{\eta} = ht$. Expanding (106) and using (105) gives

$$\bar{P}_A := (N_A - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_A - \bar{\eta}\nu)^2 = h^2 N_A Q_A, \tag{107}$$

$$\bar{P}_C := (N_C - \bar{\eta}^2)h^2\Delta_0^2 - (\bar{\tau}N_C - \bar{\eta}\nu)^2 = h^2 N_C Q_C. \tag{108}$$

The signs $Q_A, Q_C \ge 0$ are exactly

$$\|(1-2e)A - eC'\| \le \Delta_0, \qquad \|(1-2e)C' - eA\| \le \Delta_0.$$

Since $0 < e = \bar{\eta}/h \le \eta/h < 1/3$, lemma E.9 gives the same inequalities at η/h. The identical Gram factors at that radius prove $P_A(\eta), P_C(\eta) \ge 0$. Since $h\Delta_0 > 0$, the unsquared bounds following equation (101) show that both tangent contact sums are at least h. Transferring the pair avoids any assumption that a single squared residual is monotone in the disk radius. □

F.4. **Eight exact integer-polynomial signs.** The Newton witnesses A, C are rational functions of U, z, D. Substituting (104) into (106), clearing denominators, and reducing by

$$D^2 - K(U,z) = 0$$

gives, for $B \in \{L, T\}$ and $X \in \{A, C\}$,

$$\mathrm{num}(Q_{B,X}) = 32(K+3)^2\{A_{B,X}(U,z) + DB_{B,X}(U,z)\}, \tag{109}$$

where all eight core polynomials lie in $\mathbb{Z}[U,z]$. Every omitted denominator is strictly positive on the strict domain: its factors are positive constants, $(D+3)^4$, $G(U,z)^2$ or $(1+U+z)^2$, and squares of the two Newton-derivative numerators. Here $G = 2\kappa > 0$, and the fixed-line sign theorem proves both Newton derivatives strictly negative.

The derivation verifier constructs A, C, both rational envelopes, and all four residuals directly in the exact field $\mathbb{Q}(U, z, D)$. It performs the reduction modulo $D^2 - K$, removes the common positive factor, primitive-normalizes the eight integer polynomials, and compares their sparse term lists coefficient by coefficient with the authenticated transcript. Thus the data are checked against the geometric formulas rather than trusted as an independent precomputation.

F.5. **Three global positive-basis certificates.** For integers $0 \le i \le m$, the Bernstein basis polynomial is $B_i^m(r) = \binom{m}{i} r^i (1-r)^{m-i}$. A tensor Bernstein expansion on $[0,1]^2$ has the form

$$P(r,s) = \sum_{i=0}^{m} \sum_{j=0}^{n} b_{ij} B_i^m(r) B_j^n(s).$$

Every basis term is nonnegative, and their sum is one. Therefore nonnegative coefficients certify $P \ge 0$ on the whole square; strictly positive coefficients certify strict positivity. The exact coefficient conversion formula is given below.

Set

$$\mathcal{G}(U) = 1 - 6U - 3U^2, \qquad \mathcal{P}(U) = 1 - 10U + 21U^2 - 16U^3 + 4U^4,$$
$$U_\triangle = 1 - \frac{\sqrt{3}}{2}, \qquad U_{\max} = \frac{2}{\sqrt{3}} - 1.$$

The complete reflected domain is the union of the three closed cells

cell	U-range	z^2-range
T	$0 \le U \le U_\triangle$	$0 \le z^2 \le \mathcal{P}(U)$
L_0	$0 \le U \le U_\triangle$	$\mathcal{P}(U) \le z^2 \le \mathcal{G}(U)$
L_1	$U_\triangle \le U \le U_{\max}$	$0 \le z^2 \le \mathcal{G}(U)$

For the T cell use

$$U = \frac{(x-1)(3-x)}{4x}, \qquad z = y\frac{9-x^4}{8x^2}, \qquad 1 \le x \le \sqrt{3}, \quad 0 \le y \le 1,$$

and then $x = 1 + (\sqrt{3}-1)r$. This maps $[0,1]^2$ onto the complete cell. After multiplication by positive powers of 2 and x, the four transformed core polynomials have global tensor Bernstein expansions with data

	bidegree	coefficient count	zero count
$A_{T,A}$	$(88, 22)$	2047	2
$B_{T,A}$	$(80, 20)$	1701	0
$A_{T,C}$	$(88, 22)$	2047	2
$B_{T,C}$	$(80, 20)$	1701	0.

Every nonzero coefficient is strictly positive.

For an L-cell polynomial write

$$F(U,z) = E_F(U,w) + zO_F(U,w), \qquad w = z^2,$$

and

$$R_F(U,w) = E_F(U,w)^2 - wO_F(U,w)^2.$$

The implications $E_F \ge 0$ and $R_F \ge 0$ give $E_F \ge \sqrt{w}\,|O_F|$, hence $F \ge 0$ for $z \ge 0$. Use the exact charts

$$U = U_\triangle r, \qquad w = \mathcal{P}(U) + s\{\mathcal{G}(U) - \mathcal{P}(U)\}$$

for L_0, and

$$U = U_\triangle + (U_{\max} - U_\triangle)r, \qquad w = s\mathcal{G}(U)$$

for L_1. The positivity verifier forms one global Bernstein expansion for each of E_F, R_F on each square. The authenticated transcript records the bidegrees and coefficient counts of all sixteen expansions; every coefficient is strictly positive.

The verifier implements polynomial addition, multiplication, substitution, and power-to-Bernstein conversion over exact integers. Coefficients in $\mathbb{Q}(\sqrt{3})$ are represented as rational pairs and signed by exact rational comparisons. For $x + y\sqrt{3}$ with rational x, y, equal signs are immediate. When $x > 0 > y$, its sign is that of $x^2 - 3y^2$; when $y > 0 > x$, it is that of $3y^2 - x^2$. Zero components are handled directly. Thus no numerical approximation to $\sqrt{3}$ is needed. If $P(r,s) = \sum a_{ij} r^i s^j$ has bidegree (m,n), its exact tensor Bernstein coefficients are

$$b_{k\ell} = \sum_{i \le k} \sum_{j \le \ell} a_{ij} \frac{\binom{k}{i}}{\binom{m}{i}} \frac{\binom{\ell}{j}}{\binom{n}{j}}.$$

This is precisely the conversion implemented by the verifier.

Theorem F.3 (Exact tangent-residual certificate)**.** *On the complete strict witness domain with $c_* > 2/3$, all eight polynomials in* (109) *are nonnegative. Hence $Q_A, Q_C \ge 0$, and the two actual-radius tangent contact bounds in equation (100) hold.*

Proof. The T chart proves the four T-cell core signs by nonnegative Bernstein coefficients. On each L chart the positive Bernstein certificates for E_F and R_F prove the four L-cell signs. Since $D \ge 0$, (109) gives $Q_A, Q_C \ge 0$ for $z \ge 0$; reflection supplies $z \le 0$. Lemma F.2 then gives the two tangent contact bounds. □

Remark F.4 (Reproduction). The authenticated data and the two verifiers constitute the exact electronic certificate used by the proof. Reproduce it from its package directory by executing

```
python verify_mixed_overlap_core_derivation.py
python verify_global_core_positivity.py
```

Both commands use exact arithmetic and must reproduce the transcript digest above and the asserted sign checks.